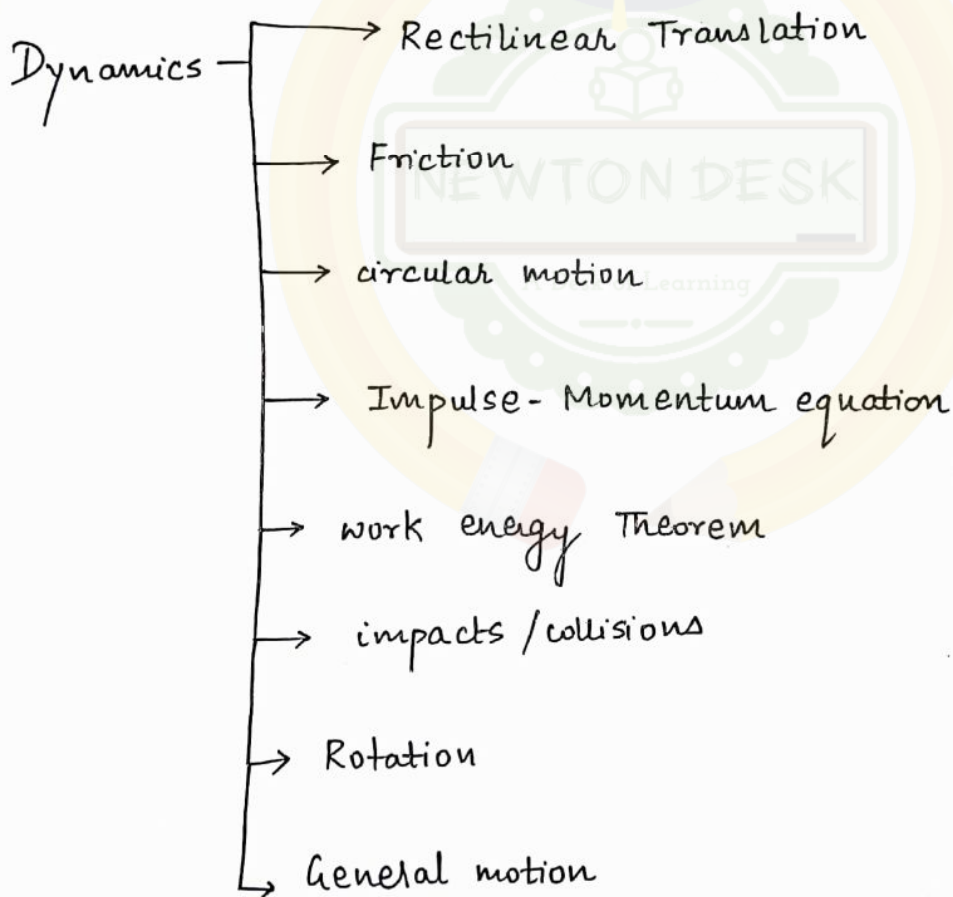
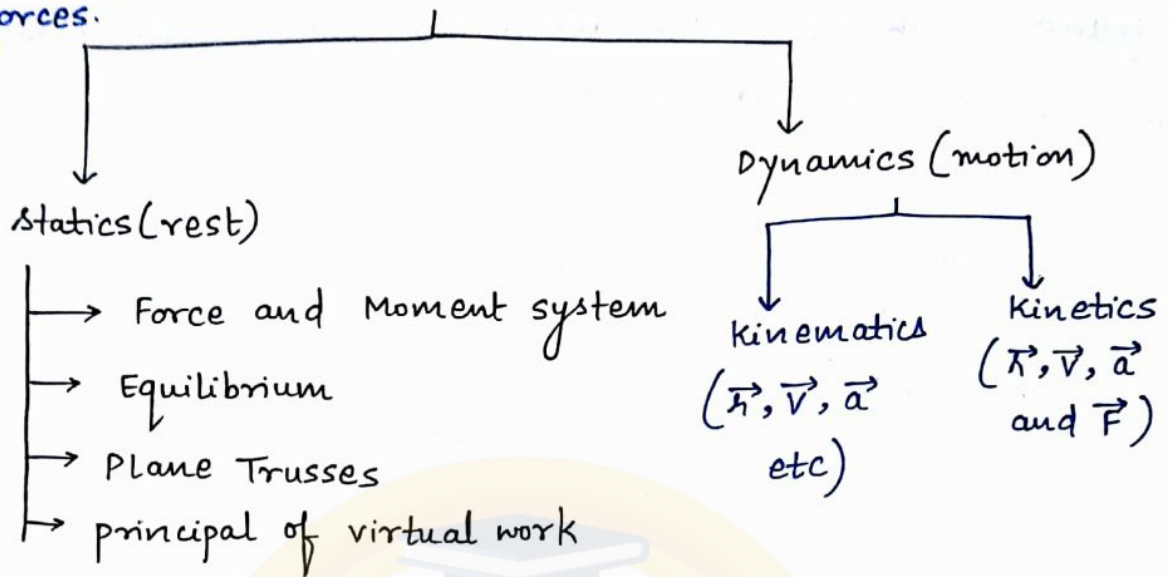


Engineering Mechanics

Study of motion of rigid bodies under the action of forces.



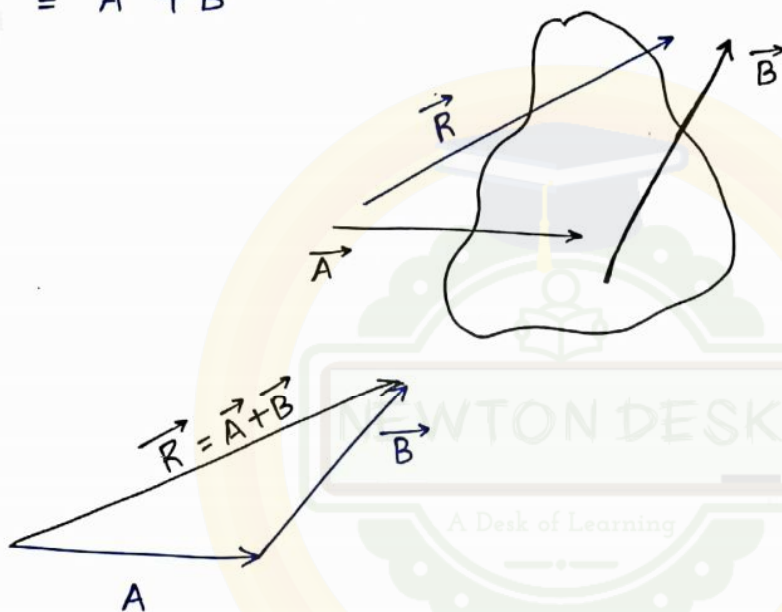
Vector Algebra :-

Vector $\rightarrow$ Those physical quantity which have magn. as well as dirn. & follows a law of vector addn. are called vector quantities.

Ex:- velocity vector $\vec{v}$, $\vec{a}$, $\vec{T}$ etc.

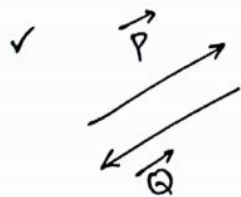
Triangle Law of Vector addition $\rightarrow$

$$\vec{R} = \vec{A} + \vec{B}$$



$$\checkmark \vec{A} + \vec{B} + \vec{C} = 0$$

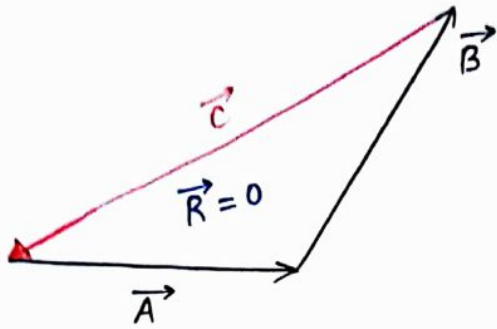
$$\vec{C} = -(\vec{A} + \vec{B})$$



$$|\vec{P}| = |\vec{Q}| \quad (P = Q)$$

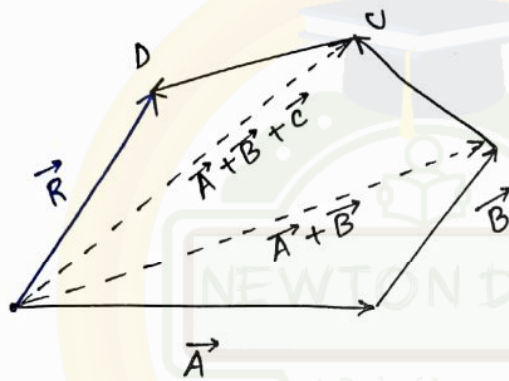
$$\vec{P} = -\vec{Q}$$

$$\vec{P} + \vec{Q} = \mathbf{0}$$

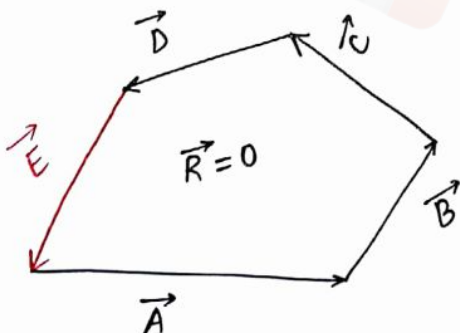


Polygon Law →

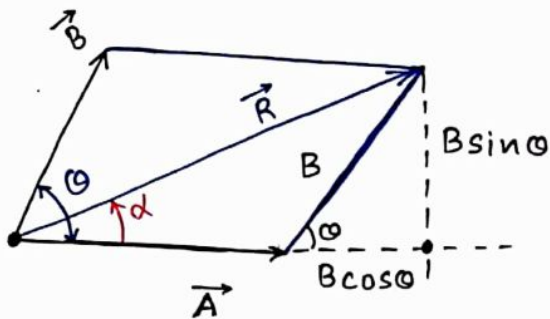
$$\vec{R} = \vec{A} + \vec{B} + \vec{C} + \vec{D}$$



$$\vec{A} + \vec{B} + \vec{C} + \vec{D} + \vec{E} = 0$$



Parallelogram law :-



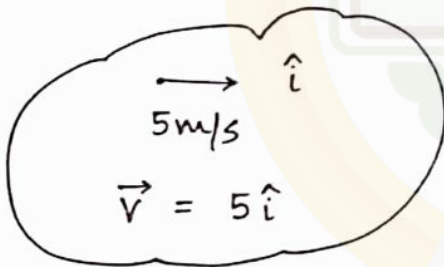
$$R^2 = (A + B \cos \theta)^2 + (B \sin \theta)^2$$

$$R = \sqrt{A^2 + B^2 + 2AB \cos \theta}$$

$$\tan \alpha = \frac{B \sin \theta}{A + B \cos \theta}$$

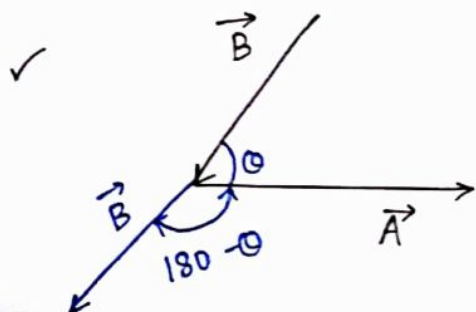
✓ vector = mag(dirn.)

✓ unit vector = $1(\text{dirn.}) \equiv \text{dirn.}$



$$\checkmark \quad \vec{A} = A_x \hat{i} + A_y \hat{j} + A_z \hat{k}$$

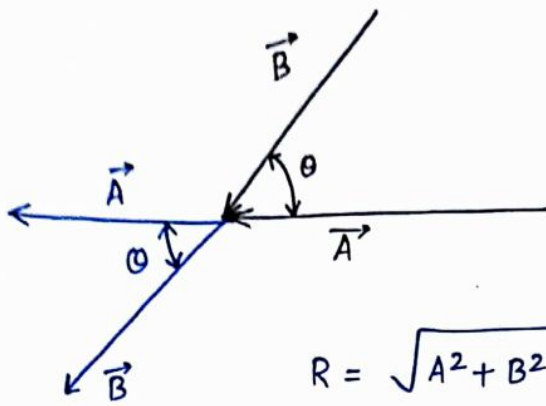
$$A = \sqrt{A_x^2 + A_y^2 + A_z^2}$$



$$R = \sqrt{A^2 + B^2 + 2AB \cos \theta}$$

$$R = \sqrt{A^2 + B^2 + 2AB \cos(180 - \theta)}$$

$$R = \sqrt{A^2 + B^2 - 2AB \cos \theta}$$



$$R = \sqrt{A^2 + B^2 + 2AB \cos \theta}$$

Product of Two Vectors:-

$\vec{A}$ and $\vec{B}$

1. Dot Product (Scalar Product) →

$$\vec{A} \cdot \vec{B} = C$$

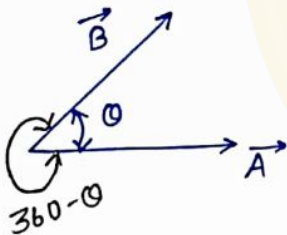
$$C = \underline{AB \cos \theta}$$

$$\vec{B} \cdot \vec{A} = D$$

$$D = \underline{BA \cos \theta}$$

$$C = D$$

$$\vec{A} \cdot \vec{B} = \vec{B} \cdot \vec{A}$$



$$\vec{A} = A_x \hat{i} + A_y \hat{j} + A_z \hat{k}$$

$$\vec{B} = B_x \hat{i} + B_y \hat{j} + B_z \hat{k}$$

$$\vec{A} \cdot \vec{B} = A_x B_x + A_y B_y + A_z B_z$$

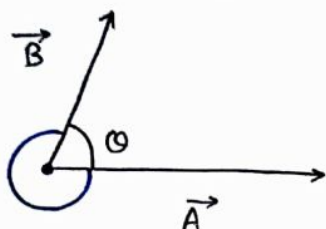
(li) Cross Product (vector Product) $\rightarrow$

$$\vec{A} \times \vec{B} = \vec{C}$$

$$\vec{C} = |\vec{C}| \hat{c}$$

$$C = AB \sin \theta$$

$$\theta < 180^\circ$$



Right hand thumb Rule,

$\vec{C} \rightarrow$ $\perp$ to the plane, outward

$$\vec{B} \times \vec{A} = \vec{D}$$

$$\vec{D} = |\vec{D}| \hat{D}$$

$$|\vec{D}| = BA \sin \theta$$

$\hat{D} \rightarrow$ $\perp$ to the plane, inward

$$\vec{C} = -\vec{D}$$

$$(\vec{A} \times \vec{B}) = -(\vec{B} \times \vec{A})$$

$$\vec{A} \times \vec{B} =$$

$\hat{i}$	$\hat{j}$	$\hat{k}$
A_x	A_y	A_z
B_x	B_y	B_z

Direction of $\vec{C}$ will be $\perp$ to the plane containing $\vec{A}$ and $\vec{B}$ and will be decided by Right hand thumb rule.

Q The Resultant of 2 vectors $\vec{A}$ & $\vec{B}$ is R . If $\vec{B}$ is doubled, then the new Resultant is $\perp$ to $\vec{A}$, then:-

(a) $A = B$

(b) $A = R$

(c) $B = R$

(d) None.

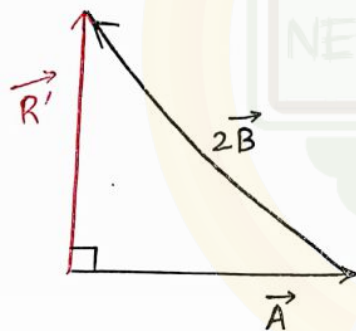
Sol:

$A \text{ \& } B \longrightarrow R$
 $\angle \theta$

$$A^2 + B^2 + 2AB\cos\theta = R^2 \quad \text{--- (1)}$$

$A \text{ and } 2B \longrightarrow R' \text{ is } \perp \text{ to } A$
 $\angle 90^\circ$

$$A^2 + 4B^2 + 4AB\cos\theta = (R')^2 \quad \text{--- (2)}$$



$$A^2 + (R')^2 = 4B^2 \quad \text{--- (3)}$$

from (2) and (3)

$$A^2 + 4B^2 + 4AB\cos\theta = 4B^2 - A^2$$

$$4AB\cos\theta = -2A^2$$

$$2AB\cos\theta = -A^2 \quad \text{--- (4)}$$

using value of eqn. (4) in eqn. (1)

$$A^2 + B^2 - A^2 = R^2$$

$$\boxed{B = R}$$

Q The Resultant of 2 vectors when acting at Right angles is 10 KN. If they act at 60° , their resultant is $5\sqrt{6}$ KN. The magnitude of individual vectors are ?

Sol

$$A \perp B ; \theta = 90^\circ ; R = 10 \text{ KN}$$

$$A^2 + B^2 = 10^2 = 100 \rightarrow (1)$$

$$A \times B ; \theta = 60^\circ ; R = 5\sqrt{6} \text{ KN}$$

$$A^2 + B^2 + AB = (5\sqrt{6})^2 = 150 \rightarrow (2)$$

from (1) and (2)

$$AB = 50 \rightarrow (3)$$

$$\begin{aligned}(A+B)^2 &= A^2 + B^2 + 2AB \\ &= 100 + 2(50) \\ &= 200\end{aligned}$$

$$A+B = \sqrt{200}$$

$$A+B = 10\sqrt{2}$$



$$2A = 10\sqrt{2}$$

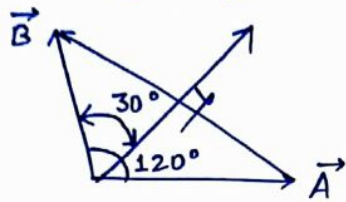
$$A = 5\sqrt{2} \text{ KN} = B$$

$$\begin{aligned}(A-B)^2 &= A^2 + B^2 - 2AB \\ &= 100 - 100 \\ &= 0\end{aligned}$$

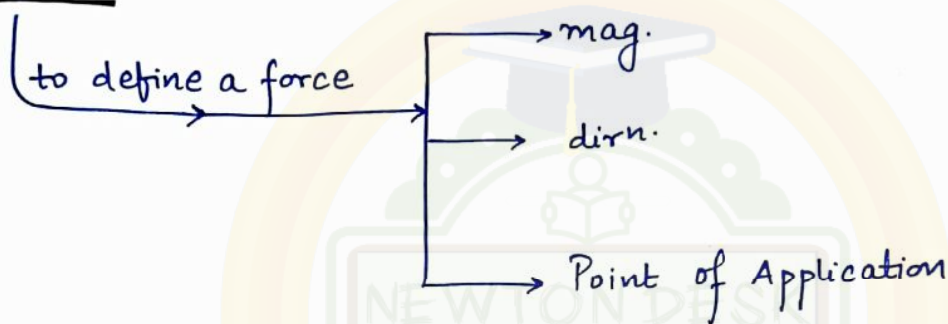
$$A-B = 0$$

$$A = B$$

Q. Two forces are acting at a point making an angle of 120° with each other. If the Resultant force is $\perp$ to the smaller force (smaller force = 50 N), the larger force is (100 N)



* FORCE :-



* MOMENT of a force (Torque) :-

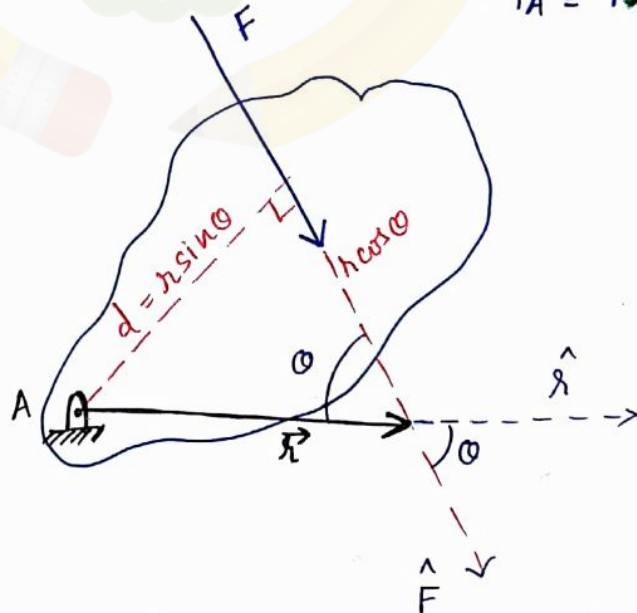
$$\vec{T}_A = \vec{r} \times \vec{F}$$

$$T_A = r F \sin \theta$$

Dirn. $\rightarrow \perp$ inward through A

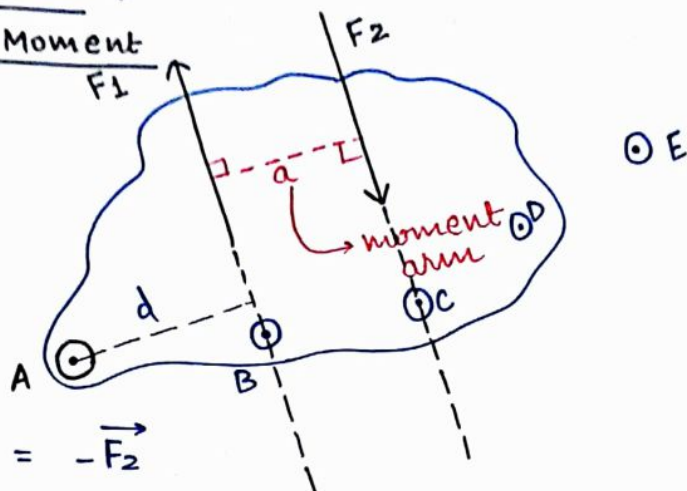
$$T_A = F \cdot d \text{ (ccw)}$$

clockwise



→ here 1st inward dirⁿ through A → $\curvearrowright$ ← fingers dirⁿ.

* COUPLE →
& its Moment



$$\vec{F}_1 = -\vec{F}_2$$

$$F_1 = F_2 = F$$

$$\boxed{\sum \vec{F} = 0} \leftarrow \text{only net force} = 0 \text{ but not moment}$$

$\curvearrowright$ (+)
inward

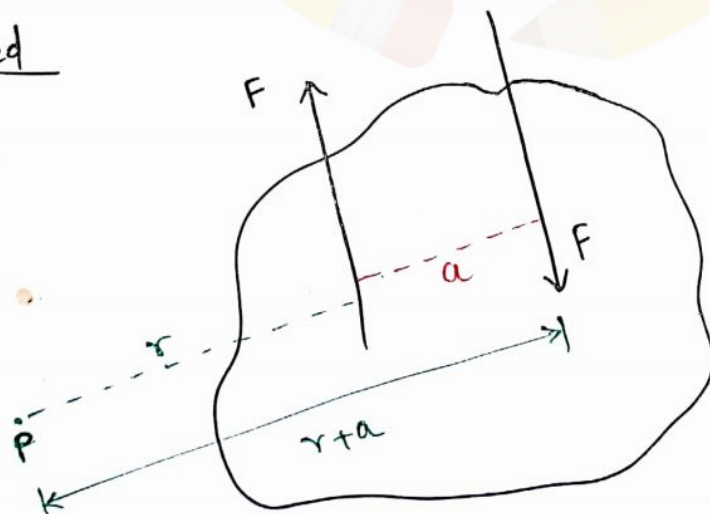
$\curvearrowleft$ (-)
outward

$$M_A = -Fd + F(a+d) = Fa ; \text{cw}$$

$$M_B = Fa ; \text{cw}$$

$$M_C = Fa ; \text{cw} = M_D = M_E$$

Simplified



Moment of a couple = one of the force $\times$ moment arm
 $= Fa ; \text{cw} \rightarrow (\text{clockwise})$

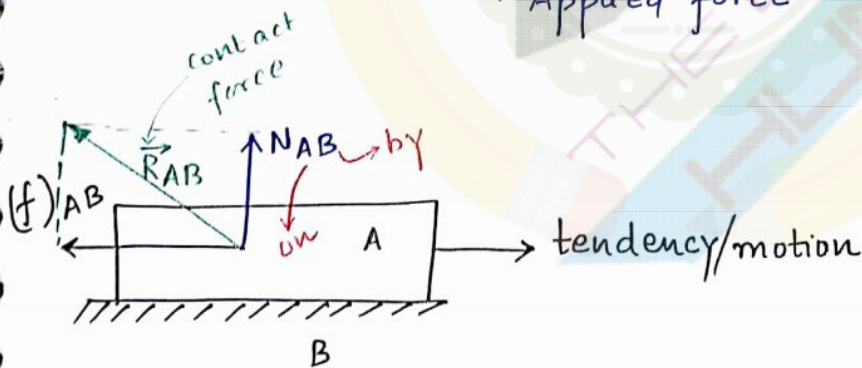
Couple vector $\rightarrow$ $\perp$ to the plane of couple and will be decided by Right hand thumb rule.

✓ couple is an arrangement of two equal and opposite forces acting $\parallel$ to each other and whose moment.

remains uniform throughout on the body on which it is acting.

* Types of Forces :-

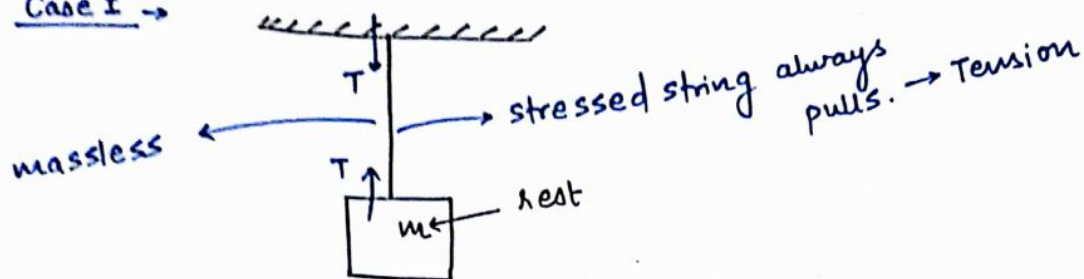
- $\rightarrow$ Gravity $\rightarrow [wt = mg \downarrow]$
- $\rightarrow$ contact force $\left\{ \begin{array}{l} \text{Normal } R_{xn} \\ \text{friction} \end{array} \right.$
- $\rightarrow$ Tension
- $\rightarrow$ Spring force $\rightarrow F = kx$
- $\rightarrow$ Applied force $\rightarrow$ given.



Non A by B

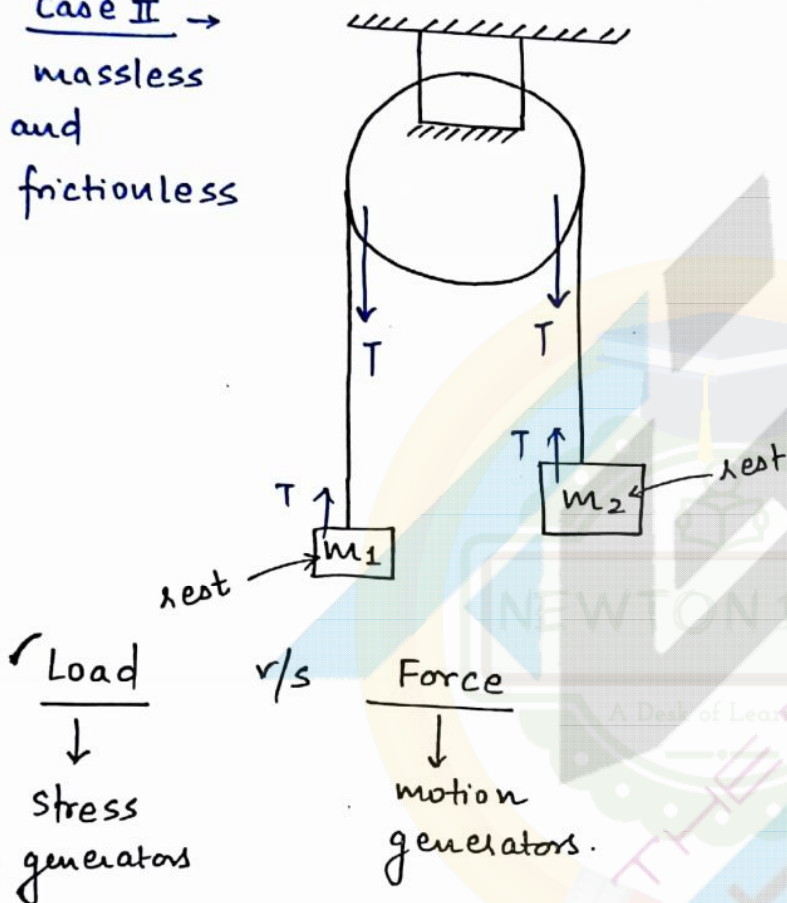
Tension $\left\{ \begin{array}{l} \text{case 1.} \\ \text{case 2.} \end{array} \right.$

Case I →



Case II →

massless
and
frictionless



String offered Tension

mass of both not concern

String pulls in stressed

condn.

stretched

* NEWTON'S 1st LAW :-

for a particle → if Resultant of forces

i.e. if $\vec{F}_R = 0$

then $\vec{a} = 0$

Rest

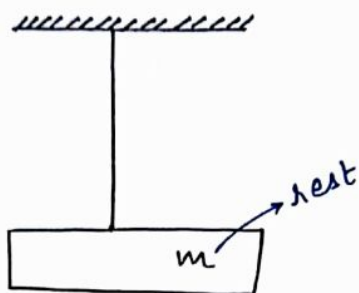
uniform velocity

for a Rigid Body $\rightarrow$

[CM $\rightarrow$ centre of mass]

if $(\vec{F}_R)_{\text{ext}} = 0$ then $\underline{\vec{a}_{\text{CM}}} = 0$

✓ Case I :-

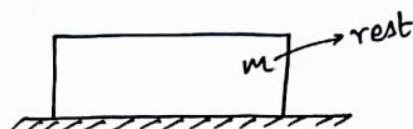


(m)
 $\vec{a}_{\text{CM}} = 0$
 $\Rightarrow (\vec{F}_R)_{\text{ext}} = 0$

$T = mg \rightarrow 1^{\text{st}} \text{ law}$



✓ Case II :-



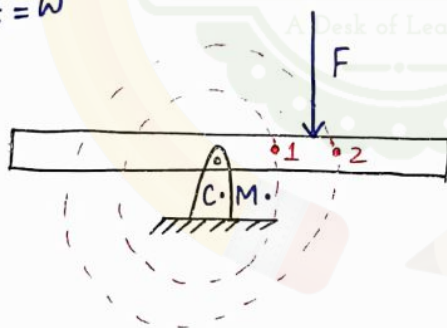
(m)
 $\vec{a}_{\text{CM}} = 0 \Rightarrow (\vec{F}_R)_{\text{ext}} = 0$

$N = mg \rightarrow 1^{\text{st}} \text{ law}$



✓ Case III :- $\omega t = \omega$

Rxn at hinge for the instant shown = ?



$a_1 \neq 0$
 $a_2 \neq 0$ } since N-1st law guarantees for only acc. of C.M. not for particle
 $a_1 \neq a_2$

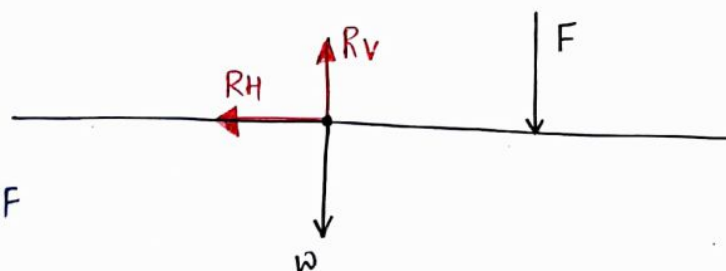
$\underline{\vec{a}_{\text{CM}}} = 0$ [Bcoz C.M. is lying on axis of rotation]

$\sum \vec{F}_H = 0$

$R_H = 0$

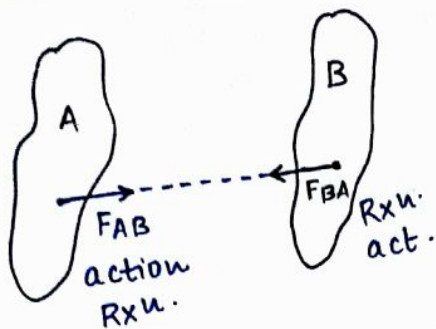
$\sum \vec{F}_V = 0$

$R_V = W + F$



R_V independent
 R_H independent
 $R \times \omega$

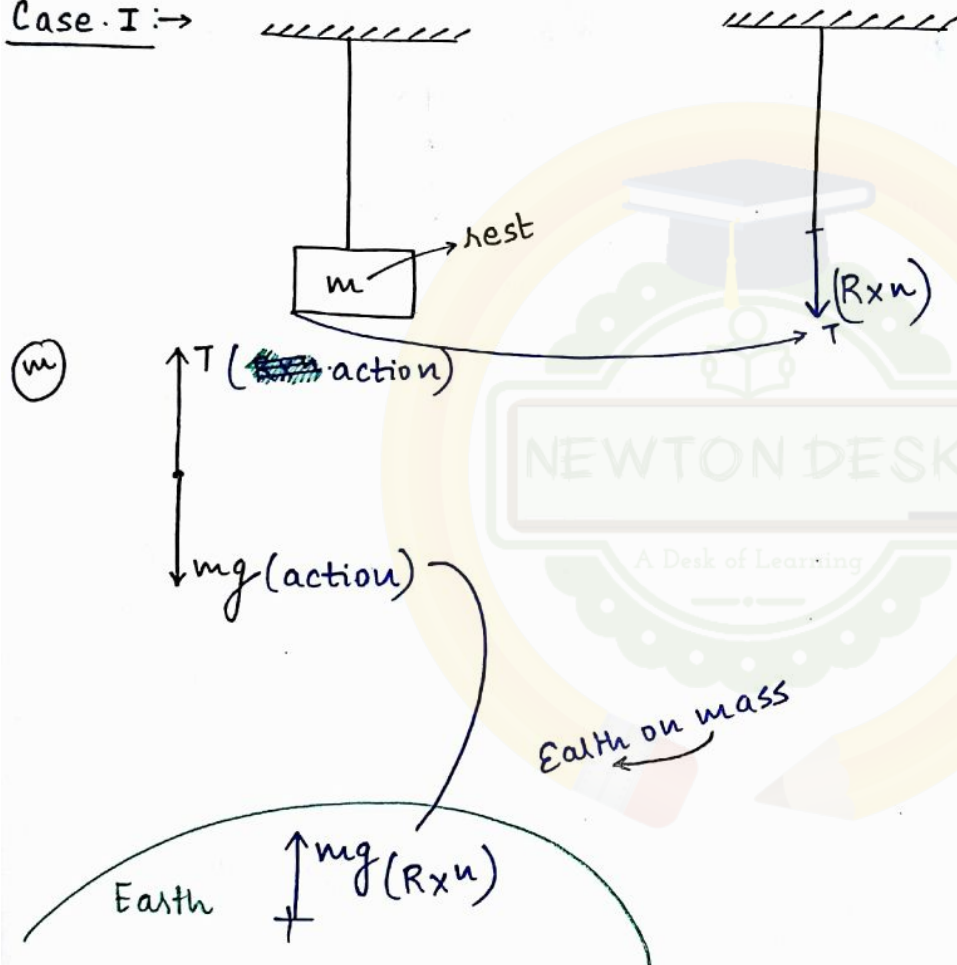
* NEWTON'S 3rd LAW :-



same in nature and
 ↓
 pulling or pushing in both
 opp. dirn.
 ↓
 dirn.
 +, -

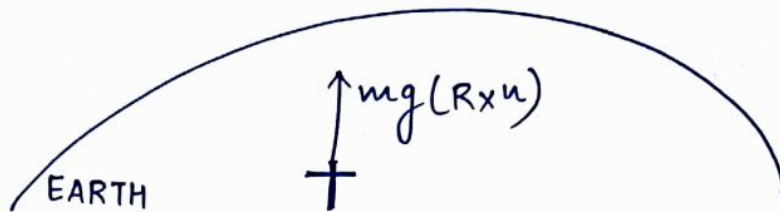
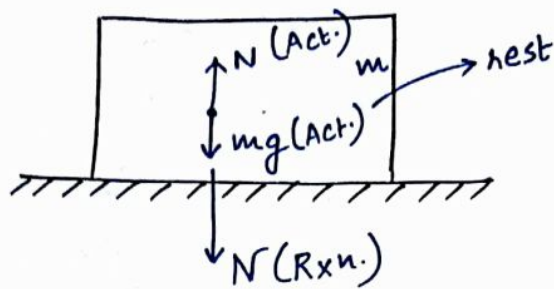
$$\vec{F}_{AB} = -\vec{F}_{BA}$$

Case I :-



mass pulls the string

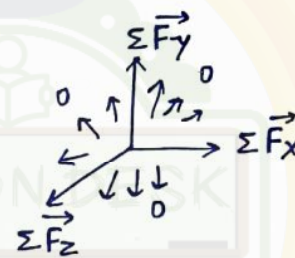
Case II :-



* EQUILIBRIUM :- $\rightarrow$ rest $\checkmark$ (mechanics)
 $\rightarrow$ uniform velocity $\rightarrow$ (FM)
 for a Rigid body

1. $\Sigma \vec{F} = 0$

$$\Sigma \vec{F}_x = \Sigma \vec{F}_y = \Sigma \vec{F}_z = 0$$

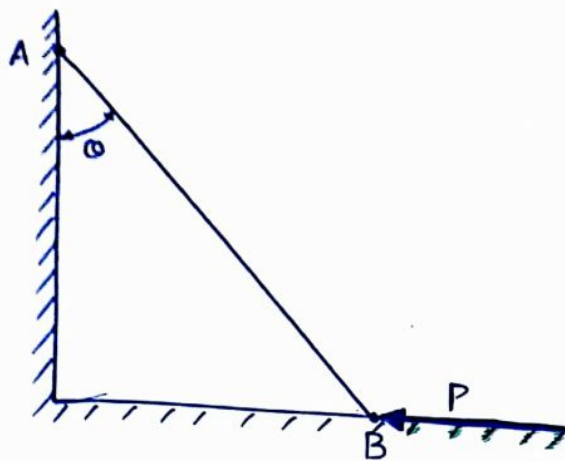


2. $\Sigma \vec{M} = 0$

$\hookrightarrow$ about any point or line in space.

Q. A ladder ^{AB} of weight w and length L is held in eqm. by a horizontal force P as shown in figure. assume ladder to be uniform body and all surfaces smooth. Find P .

sol



$$\sum \vec{F}_H = 0$$

$$P = N_A \rightarrow (1)$$

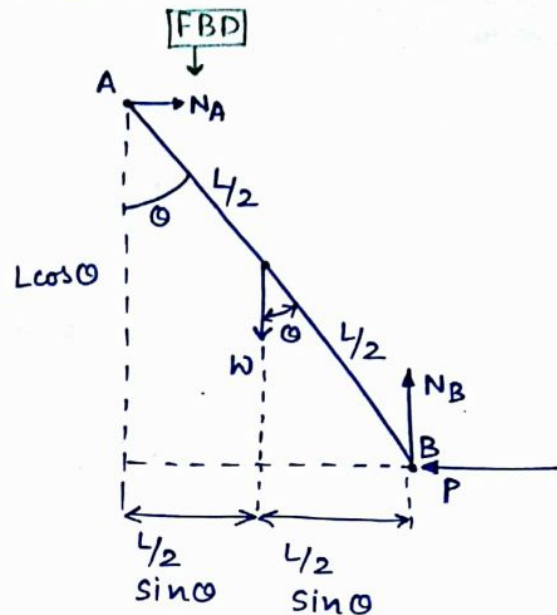
$$\sum \vec{F}_V = 0$$

$$W = N_B \rightarrow (2)$$

$$\sum \vec{M}_B = 0$$

$$N_A \times L \cos \theta = W \times \frac{L}{2} \sin \theta$$

$$N_A = \frac{W}{2} \tan \theta = P$$



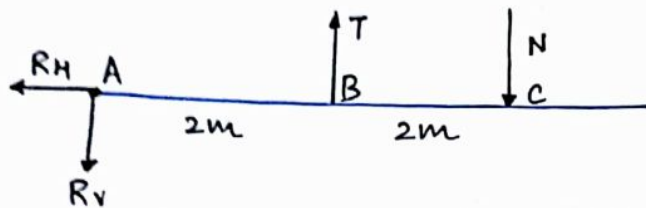
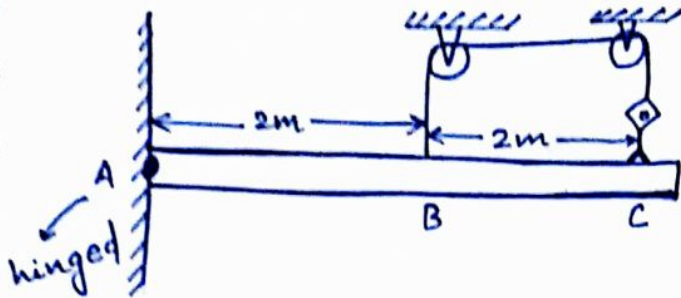
2nd method :-

Moment of cw couple = Moment of ACW couple

$$P \times L \cos \theta = \frac{W \times L}{2} \sin \theta$$

$$P = \frac{W}{2} \tan \theta$$

Q. A man weighing 600N stands on a horizontal beam of negligible weight at 'C' and holds a string passing over 2 smooth pulleys and attached to the point 'B' on the beam as shown in figure. The tension in the string is?



$$\sum \vec{M}_A = 0$$

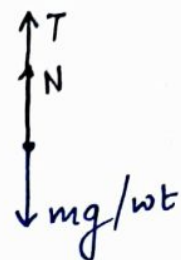
$$T \times 2 = N \times 4$$

$$T = 2N \quad \text{--- (1)}$$

$$T = 2(600 - T)$$

$$T = 1200 - 2T$$

$$T = 400N$$



$$T + N = wt$$

$$T + N = 600$$

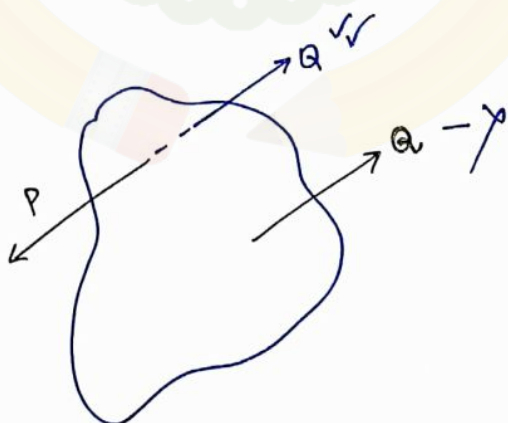
$$N = 600 - T \quad \text{--- (2)}$$

* Equilibrium $\rightarrow$ CASE I $\rightarrow$ Two-force system $\rightarrow$

$\vec{P}$ and $\vec{Q}$

$$1. \vec{P} + \vec{Q} = 0$$

$$\vec{P} = -\vec{Q}$$

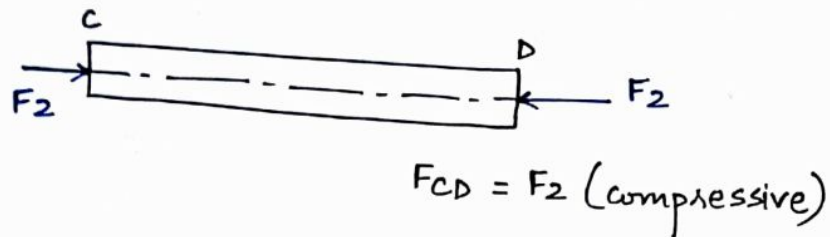
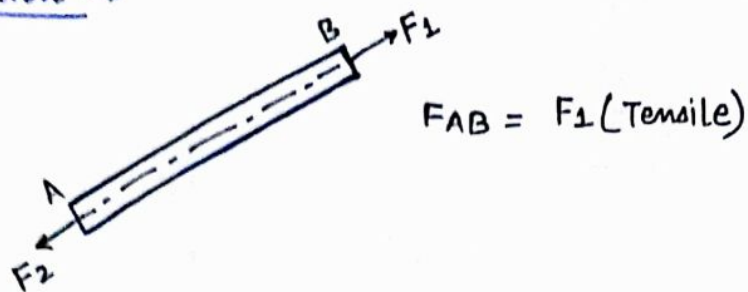


$$2. \sum \vec{M} = 0$$

$\hookrightarrow$ collinear (same line & same points)

To keep 2 forces in equilibrium, they must be equal in magnit, opposite in dirⁿ. and collinear in action.

Application →



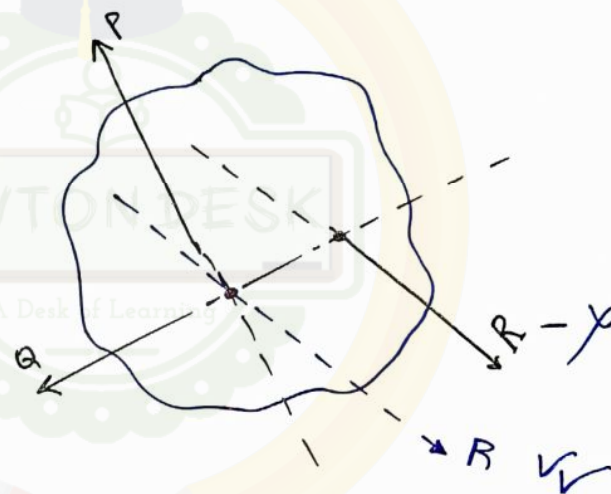
Case II :→ Three-force system →
 $\vec{P}$, $\vec{Q}$ and $\vec{R}$

1. $\vec{P} + \vec{Q} + \vec{R} = 0$.

↳ coplanar (same plane $\vec{H}$)

2. $\sum \vec{M} = 0$

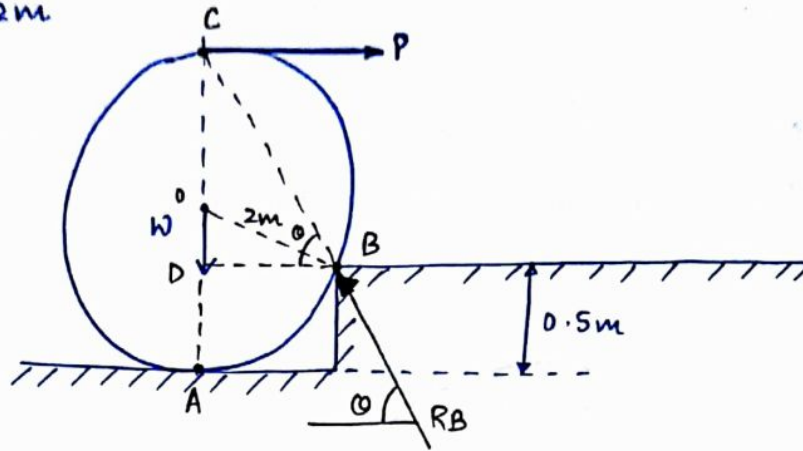
↳ concurrent (same point $\vec{H}$)



To keep three-forces in equilibrium, they must be coplanar and concurrent.

Q: find the horizontal force 'P' required to move the cylinder out of the disk^{tch} as shown in fig?

Sol) $W = 1000\text{N}$
 $\lambda = 2\text{m}$



$$\vec{W} + \vec{P} + \vec{R}_B = 0$$

In $\triangle ODB$

$$OD^2 + DB^2 = AB^2$$

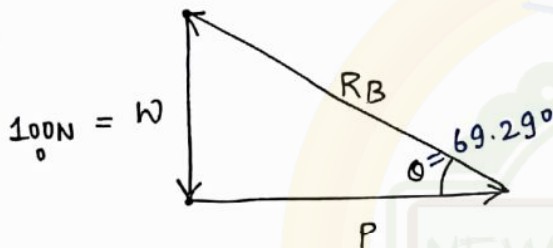
$$DB = \sqrt{2^2 - 1.5^2}$$

$$DB = 1.3228\text{m}$$

In $\triangle ABCD$

$$\tan \theta = \frac{CD}{DB} = \frac{3.5}{1.3228}$$

$$\theta = 69.29^\circ$$



$$\frac{W}{P} = \tan \theta = \frac{3.5}{1.3228}$$

$$P = \frac{1000 \times 1.3228}{3.5} = 377.9\text{N}$$

$$\frac{W}{R_B} = \sin 69.29^\circ$$

$$R_B = \frac{1000}{\sin(69.29^\circ)}$$

$$R_B = 1069.08\text{N}$$

$$(or) \sum \vec{M}_B = 0$$

$$P \times CD = W \times DB$$

$$P = \frac{1000 \times 1.3228}{3.5} = 377.9\text{N}$$

Since B is about $R_B = 0$ ✓.

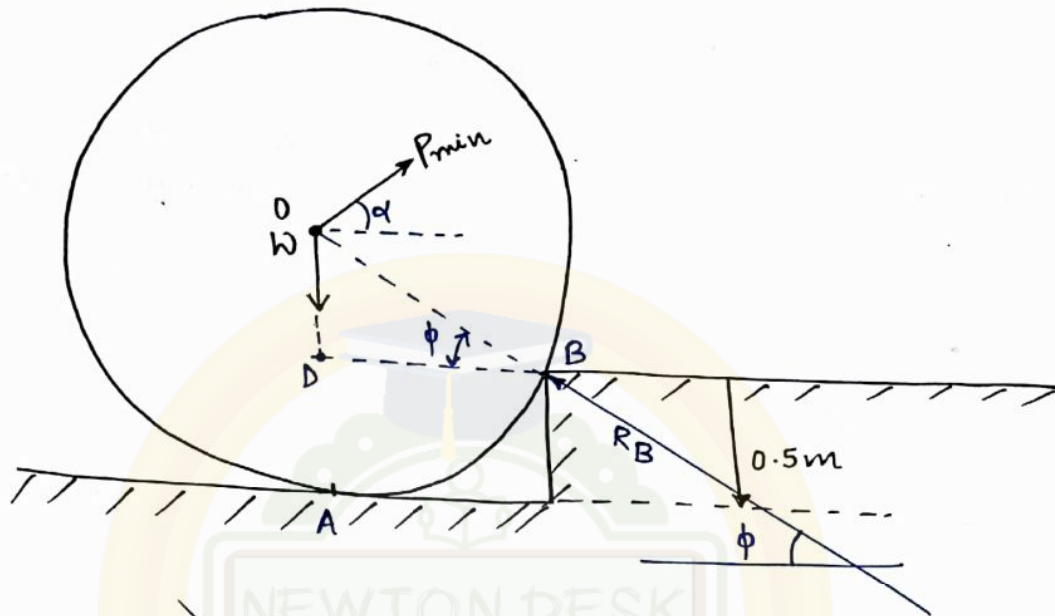
Note → When the cylinder is about to move out of the ditch it will lose its contact at point A. The only contact will be at point B.

- To keep the cylinder under 'W', 'P' and 'R_B', they must meet at a same point C.

Q Find the minimum force 'P' required to move the cylinder out of the ditch (small pit) as shown in the figure.

$$W = 1000\text{ N}$$

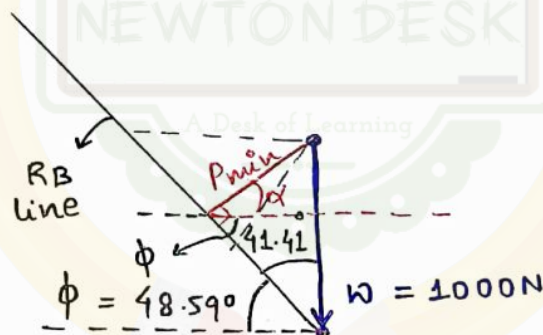
$$r = 2\text{ m}$$



In $\triangle ODB$

$$\sin \phi = \frac{OD}{OB} = \frac{1.5}{2}$$

$$\phi = 48.59^\circ$$



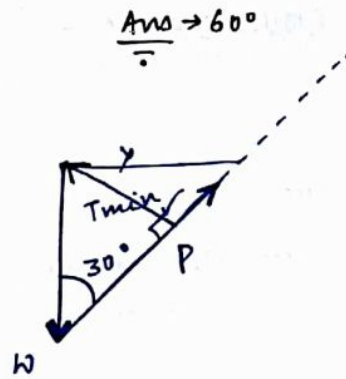
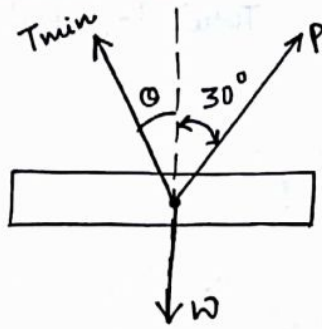
$$\frac{P_{\min}}{W} = \sin 41.41^\circ$$

$$P_{\min} = 1000 \times \sin 41.41^\circ$$

$$P_{\min} = 661.44\text{ N}$$

$$\alpha = 90 - \phi = 41.41^\circ$$

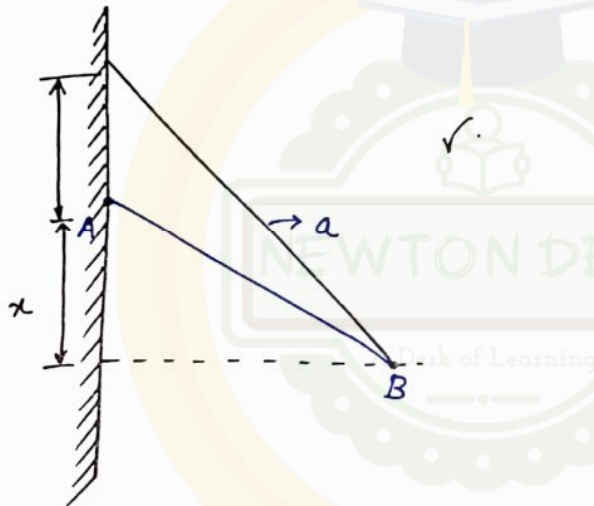
Q. A Block of weight 300N is supported as shown in figure, for T to be minimum, the value of θ is



$$\frac{Ans \rightarrow 60^\circ}{\cdot}$$

Q> A uniform bar AB of length L is supported against a smooth vertical wall and by a wire of length ' a ' as shown in figure.

The value of x to keep the ~~sys~~ Bar in eqm. is:- ?



$$AE = CD = DB$$

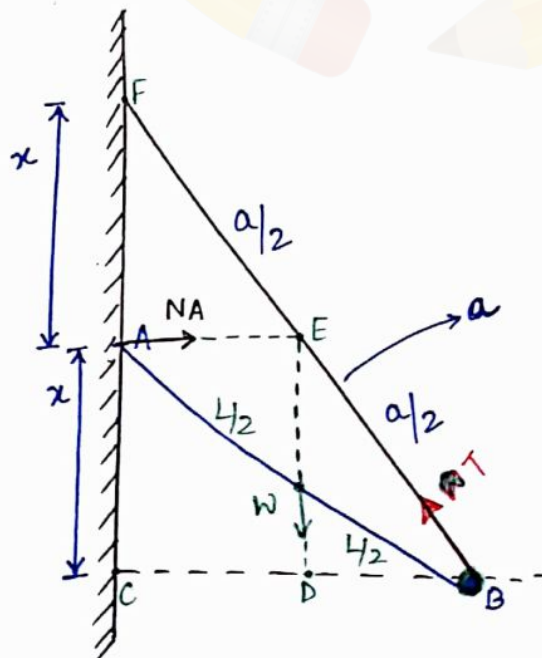
$$CB = 2AE$$

$$\sqrt{L^2 - x^2} = 2\sqrt{a^2/4 - x^2}$$

$$L^2 - x^2 = 4(a^2/4 - x^2)$$

$$x = \sqrt{\frac{a^2 - L^2}{3}}$$

Sol.>



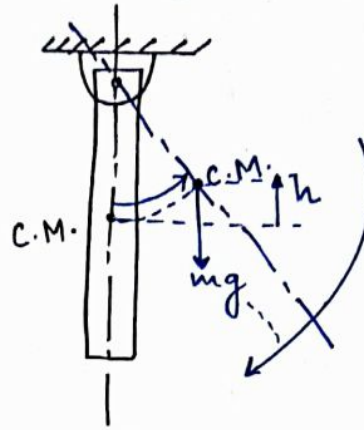
* TYPES OF EQUILIBRIUM :-

$$\text{Total Potential Energy} = V$$

(i) Stable →

$$V = \text{minimum}$$

Restoring moments > Disturbing moments



(ii) Unstable →

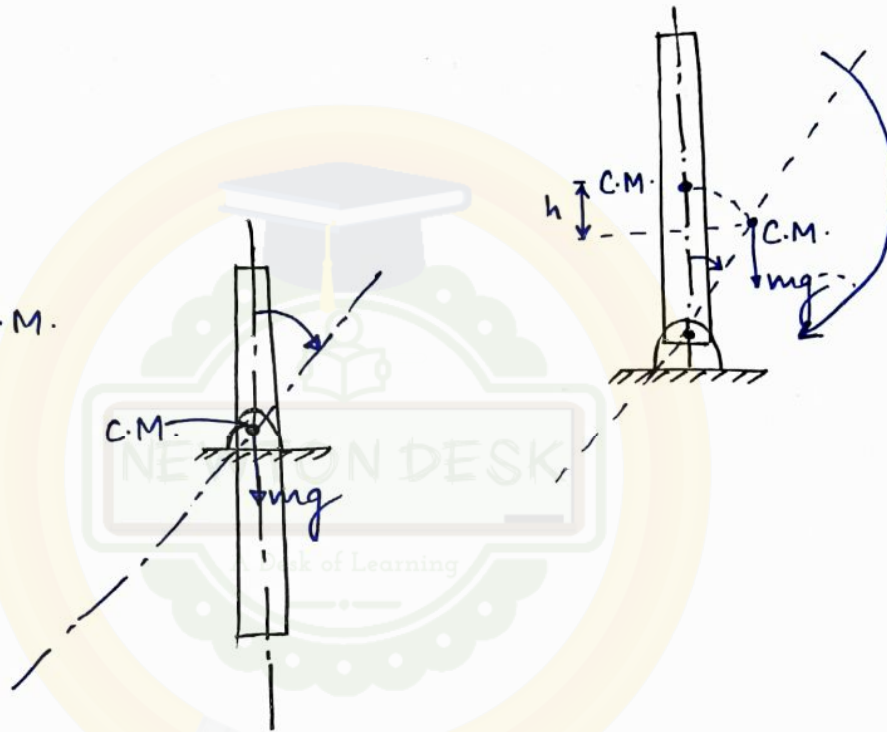
$$V = \text{maximum}$$

$$R.M. < D.M.$$

(iii) Neutral →

$$V = \text{const.}$$

$$R.M. = D.M.$$

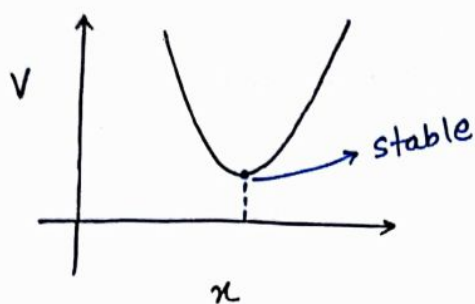


Case I → if system has single DOF

no. of independent variable = 1 (say x)

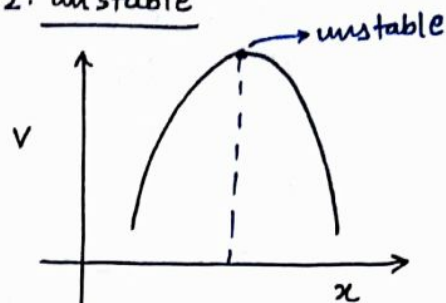
$$V = F(x)$$

1. Stable →



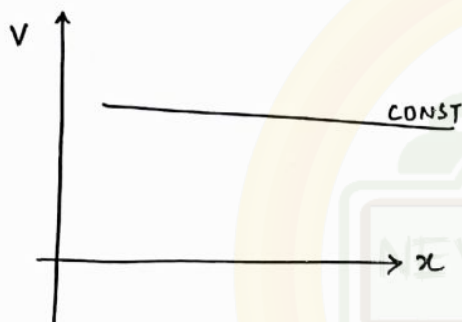
$$\frac{dv}{dx} = 0 ; \frac{d^2v}{dx^2} > 0$$

2. Unstable



$$\frac{dv}{dx} = 0 ; \frac{d^2v}{dx^2} < 0$$

3. Neutral



$$\frac{dv}{dx} = \frac{d^2v}{dx^2} = \frac{d^3v}{dx^3} = \dots = 0.$$

Case II → If system has multiple DOF

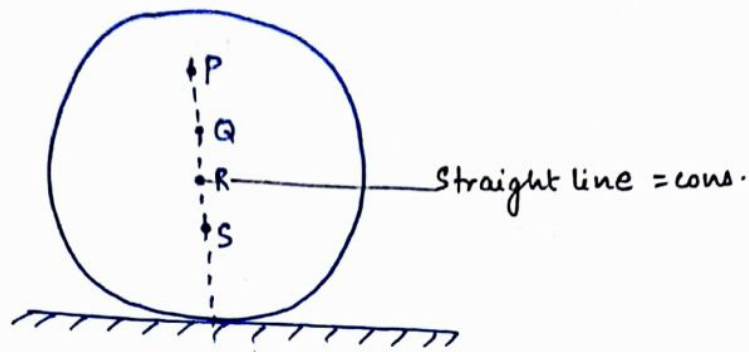
no. of independent variable > 1 (say $x, y, z, \dots$)

$$V = f(x, y, z, \dots)$$

$$\frac{\partial V}{\partial x} = \frac{\partial V}{\partial y} = \frac{\partial V}{\partial z} = 0 \dots \dots = 0 \text{ [for eqn}^m]$$

Q. for the sphere shown in fig, its C.G. must lie at which point so as to keep the sphere in stable eqn.?
where R is geometric centre.

- a) P
- b) Q
- c) R
- d) S



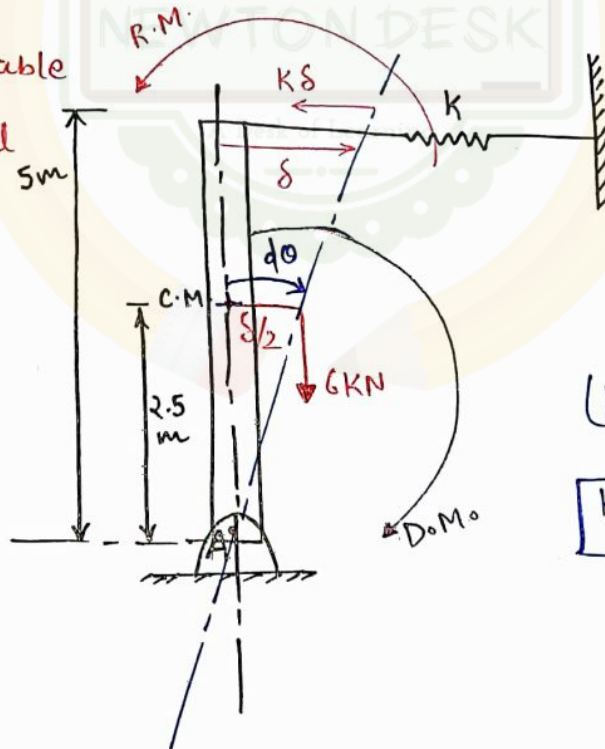
S $\rightarrow$ stable ($V = \min$)

R $\rightarrow$ neutral ($V = \text{const.}$)

P, Q $\rightarrow$ unstable

Q. A column of weight 6 kN and length 5m is hinged at point 'A' and supported by a spring of stiffness 'k' as shown in fig. The value of 'k' to keep it in equilibrium is

- a) $k > 0.6 \text{ kN/m}$
- b) $k < 0.6 \text{ kN/m} \rightarrow \text{unstable}$
- c) $k = 0.6 \text{ kN/m} \rightarrow \text{Neutral}$
- d) None



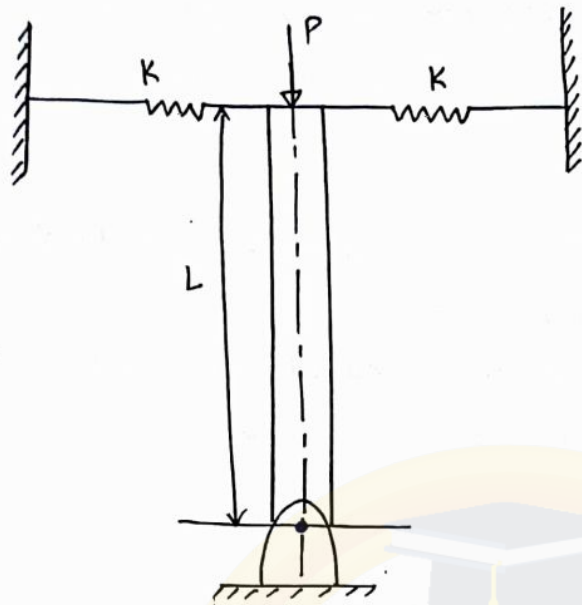
for stable eqm.

$$R.M. > D.M.$$

$$(k\delta)5 > 6 \times \frac{5}{2}$$

$$k > 0.6 \text{ kN/m}$$

Q. Find the limiting value of P to keep the system in equilibrium



(a) KL

(b) $2KL$

(c) 0

(d) $\frac{KL}{2}$

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PLANE TRUSSES

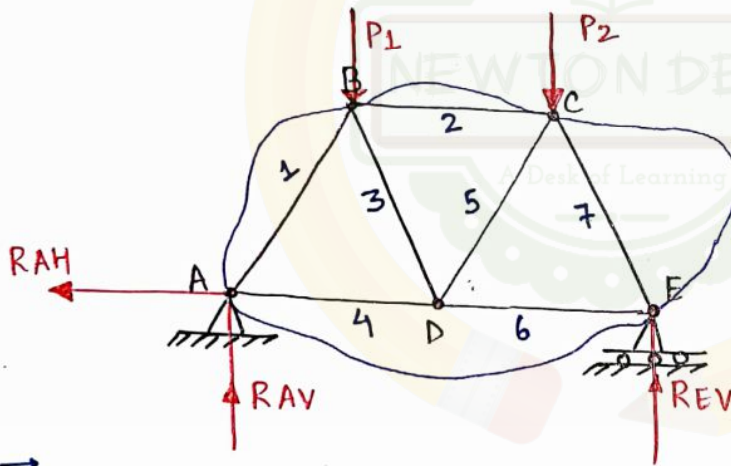
Truss is a Rigid structure in which members are subjected to either axial tensile or axial compressive load only.

Bending moment is 0 everywhere in the structure.

- * Conditions of Truss:-
- (a) Members should be pin jointed or hinged only.
 - (b) Only concentrated point loads should be apply.
 - (c) Load should be applied at the joints only.

Note:- A member should make joints at its ends only.

* Truss system = members + Pins/joints.

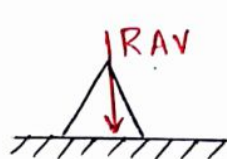


$$\sum \vec{F}_H = 0$$

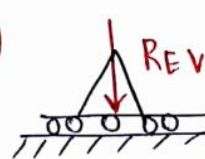
$$R_{AH} = 0$$

$$\sum \vec{F}_V = 0$$

$$R_{AV} + R_{EV} = P_1 + P_2$$



$$R_{AV} + R_{EV} (\downarrow)$$



$m \rightarrow$ no. of members.

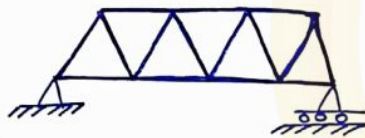
$j \rightarrow$ no. of joints/pins.

* Types of Plane Trusses [2-D]

Stable/Perfect/-
determinate

if $m = 2j - 3$

then we will get
a stable triangulated
truss.



unstable/
collapsible

if $m < 2j - 3$



Indeterminate

(or)

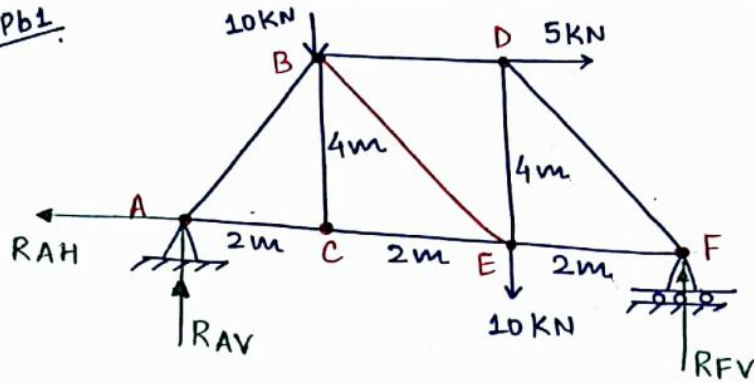
Indestructible

if $m > 2j - 3$

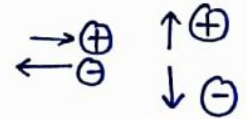
* Determination of Reactions

Reactions at the supports are calculated by considering equilibrium of the entire truss.

Pbl.



Reactions at the supports?



Sol.

$$\sum F_H = 0$$

$$5 - R_{AH} = 0$$

$$\Rightarrow R_{AH} = 5 \text{ kN} (\leftarrow)$$

$$\sum F_V = 0$$

$$R_{AV} + R_{FV} - 20 = 0$$

$$R_{AV} + R_{FV} = 20 \text{ kN} (\uparrow)$$

$$\sum M_A = 0$$

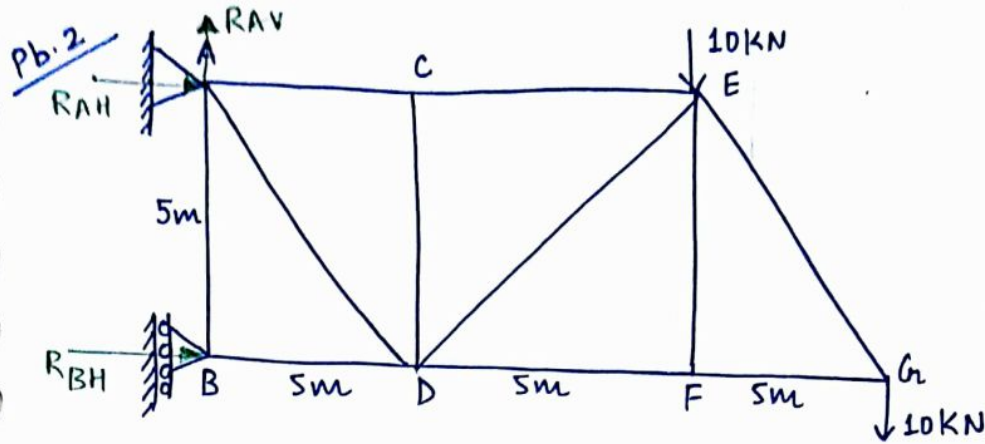
$$\Rightarrow 10 \times 2 + 5 \times 4 + 10 \times 4 - R_{FV} \times 6 = 0$$

$$\Rightarrow R_{FV} = 13.33 \text{ kN} (\uparrow)$$

Now,

$$R_{AV} + 13.33 = 20$$

$$R_{AV} = 6.67 \text{ kN} (\uparrow)$$



Sol $\sum \vec{F}_H = 0$

$$\Rightarrow R_{AH} + R_{BH} = 0$$

$$\Rightarrow R_{AH} = -R_{BH}$$

means
assumption
kisi ek ka
wrong hoga



$$\sum \vec{F}_V = 0$$

$$\Rightarrow R_{AV} = 20 \text{ kN } (\uparrow)$$

$$\sum \vec{M}_A = 0$$

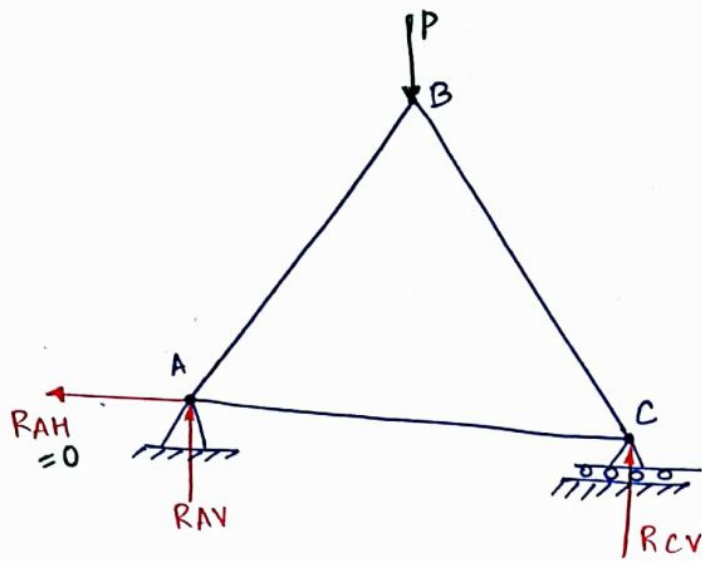
$$\Rightarrow 10 \times 10 + 10 \times 15 - R_{BH} \times 5 = 0$$

$$\Rightarrow R_{BH} = 50 \text{ kN } (\rightarrow)$$

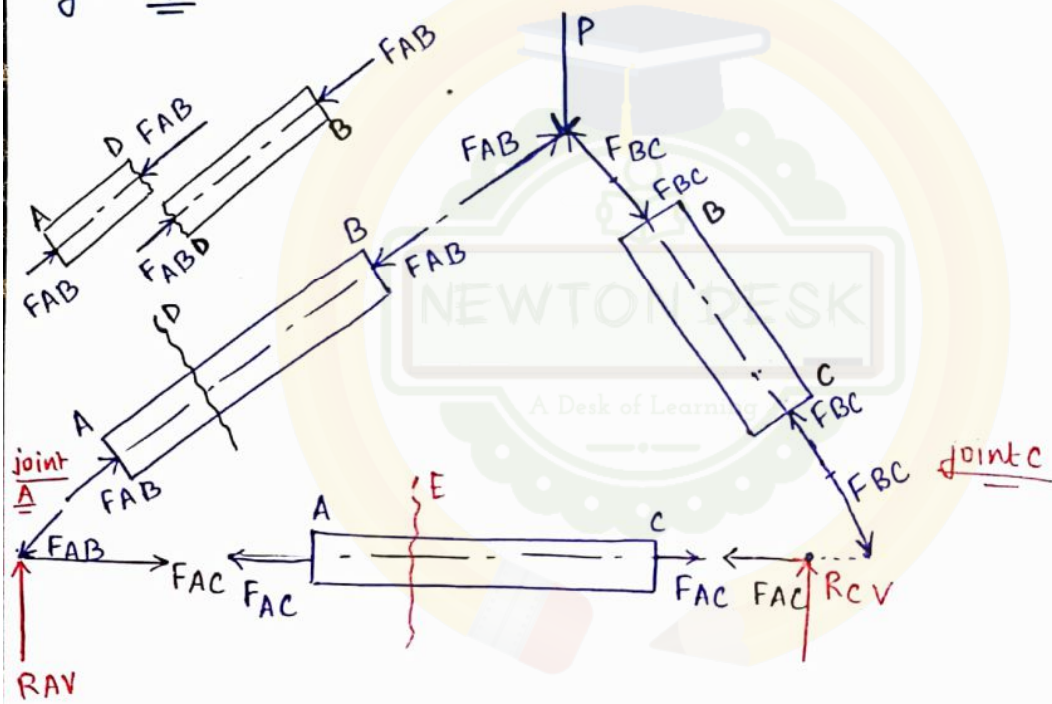
$$R_{AH} = 50 \text{ kN } (\leftarrow)$$

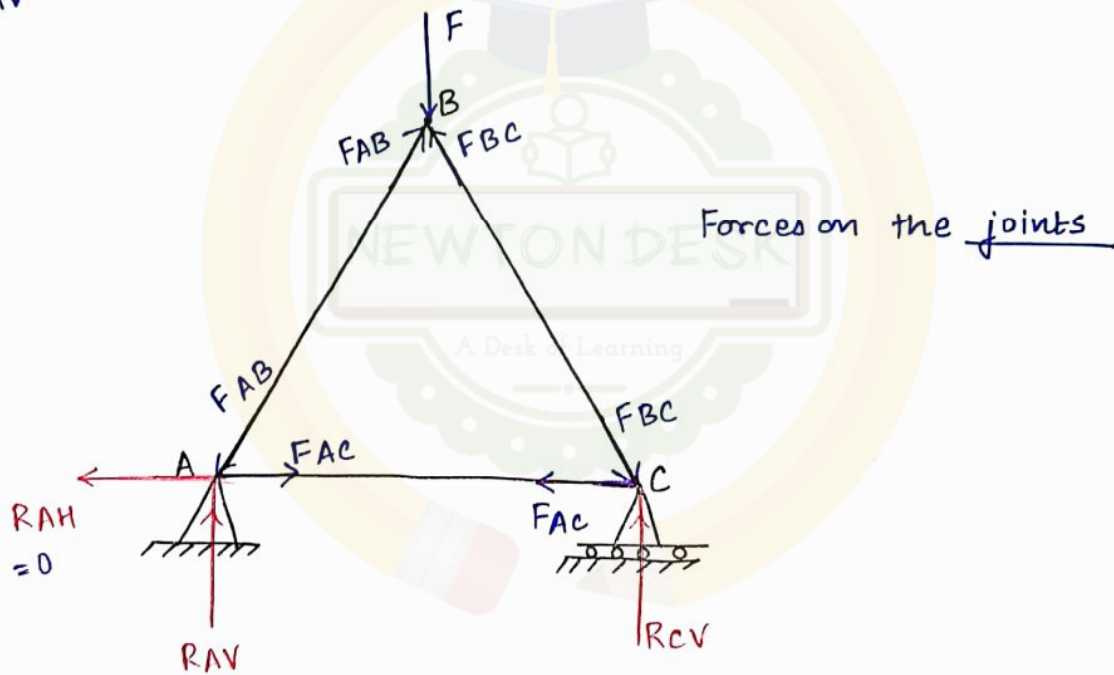
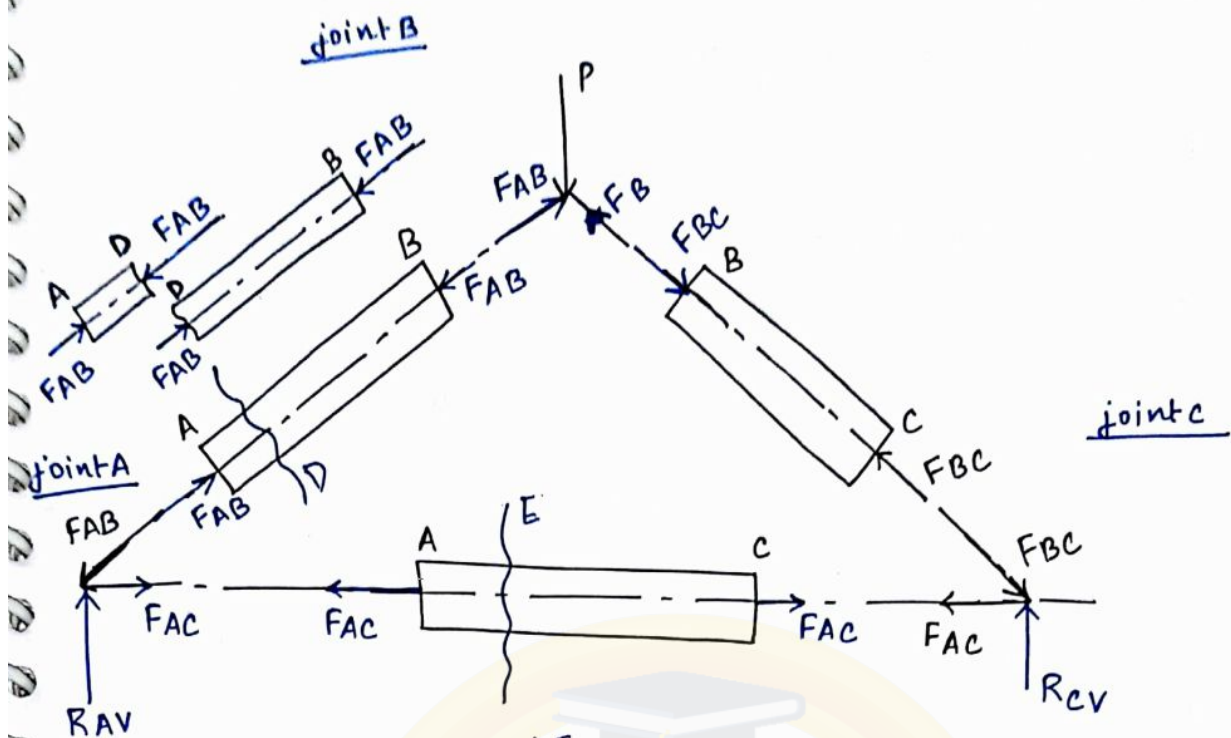
* Interaction of loads, internal forces and Reactions $\Rightarrow$

"N.P."



joint B





$$P = R_{AV} + R_{cv}$$

Method of joint :- Concept → equilibrium of a joint is considered in method of joint.

Procedure → ① find Reactions at supports.

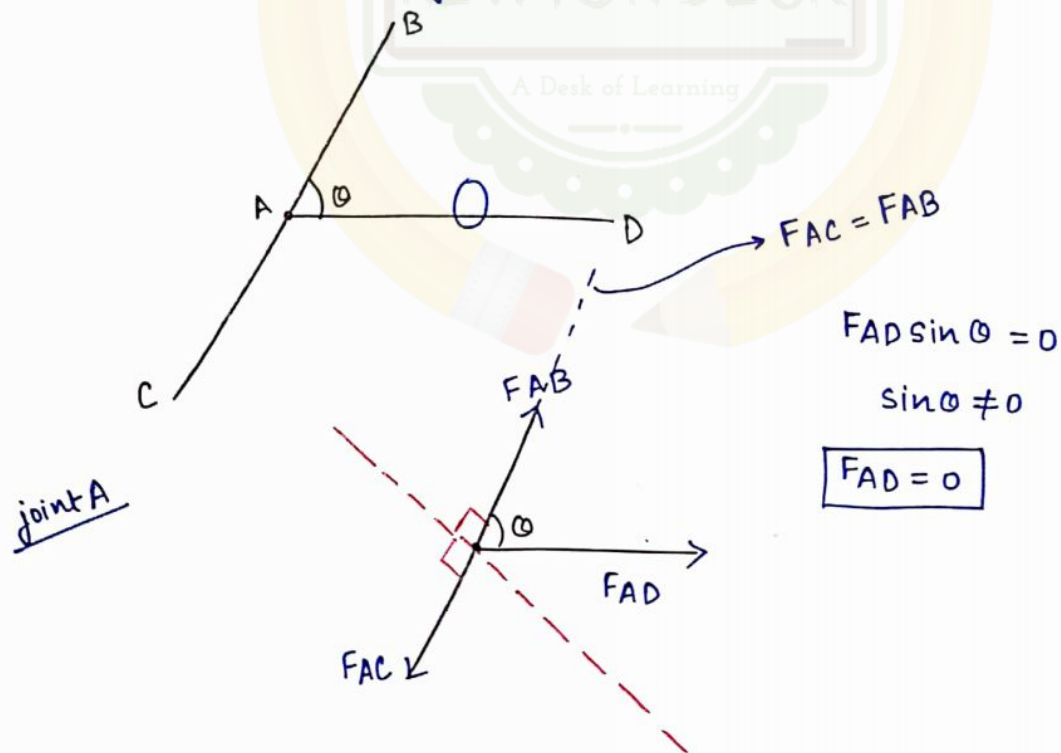
② Consider the equilibrium of a joint where only 2 unknown members are meeting and then use

$$\sum \vec{F}_x = \sum \vec{F}_y = 0 \text{ to find the unknowns.}$$

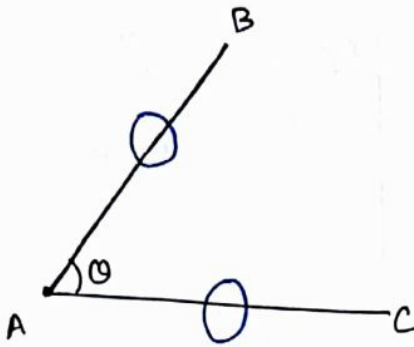
Note :- ① If a member pulls a joint then the member itself will be in tension with the same intensity.

② If a member pushes a joint then the member itself will be in compression with the same intensity.

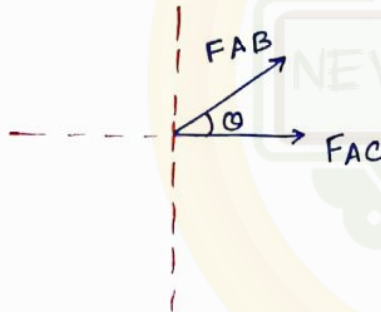
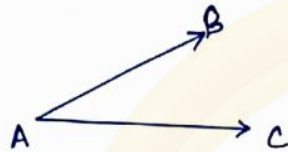
③ If at a joint 3 members are meeting and 2 are collinear then the force in the 3rd member will be zero (if there is no load/reaction at that joint).



④ If at a joint 2 members are meeting and they are non-collinear then force in both the members are zero (there is no load or reaction at that joint).



Joint A



$$\sum \vec{F}_V = 0$$

$$F_{AB} \sin \theta = 0$$

$$\sin \theta \neq 0$$

$$\boxed{F_{AB} = 0}$$

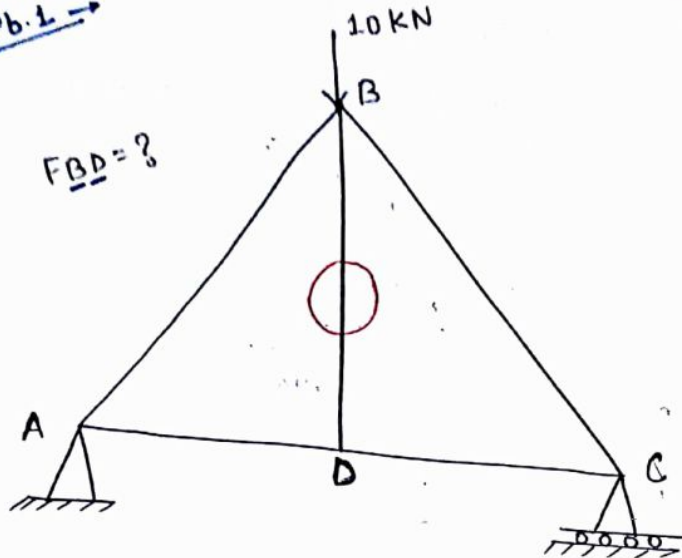
$$\sum \vec{F}_H = 0$$

$$F_{AC} + F_{AB} \cos \theta = 0$$

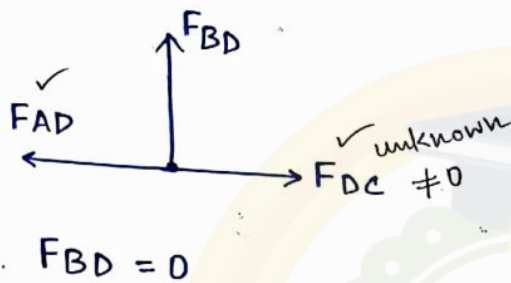
$$\boxed{F_{AC} = 0}$$

Pb. 1 →

$F_{BD} = ?$



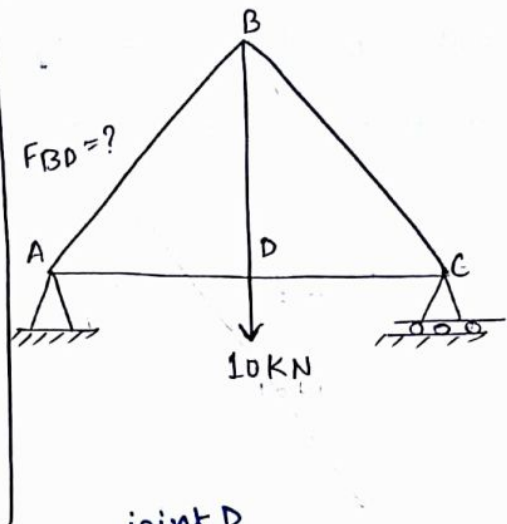
joint D



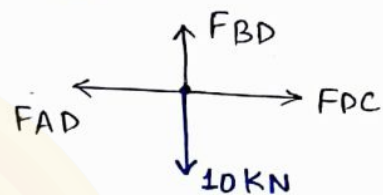
$F_{BD} = 0$

Pb 2 →

$F_{BD} = ?$

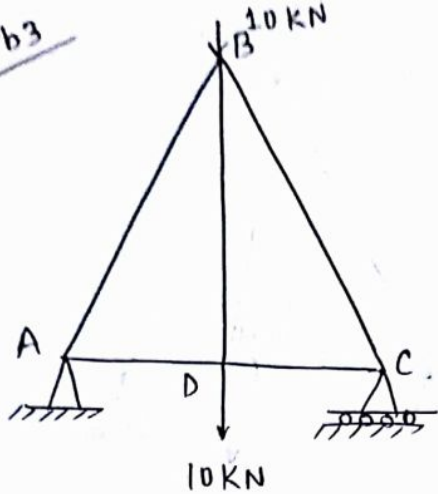


joint D



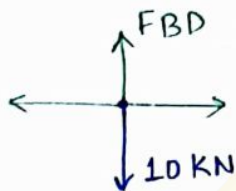
$F_{BD} = 10 \text{ kN (Tensile)}$

Pb3



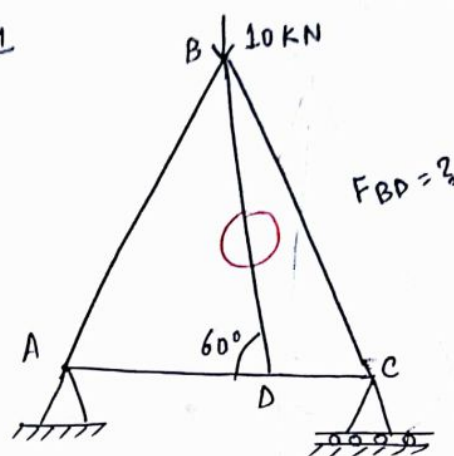
$F_{BD} = ?$

joint D



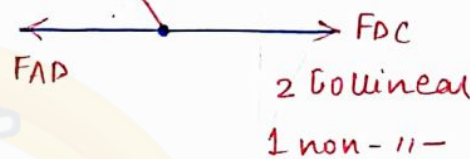
$F_{BD} = 10 \text{ kN (Tensile)}$

Pb4



$F_{BD} = ?$

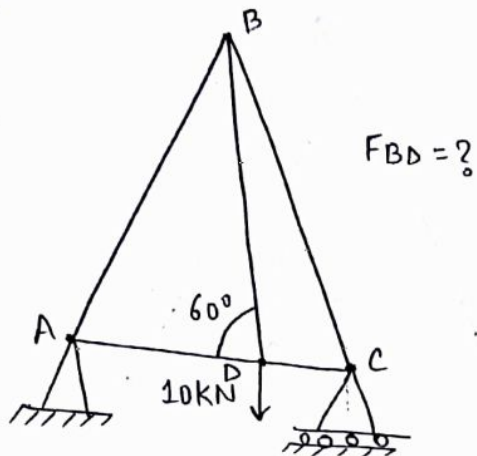
joint D $F_{BD} = 0$



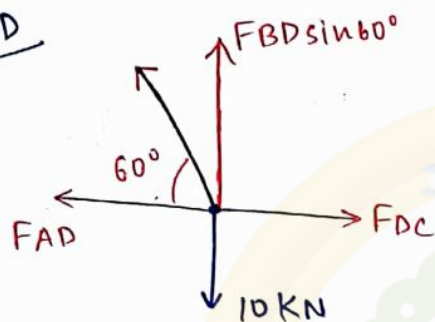
2 collinear
1 non-collinear



Pb 5



joint D



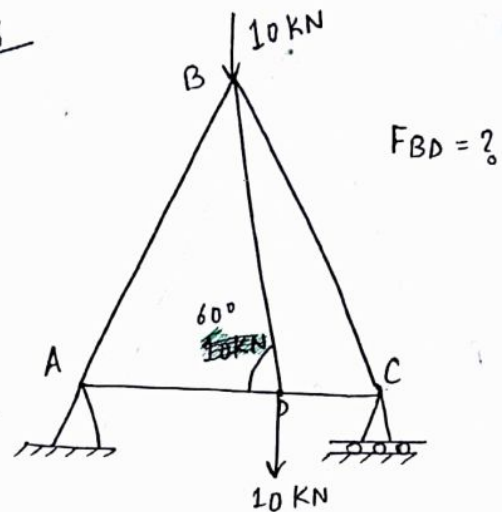
$$F_{BD} \sin 60^\circ = 10$$

$$F_{BD} \frac{\sqrt{3}}{2} = 10$$

$$F_{BD} = \frac{20}{\sqrt{3}}$$

$$F_{BD} = 11.547 \text{ kN (Tensile)}$$

Pb 6

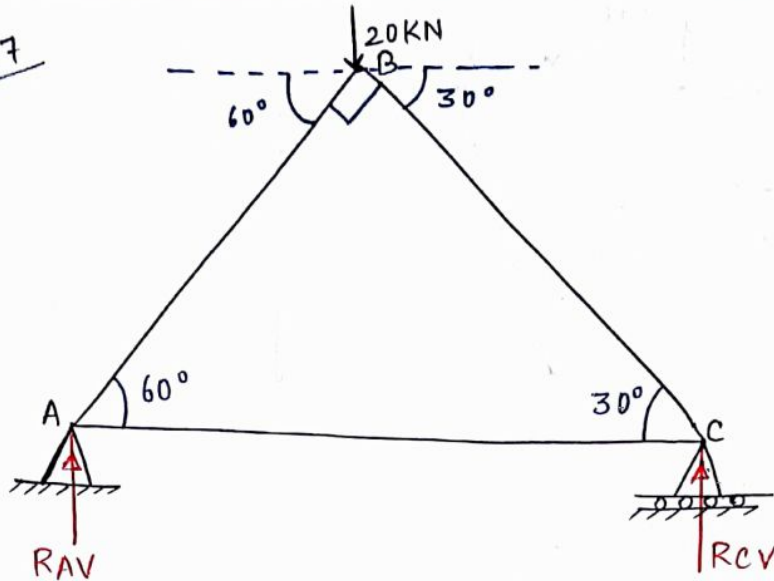


Similarly

Force on BD

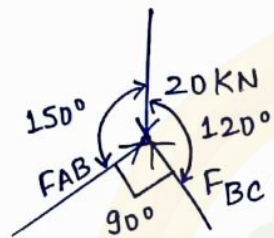
$$F_{BD} = 11.547 \text{ kN (Tensile)}$$

Pb. 7



$$F_{AC} = ?$$

joint B

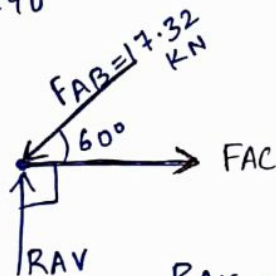


$$\frac{F_{AB}}{\sin 120^\circ} = \frac{F_{BC}}{\sin 150^\circ} = \frac{20 \text{ kN}}{\sin 90^\circ}$$

$$F_{AB} = \frac{20 \times \sin 120^\circ}{\sin 90^\circ} = 17.32 \text{ kN (comp.)}$$

$$F_{BC} = \frac{20 \times \sin 150^\circ}{\sin 90^\circ} = 10 \text{ kN (compr.)}$$

joint A



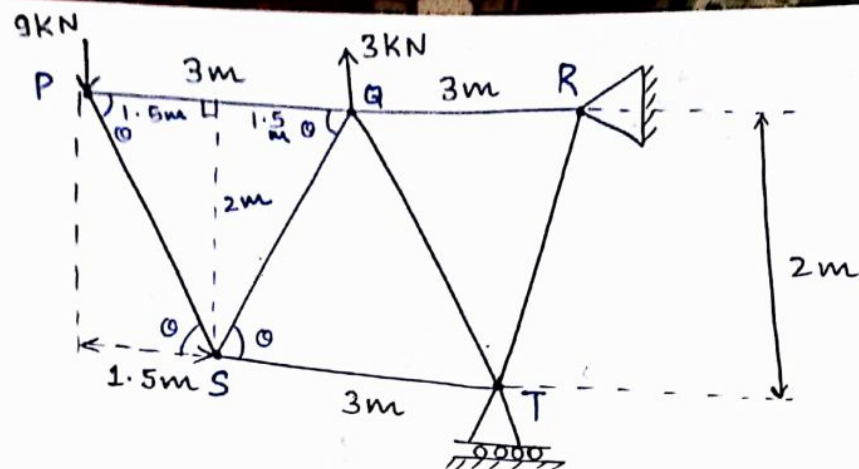
$$R_{AV} = F_{AB} \sin 60^\circ = 15 \text{ kN}$$

$$F_{AC} = F_{AB} \cos 60^\circ = 8.66 \text{ kN (Tensile)}$$

$$R_{CV} = 20 - R_{AV} = 5 \text{ kN (A)}$$

Prob. 8. →

$F_{QS} = ?$



Sol

joint P



$$\tan \theta = \frac{2}{1.5}$$

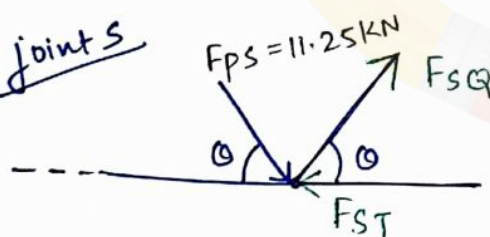
$$\theta = 53.13^\circ$$

By Lami's theorem ← no need.

$$9 = F_{PS} \sin 53.13^\circ$$

$$F_{PS} = 11.25 \text{ kN (comp.)}$$

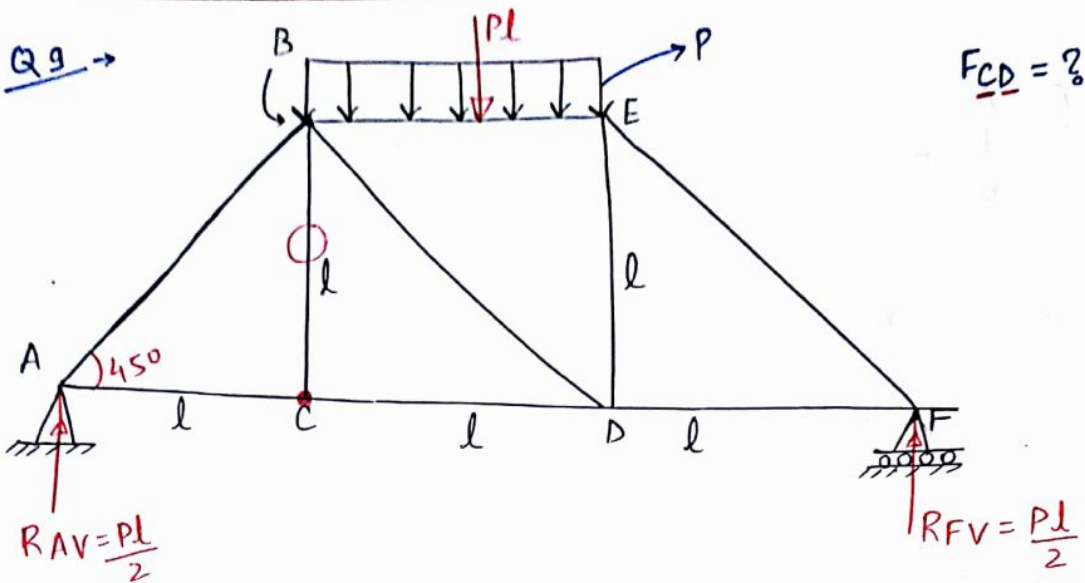
joint S



$$F_{PS} \sin \theta = F_{SQ} \sin \theta$$

$$F_{SQ} = 11.25 \text{ kN (Tensile)}$$

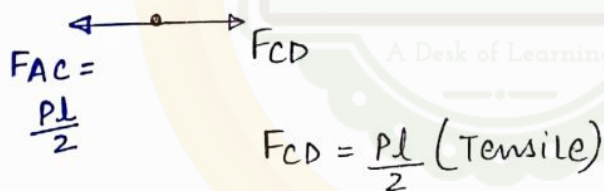
Q9 →



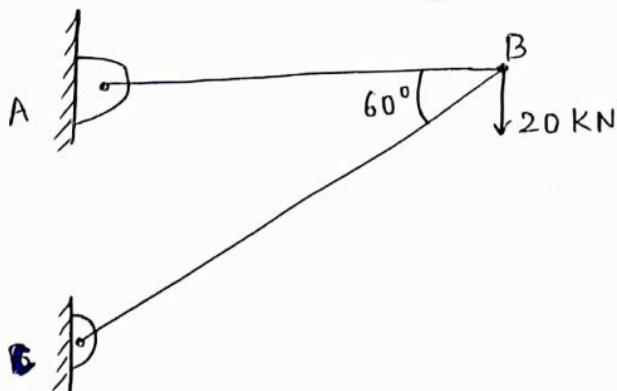
joint A



joint C

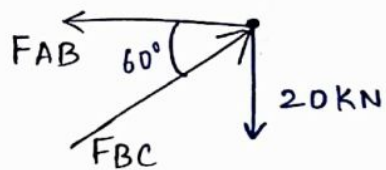


Pb 10



$(R_C)_H = ?$

Joint B

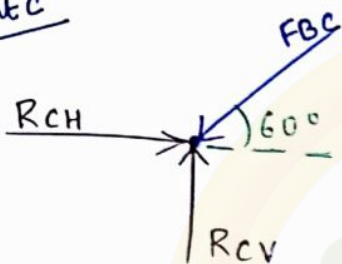


$$F_{BC} \sin 60^\circ = 20 \text{ kN}$$

$$F_{BC} = \frac{20}{\sqrt{3}/2} = \frac{40}{\sqrt{3}}$$

$$F_{BC} = 23.09 \text{ kN (comp.)}$$

Joint C



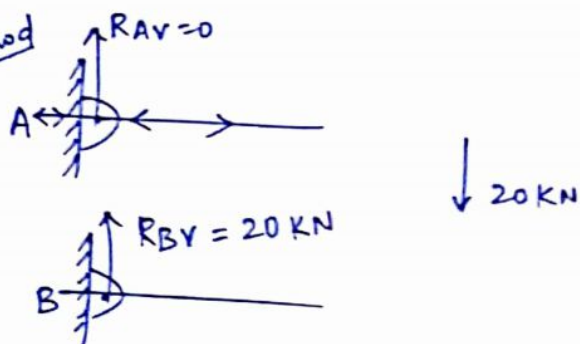
$$F_{BC} \sin 60^\circ = 20 \text{ kN} = R_{CV}$$

$$R_{CV} = 20 \text{ kN} (\uparrow)$$

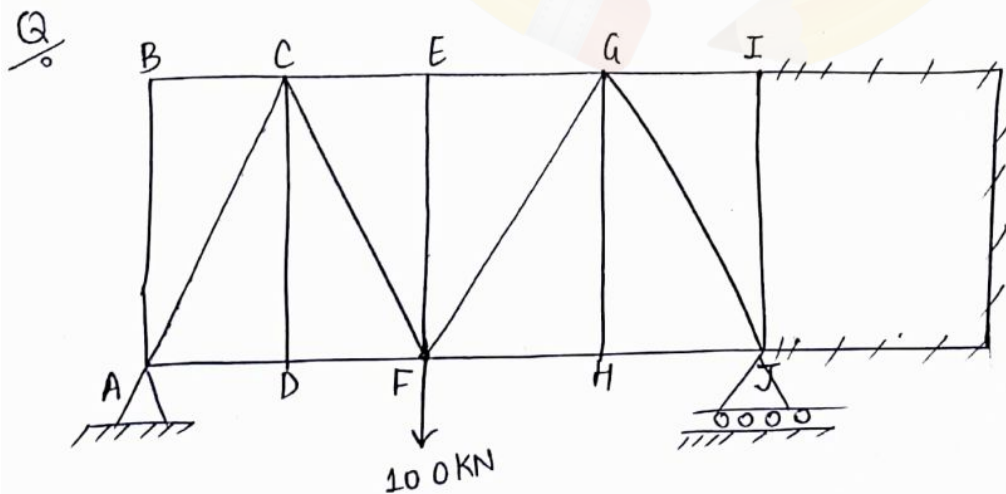
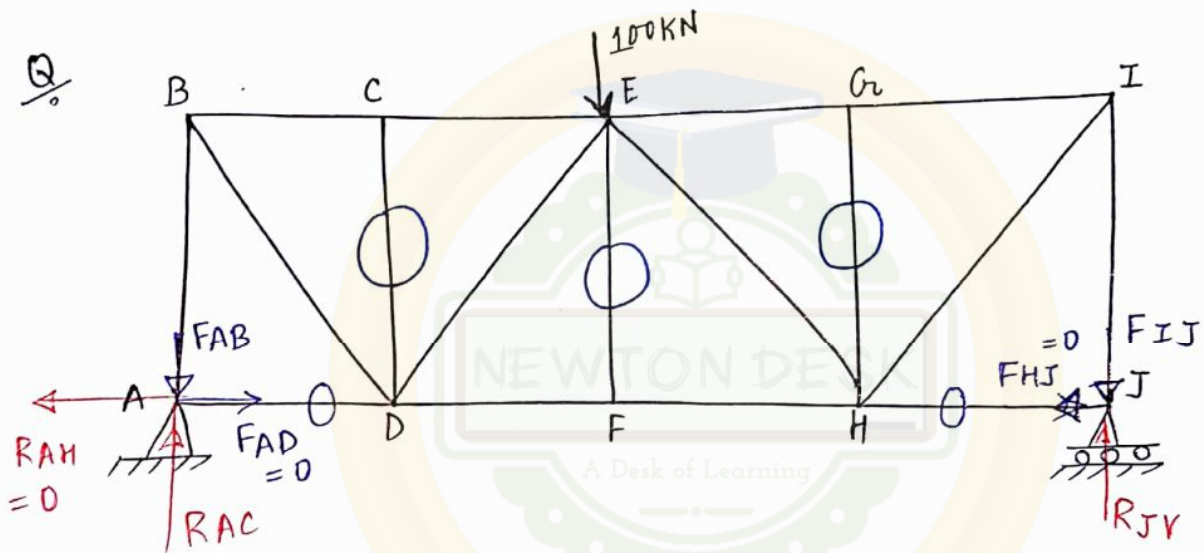
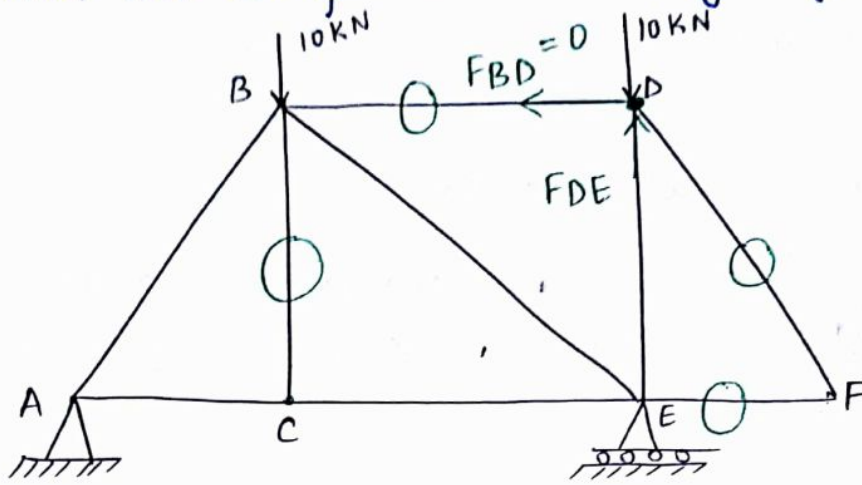
$$R_{CH} = F_{BC} \cos 60^\circ$$

$$R_{CH} = 11.5 \text{ kN} (\rightarrow)$$

2nd method



Q. Find the No. of members having 0 force.



29/7/16

Method of Section

Concept → eqn. of a section of truss is considered in method of section.

Procedure → (a) find Reactions and supports.

(b) cut the member under consideration by a section

①-----① and consider eqn. of either Left hand side or RHS of section ①---① for applied loads, Reactions and forces in the cut members and use

$$\boxed{\sum \vec{F}_x = \sum \vec{F}_y = \sum \vec{Z} = 0}$$

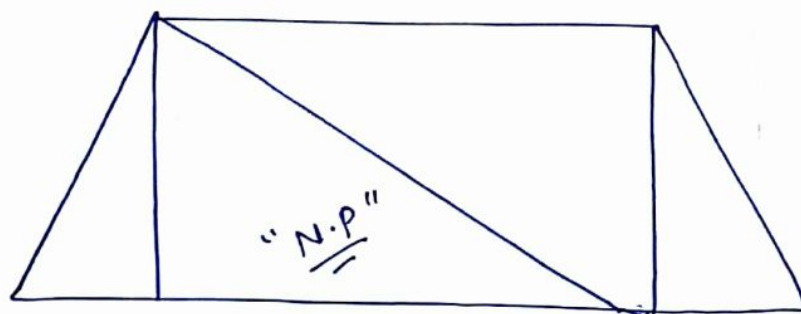
and $\boxed{\sum \vec{M} = 0}$ ↑ not applied
(since 2D Truss)

Note :- (1) Advantage of method of section is that force in any intermediate member can be found directly without finding force in any other member.

(2) Cut the members such that entire truss is divided into two separate parts.

(3) Preferably do not cut more than 3 members because in method of section we have only 3 equations of equilibriums.

(4) Cut the member such that all the cut members do not meet at same joint (If they meet at same joints it becomes of method of joints only).



FBD = ?



$$R_{AV} = 10 \text{ kN}$$

$$\tan \theta = \frac{2}{3}$$

$$F_{BD} = ?$$

$$\sum \vec{M}_E = 0$$

$$\Rightarrow (F_{BD} \times 3) + (-10 \times 4) - (5 \times 2) = 0$$

$$\Rightarrow F_{BD} = -10 \text{ kN}$$

$$\Rightarrow F_{BD} = 10 \text{ kN (compressive)}$$

$$F_{CE} = ?$$

$$\Rightarrow \sum \vec{M}_B = 0$$

$$\Rightarrow (10 \times 2) - (F_{CE} \times 3) = 0$$

$$\Rightarrow \boxed{F_{CE} = 6.67 \text{ kN (Tensile)}}$$

$$F_{BE} = ?$$

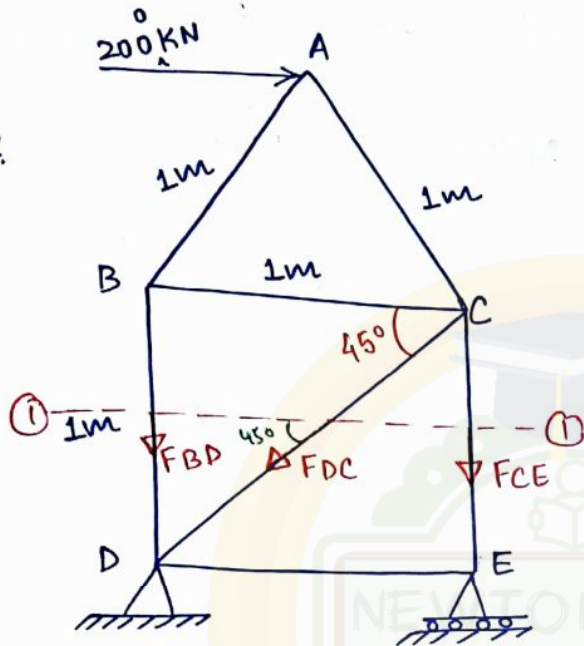
$$\sum \vec{F}_V = 0$$

$$\Rightarrow 10 - 5 - F_{BE} \cos 30^\circ = 0$$

$$\Rightarrow F_{BE} = 6 \text{ kN (Tensile)}$$

Q.2. →

$$F_{CD} = ?$$



Considering upper side of

①-----①

$$\sum \vec{F}_H = 0$$

$$2000 - F_{DC} \cos 45^\circ = 0$$

$$F_{DC} = 2000\sqrt{2} \text{ (Tensile)}$$

soln. →

Considering upper side of ①-----①

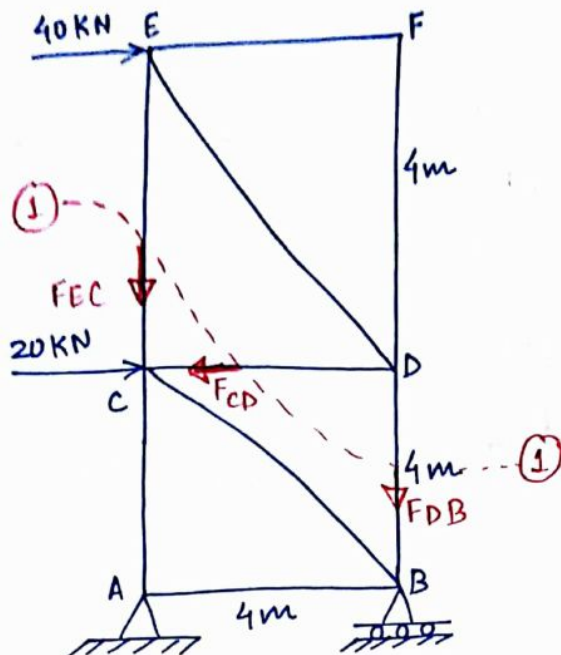
$$\sum \vec{F}_H = 0$$

$$40 - F_{CD} = 0$$

$$F_{CD} = 40 \text{ kN (Tensile)}$$

Pb.3 →

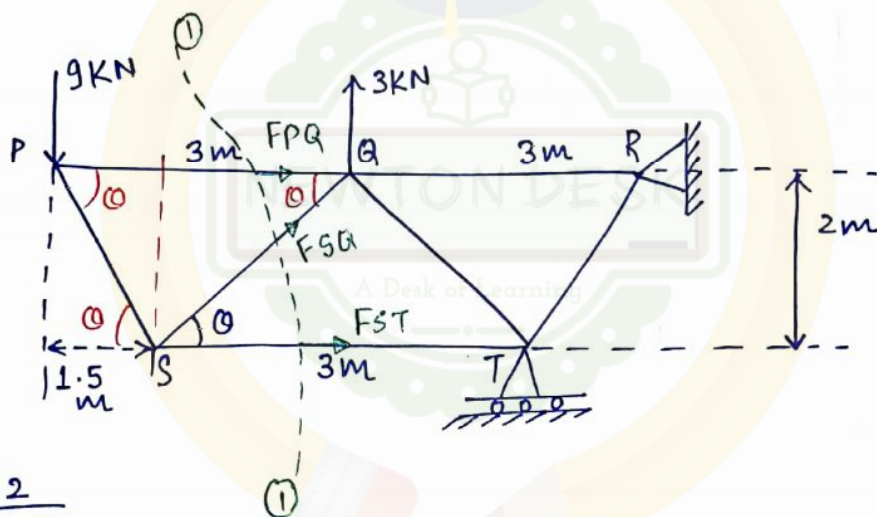
$F_{CD} = ?$



Sol

Pb.4 →

$F_{QS} = ?$



$$\tan \theta = \frac{2}{1.5}$$

$$\theta = 53.13^\circ$$

Considering LHS of (1) ----- (1)

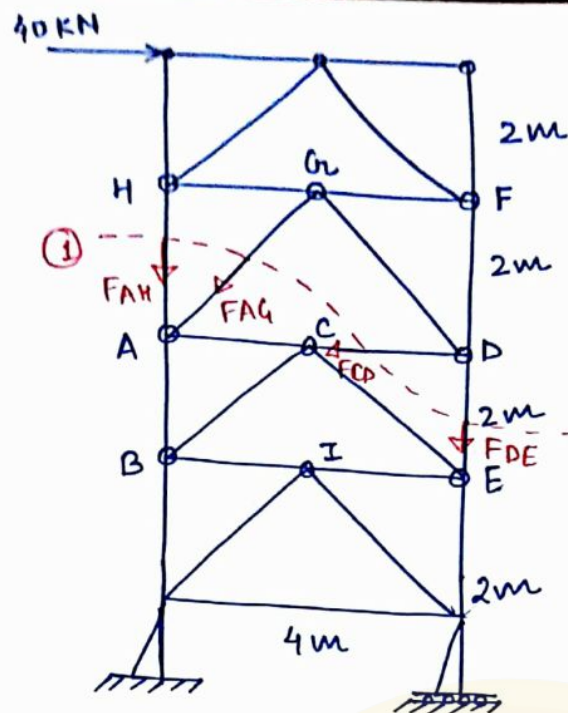
$$\sum \vec{F}_V = 0$$

$$\Rightarrow -9 + F_{SQ} \sin \theta = 0$$

$$\Rightarrow F_{SQ} = 11.25 \text{ kN (Tensile)}$$

Pb5

$F_{DE} = ?$



Considering upper side
of ①-----①

$$\sum \vec{M}_A = 0$$

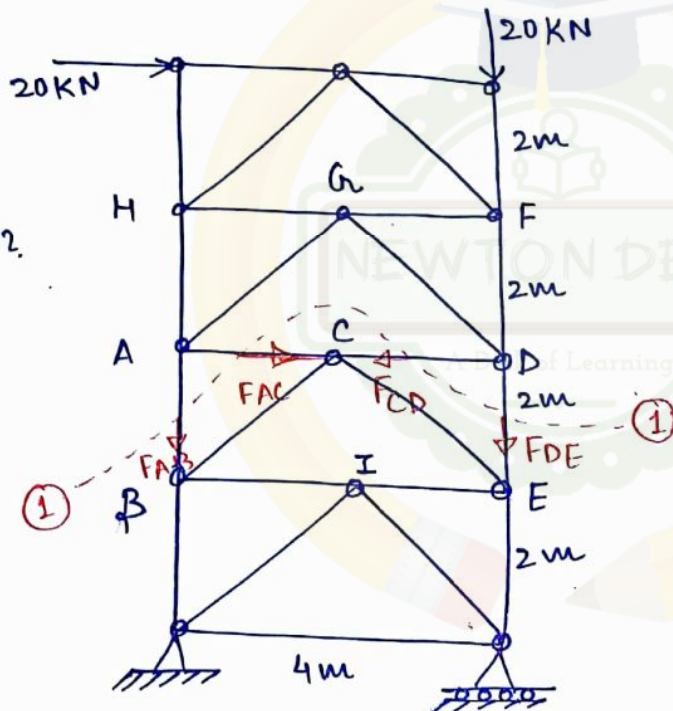
$$40 \times 4 + F_{DE} \times 4 = 0$$

$$F_{DE} = -40 \text{ kN}$$

$$F_{DE} = 40 \text{ kN (compressive)}$$

Pb6

$F_{AB} = ?$



Considering upper side
of ①-----①

$$\sum \vec{M}_D = 0$$

$$20 \times 4 - F_{AB} \times 4 = 0$$

$$F_{AB} = 20 \text{ kN (Tensile)}$$

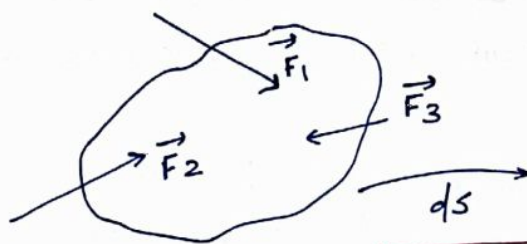
* WORKDONE

$$\text{Workdone} = \vec{F} \cdot d\vec{s} = T \cdot d\theta$$

$$W_{F_1} = \vec{F}_1 \cdot d\vec{s}$$

$$W_{F_2} = \vec{F}_2 \cdot d\vec{s}$$

$$W_{F_3} = \vec{F}_3 \cdot d\vec{s}$$



$$W_{F_1} = \vec{F}_1 \cdot d\vec{s}$$

independent

$$\odot \downarrow \downarrow T \rightarrow +ve$$

$$\odot \uparrow \uparrow T \rightarrow -ve$$

$$\begin{aligned} \text{Power} &= \frac{\vec{F} \cdot d\vec{s}}{dt} = \vec{F} \cdot \vec{v} \\ &= \frac{T d\theta}{dt} = T \omega \\ &= \frac{2\pi NT}{60} \end{aligned}$$

$$\begin{aligned} \text{TWD} = \text{Total workdone} &= W_{F_1} + W_{F_2} + W_{F_3} \\ &= (\vec{F}_1 + \vec{F}_2 + \vec{F}_3) \cdot d\vec{s} \end{aligned}$$

$$\boxed{\text{TWD} = \vec{F}_R \cdot d\vec{s}}$$

If Body is in equilibrium

$$\vec{F}_R = 0 \Rightarrow \boxed{\text{TWD} = 0}$$

(imaginary)

* Principle of Virtual Work $\rightarrow$ [POVW]

It states "if a system is in equilibrium, then the sum of virtual workdone by all the forces will be zero."

Virtual Work = Force (virtual displacement).

Note - ① Using POVW we can find forces necessary to keep a system in equilibrium.

② If using eqn's of eqm. is difficult and time taken, then use POVW to find the unknowns easily.

③ POVW is used in the system when no. of interconnected Rigid bodies have degree of freedom more than 0 but are in equilibrium.

- ④ In POVW, we do not need to find the Reaction at supports because virtual work of Reactions is always 0.

Procedure $\rightarrow$ ① Take any fixed point in problem as origin, fix coordinate axis and find the coordinates of all the points where forces are acting.

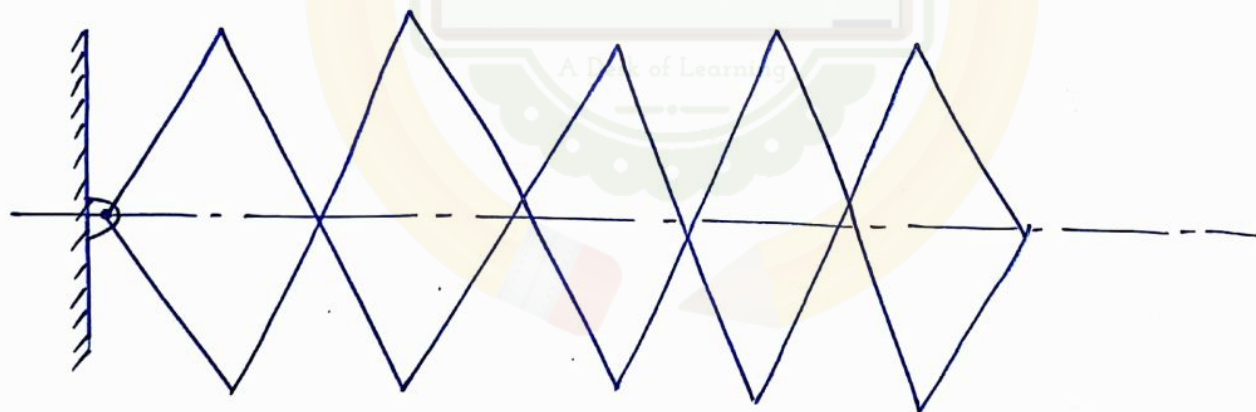
② find virtual displacements.

③ Use POVW to find the unknowns.

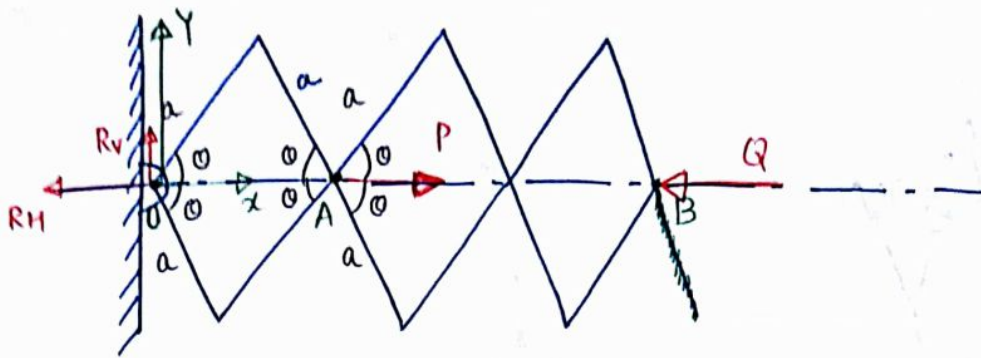
Sign Convention $\rightarrow$ ① Sign convention for the coordinates is chosen based on the quadrant in which they are lying.

② If any force is acting along +ve x-axis or +ve y-axis then take that force as +ve and vice-versa.

Q. For the Lazy tong mechanism shown in figure the Relationship b/w P and Q to keep the system in equilibrium is



" [N.P.] "



$$\begin{array}{l|l} \sum \vec{F}_V = 0 & \sum \vec{M}_A = 0 \\ R_V = 0 \checkmark & 0 \checkmark \\ \hline \sum \vec{F}_H = 0 & \\ R_H + Q = P \checkmark & \end{array} \quad \begin{array}{l} \text{NO means} \end{array}$$

POW $\rightarrow$

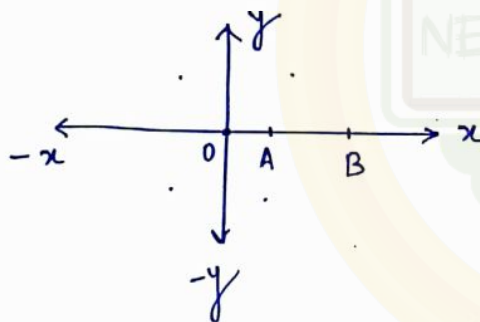
1. $x_A = +2a \cos \theta$

$x_B = +6a \cos \theta$

2. $\partial x_A = -2a \sin \theta \partial \theta$

$\partial x_B = -6a \sin \theta \partial \theta$

3. $(v \cdot w)_P + (v \cdot w)_Q = 0$

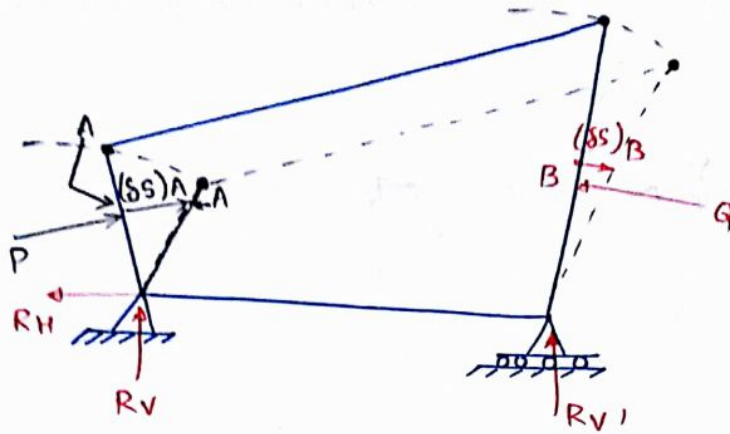


$\Rightarrow (+P)(\partial x)_A + (-Q)(\partial x)_B = 0$

$\Rightarrow +P(-2a \sin \theta \partial \theta) + Q(6a \sin \theta \partial \theta) = 0$

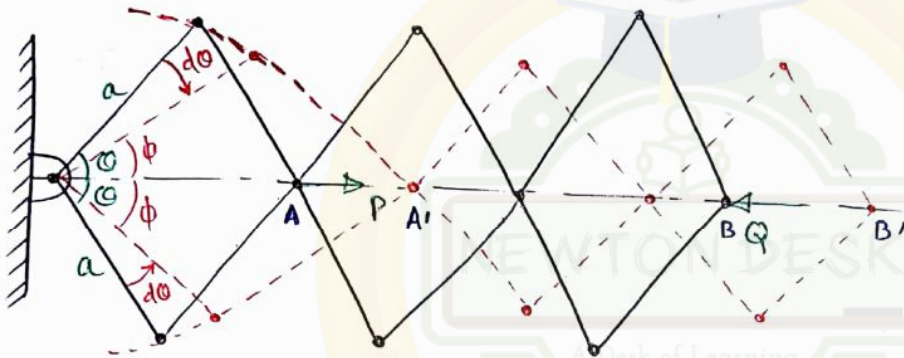
$\Rightarrow \boxed{P = 3Q}$

$\boxed{R_H = P - Q = 2Q}$



$$(vw)_P + (vw)_Q = 0$$

$$P(\delta s)_A + (-Q(\delta s)_B) = 0$$



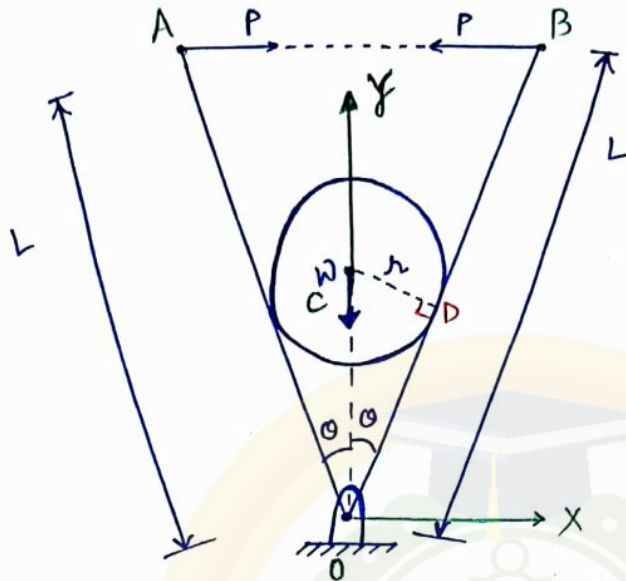
$$AA' = x$$

$$BB' = 3x$$

$$Px + (-Q \times 3x) = 0$$

$$P = 3Q$$

Q. A sphere of weight 'W' and radius 'r' is supported by 2 rods of length 'L' as shown in fig. The value of 'P' to keep it in equilibrium is.



Sol

1. $x_A = -L \sin \theta$

$x_B = +L \sin \theta$

$y_c = +L \cos \theta$

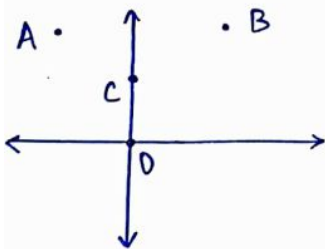
In $\triangle ODC$

$\frac{h}{OC} = \sin \theta$

$L = OC \sin \theta$

$\frac{L}{\sin \theta} = y_c$

$L \cos \theta = y_c$



2. $\partial x_A = -L \cos \theta \partial \theta$

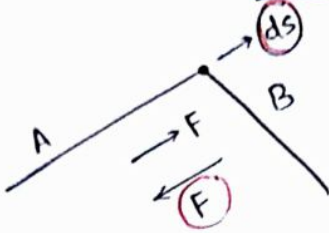
$\partial x_B = +L \cos \theta \partial \theta$

$\partial y_c = -L \sin \theta \partial \theta$

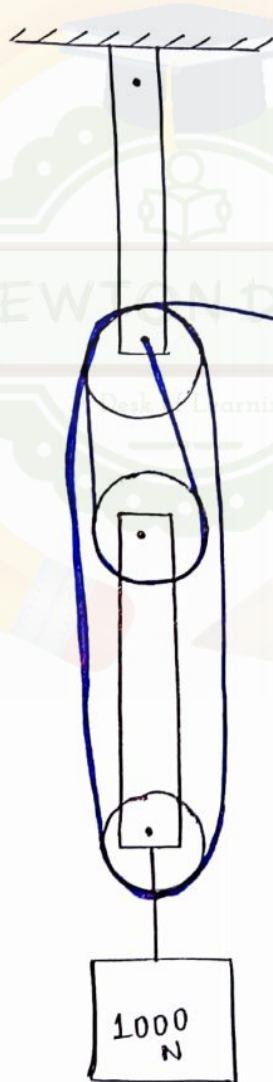
$$3. (+P)(-L \cos \theta) + (-P)(L \cos \theta) + (-W)(-L \sec \theta) = 0$$

$$2PL \cos \theta = W L \times \frac{1}{\sin \theta} \times \frac{\cos \theta}{\sin \theta}$$

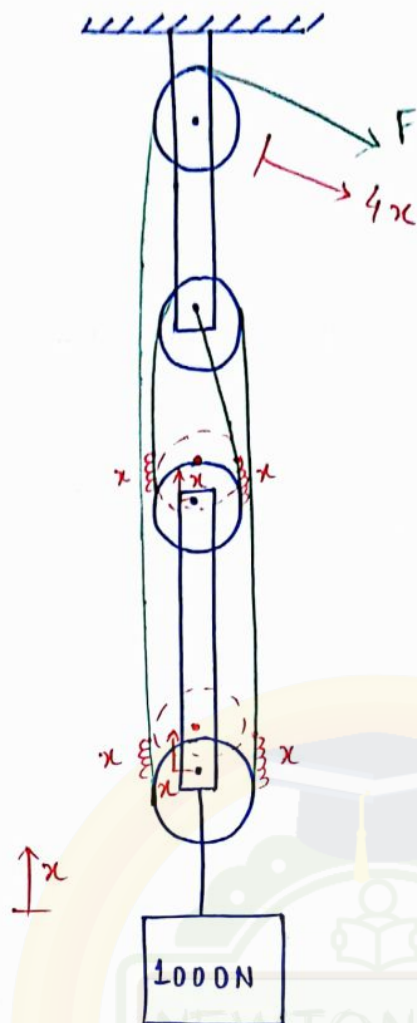
$$P = \frac{WL}{2L \sin^2 \theta}$$



Pb. 7.10



wrong diagram.

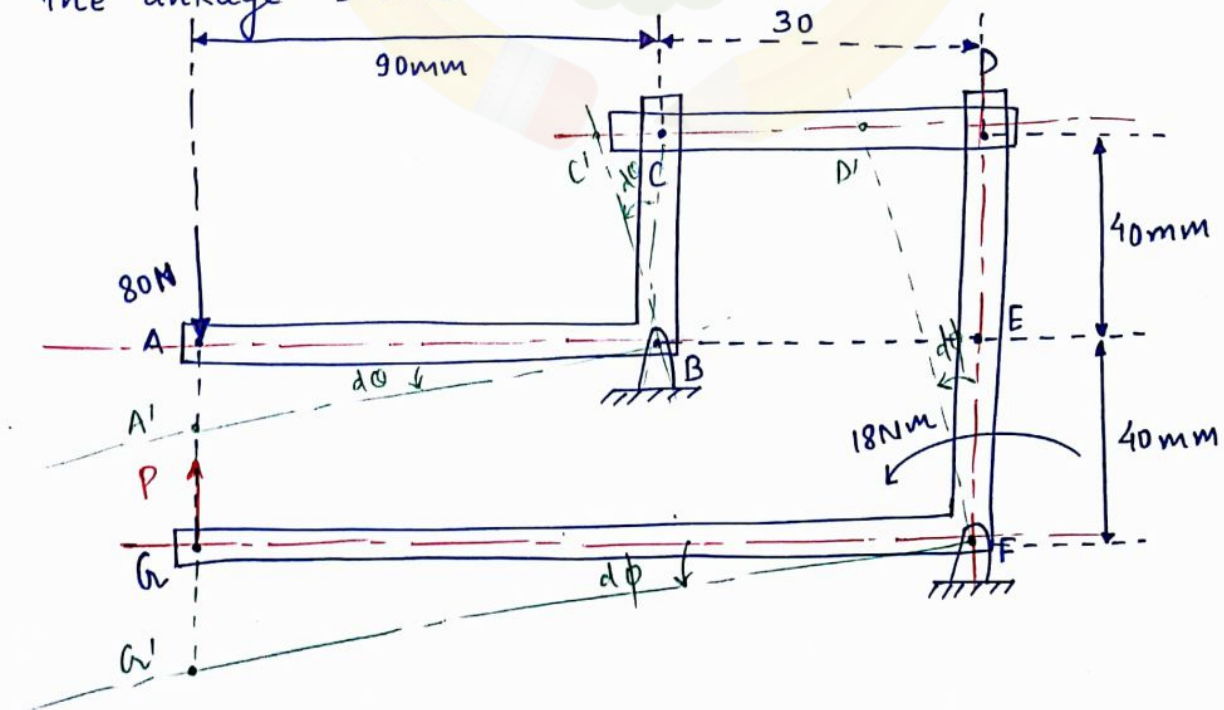


$$-1000 \times x + F \times 4x = 0$$

$$F = 250 \text{ N}$$

Mechanical
Advantage = 4.

Q Determine the vertical force at 'G' to maintain eqm. of the linkage shown below.



$$CC' = DD'$$

$$40d\theta = 80d\phi$$

$$\boxed{d\theta = 2d\phi}$$

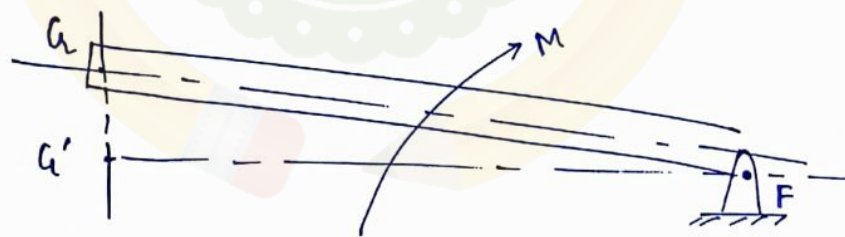
$$\text{POVW} \Rightarrow 80(AA') + 18d\phi - P(CD') = 0$$

$$\Rightarrow 80(0.9d\theta) + 18d\phi - P(0.12d\phi) = 0$$

$$\Rightarrow 80 \times 0.9 \times 2d\phi + 18d\phi - P(0.12d\phi) = 0$$

$$\Rightarrow \boxed{P = 270 \text{ N} (\uparrow)}$$

Pb Determine the couple M that must be applied on member DEFG instead of vertical force at G so as to maintain equilibrium of previous problem.



$$80(AA') + 18(d\phi) - Md\phi = 0$$

$$80(0.9 \times 2d\phi) + 18d\phi - Md\phi = 0$$

$$\underline{M = 32.4 \text{ Nm} ; \text{ CW}}$$

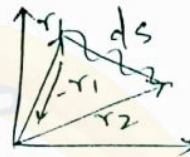
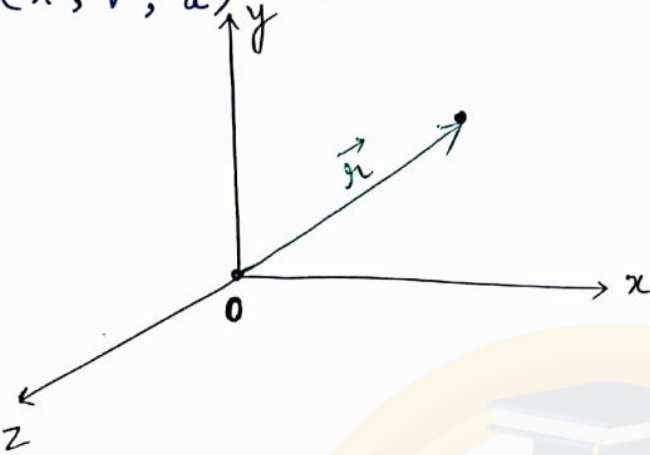
DYNAMICS

Dynamics

Kinematics

$(\vec{r}, \vec{v}, \vec{a})$

$\vec{r} \rightarrow$ position ~~velocity~~ vector



$$-r_1 + r_2 = ds$$

$$\vec{v} = \frac{d\vec{r}}{dt} = \left(\frac{d\vec{s}}{dt} \right)$$

$$\vec{a} = \frac{d\vec{v}}{dt}$$

Rectilinear Translation

Kinematics



$$(\vec{ds})_1 = (\vec{ds})_2 = (\vec{ds})_3 = (\vec{ds})_4 \rightarrow \text{during time } dt$$

$$\left. \begin{aligned} \vec{v}_1 &= \vec{v}_2 = \vec{v}_3 = \vec{v}_{cm} = \vec{v} \\ \vec{a}_1 &= \vec{a}_2 = \vec{a}_3 = \vec{a}_{cm} = \vec{a} \end{aligned} \right\} \rightarrow \text{at any instant}$$

$$\left. \begin{aligned} \vec{v}_{1/2} &= \vec{v}_1 - \vec{v}_2 = 0 \\ \vec{a}_{1/2} &= \vec{a}_1 - \vec{a}_2 = 0 \end{aligned} \right\} \rightarrow \text{Relative rest.}$$

if $\vec{a}$ is uniform.

1. $V = u + at$

2. $S = ut + \frac{1}{2}at^2$

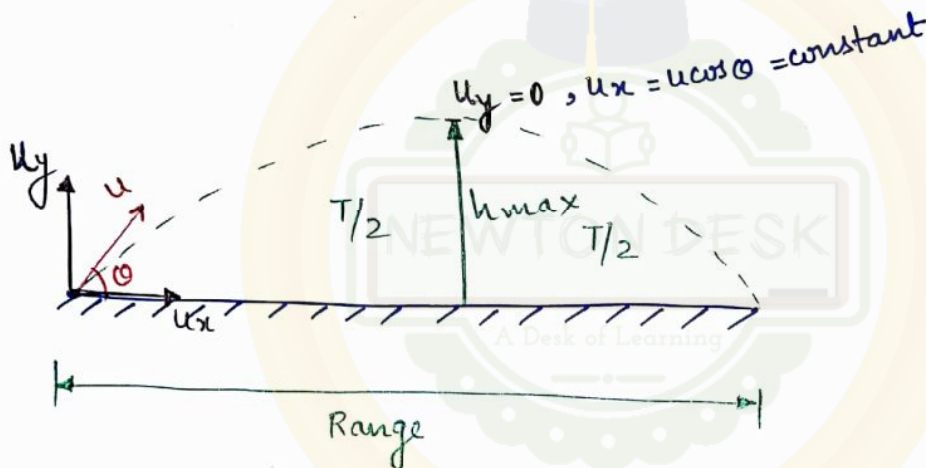
3. $V^2 = u^2 + 2as$

Ex →

Case. 1 :- Free falling bodies

$$u = 0 ; a = g(-\hat{j})$$

Case. 2 :- Projectile motion.



(i) $u_x = u \cos \theta ; a_x = 0$

$$u_x = \text{constant} = u \cos \theta$$

(ii) $u_y = u \sin \theta ; a_y = g(-\hat{j})$

⇒ if $a = F(s)$ — displacement

$$a = \frac{dv}{dt} ; v = \frac{ds}{dt}$$

$$dt = \frac{dv}{a} ; dt = \frac{ds}{v}$$

$$\boxed{\frac{dv}{a} = \frac{ds}{v}}$$

$$a ds = v dv$$

$$F(s) ds = v dv$$

Q. The initial velocity of an object is 40 m/s , acceleration $a = -0.1v$ where v is instantaneous velocity, the velocity of the object after 3 sec's is :-

Sol.

$$v = u + at$$

$$v = 40 + (-0.1v)3$$

$$v = 40 - 0.3v$$

$$v = \frac{40}{1.3} \text{ m/s}$$

unknown

SIR $a = -0.1v$ ✓

$$u_i = 40 \text{ m/s} ; t = 0$$

$$v_t = 3s = ?$$

$$v = u + \underbrace{at}_{\text{constant}}$$

$$a = -0.1v$$

$$\frac{dv}{dt} = -0.1v = a$$

$$-0.1 \int_0^3 dt = \int_{40}^{v_f} \frac{dv}{v}$$

$$\Rightarrow -0.1(3-0) = \log_e v_f - \ln 40$$

$$\Rightarrow -0.3 = \ln\left(\frac{v_f}{40}\right)$$

$$\Rightarrow \boxed{v_f = 29.63 \text{ m/s}}$$

Q if $s = \frac{t^3}{3} - 36t$ then when does the particle
reverse its direction and what is its/
the acceleration at that instant?

Sol

$$v = \frac{ds}{dt} = t^2 - 36 = 0$$

$$t = 6s$$

$$a = \frac{dv}{dt} = 2t$$

$$a_{t=6} = 12 \text{ m/s}^2$$

* Kinetics :-

Newton's 2nd Law →

for a particle →

if $\vec{F}_R \neq 0$ then $\vec{a} \neq 0$

$$\vec{a} = \frac{\vec{F}_R}{m} \rightarrow \text{dirn. in the } \vec{F}_R$$

$$\vec{F}_R = m\vec{a} \text{ — 2nd law}$$

Resultant of actual forces

Response

$m\vec{a} \rightarrow$ is not a force

for a Rigid Body →

if $(\vec{F}_R)_{\text{ext}} \neq 0$ then $\vec{a}_{\text{cm}} \neq 0$

$$\vec{a}_{\text{cm}} = \frac{(\vec{F}_R)_{\text{ext}}}{m}$$

not of
Body

$$(\vec{F}_R)_{\text{ext}} = m \vec{a}_{\text{cm}} \rightarrow \underline{\underline{2^{\text{nd}} \text{ Law}}}$$

Resultant
of actual
external
forces

effect/Response.

1st, 2nd law

✓
one Body

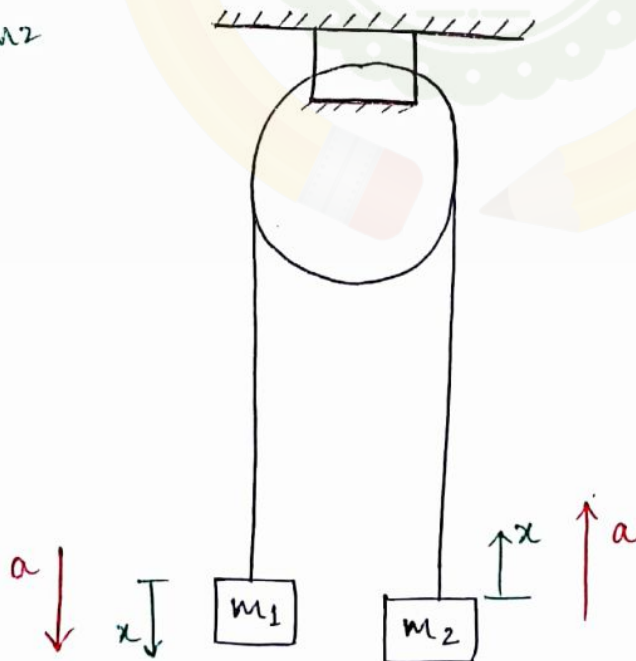
3rd

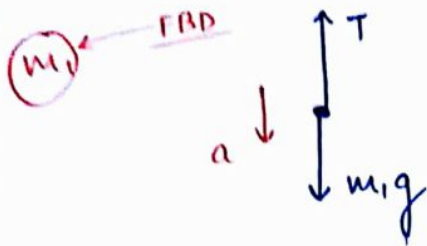
↓
2 Body

* Rectilinear Translation →

Case. I →

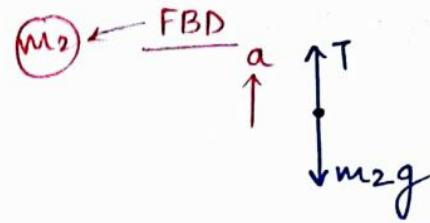
$m_1 \neq m_2$





$$m_1g - T = m_1a \quad \text{--- 2nd law}$$

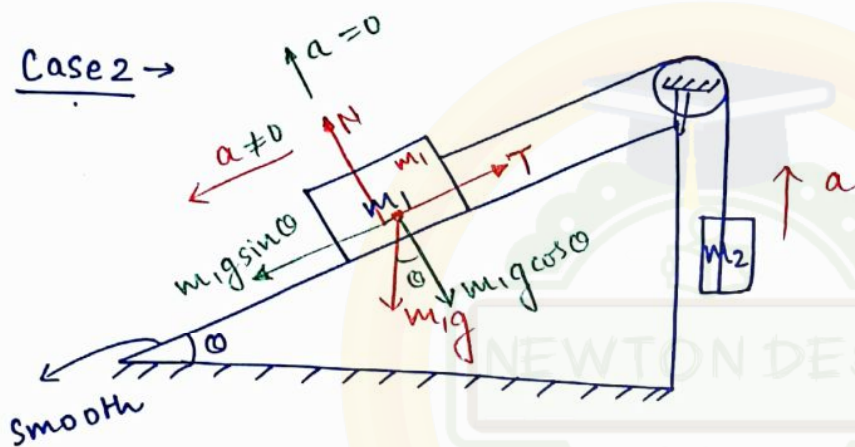
Response



$$T - m_2g = m_2a \quad \text{--- 2nd law}$$

Response

$\vec{F}_R$ $\longrightarrow$ also called effective force

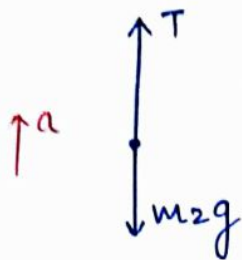


(m1) $N = m_1g \cos \theta$ --- 1st law

$$m_1g \sin \theta - T = m_1a \quad \text{--- 2nd law}$$

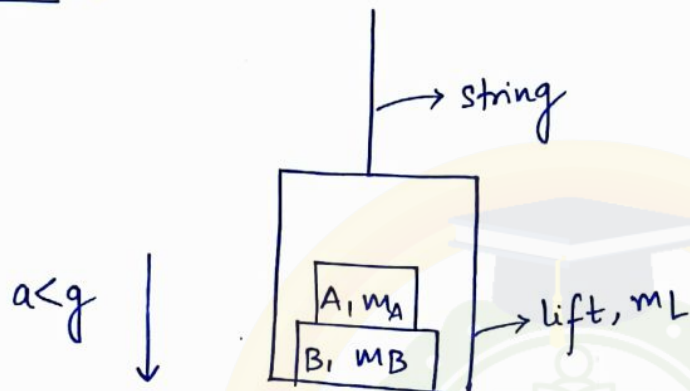
Resultant of actual forces Response

(m₂)

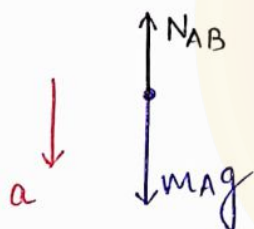


$$T - m_2 g = m_2 a \quad \text{--- 2nd law}$$

Case 3 :->

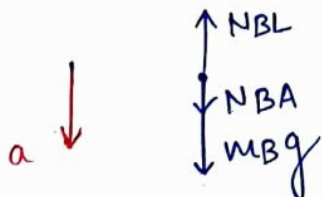


(A)



$$m_A g - N_{AB} = m_A a \quad \text{--- 2nd law}$$

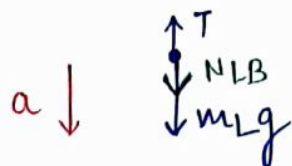
(B)



$$m_B g + N_{BA} - N_{BL} = m_B a \quad \text{--- 2nd law}$$

$$N_{BA} = N_{AB} \quad \text{--- 3rd Law}$$

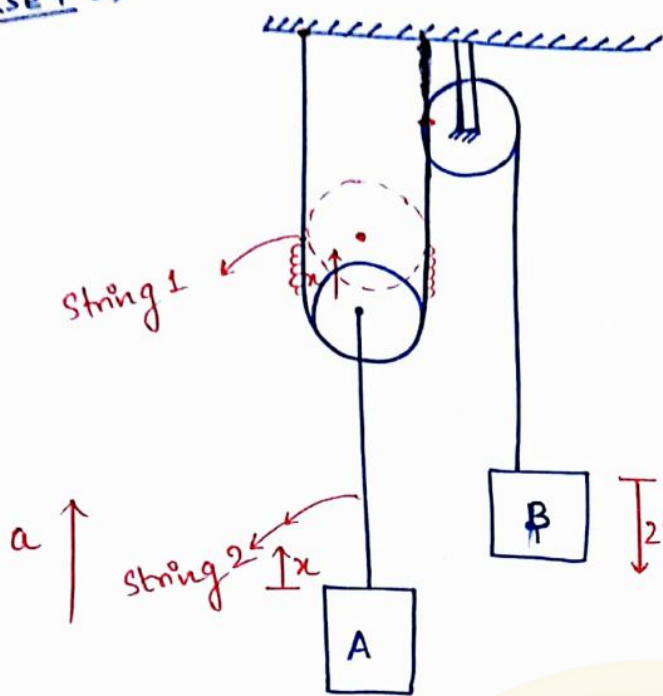
(Lift)



$$N_{LB} + m_L g - T = m_L a \quad \text{--- 2nd law}$$

$$N_{LB} = N_{BL} \quad \text{--- 3rd law}$$

Case 4 →



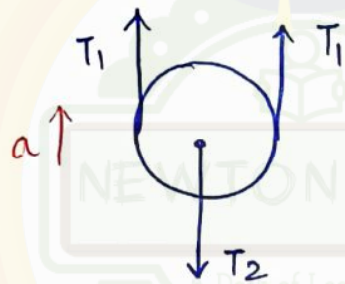
(A)

$$T_2 - m_A g = m_A a \quad \text{--- 2nd law}$$

(B)

$$m_B g - T_1 = m_B 2a \quad \text{--- 2nd law}$$

moving pulley



$$2T_1 - T_2 = (0)a \quad \text{--- 2nd law}$$

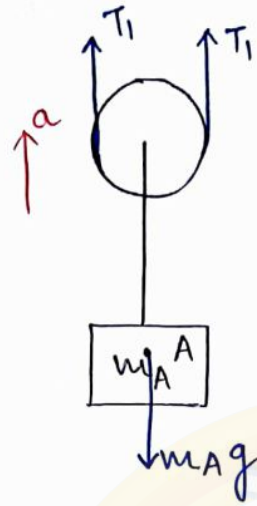
$$2T_1 = T_2$$

put T_2 in 1st equ.

$$\Rightarrow 2T_1 - m_A g = m_A a \quad \text{--- (1)}$$

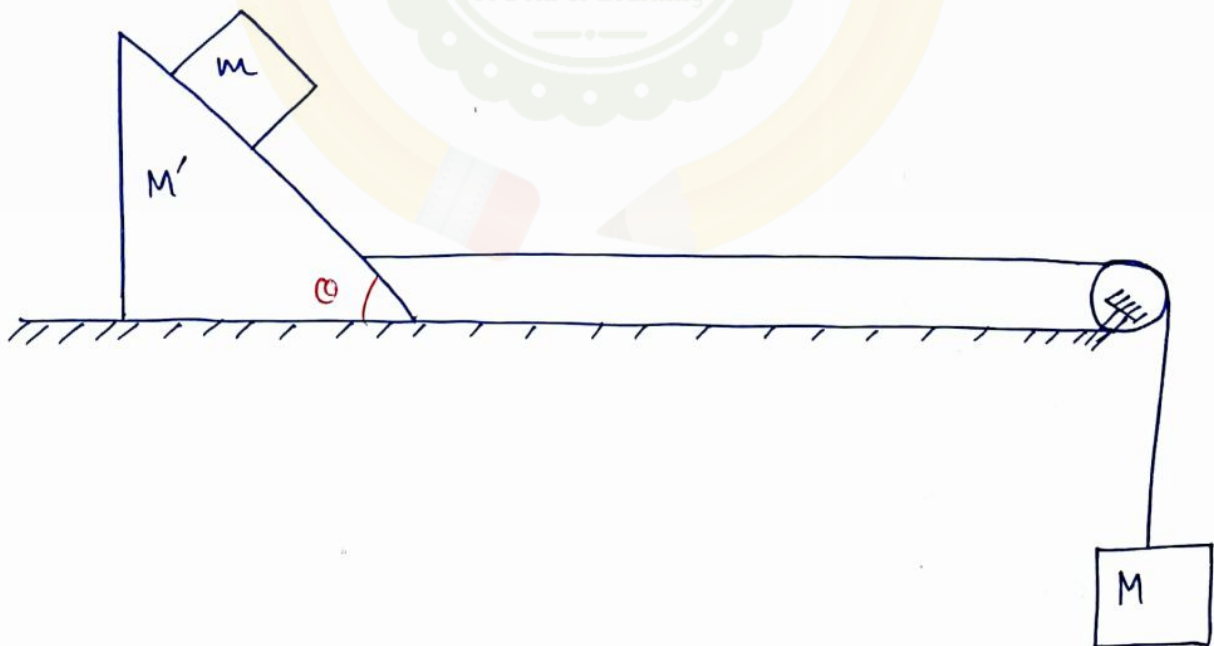
A + string 2 + moving pulley

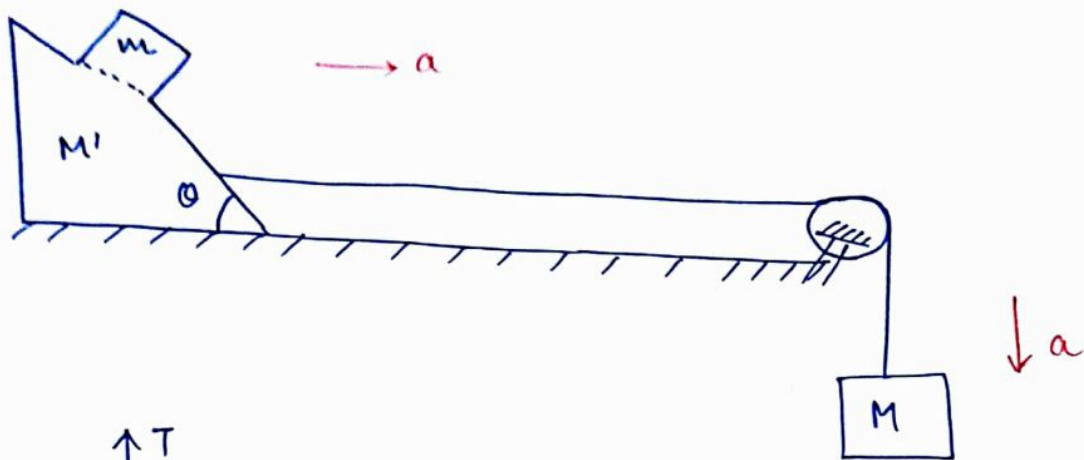
Because $\vec{a}$ vector is same for all.



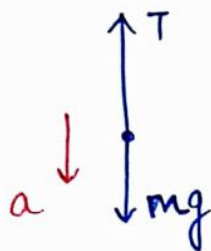
$$2T_1 - m_A g = m_A a \quad \text{--- 2nd Law}$$

Q. find the mass M of the hanging block as shown in figure so as to prevent the slipping of smaller block over the triangulated block. assume all surfaces smooth.
can motion





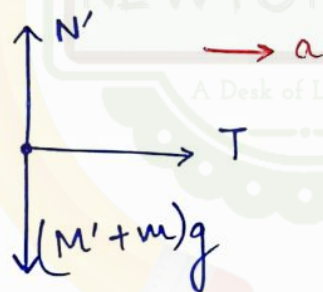
(M)



$$Mg - T = Ma \quad \text{--- 2nd Law} \quad (1)$$

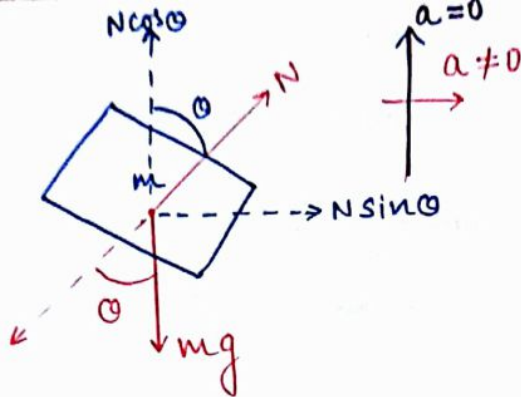
(M' + m)

$$\vec{a}_{M'} = \vec{a}_m = \vec{a}$$



$$N' = (M' + m)g \quad \text{--- 1st Law}$$

$$T = \underbrace{(M' + m)a}_{\text{Response}} \quad \text{--- 2nd Law} \quad (2)$$



$$N \cos \theta = mg \quad \text{--- 1st law}$$

$$N \sin \theta = ma \quad \text{--- 2nd law}$$

response

$$\tan \theta = \frac{a}{g}$$

$$a = g \tan \theta \quad \text{--- (3)}$$

using value of (2), (3) in equ. (1)

$$\Rightarrow Mg - (M' + m)g \tan \theta = Mg \tan \theta$$

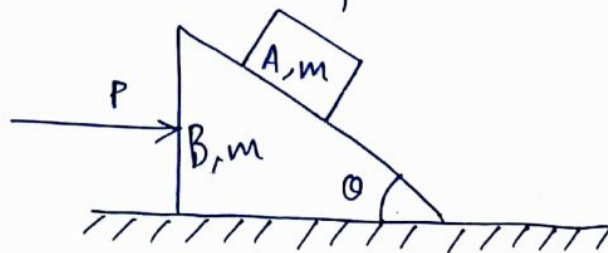
$$\Rightarrow Mg - M'g \tan \theta - mg \tan \theta = Mg \tan \theta$$

$$\Rightarrow M(1 - \tan \theta) = (M' + m) \tan \theta$$

$$\Rightarrow M = \frac{(M' + m) \tan \theta}{(1 - \tan \theta)}$$

$$\Rightarrow \boxed{M = \frac{M' + m}{\cot \theta - 1}}$$

Q Find P such that there is no slipping between A and B. assume all surfaces smooth.

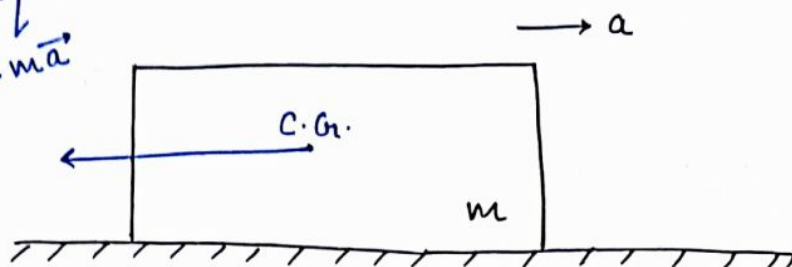


$$\text{Ans} \rightarrow \boxed{P = 2mg \tan \theta}$$

30/7/2016

D'Alembert's Principle → It states "under the action of effective force and inertia force body will be in dynamic equilibrium."

$$\vec{F}_I = -m\vec{a}$$



$$\vec{F}_R = m\vec{a}$$

$$\vec{F}_R + \vec{F}_I = 0$$

effective force
inertia force

If a body of mass 'm' moves rightward with an accel. 'a' then it means there is a resultant force

$$\boxed{\vec{F}_R = m\vec{a}} \text{ acting rightwards.}$$

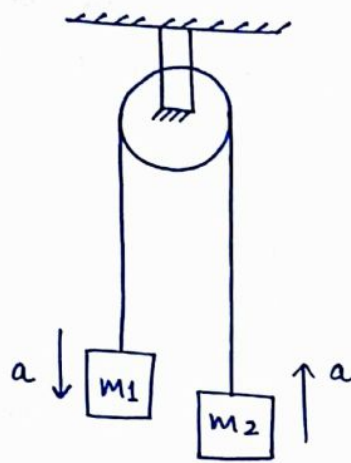
Now If we apply a force $\boxed{F_I = ma}$ acting leftwards at the C.G. of the body then it will bring the body in equilibrium called "DYNAMIC EQUILIBRIUM".

Note ①:- Inertia force acts opposite to the acceleration but not opposite to the motion.

$$\begin{array}{cc} \rightarrow a & \checkmark \rightarrow a \\ \leftarrow v & \rightarrow v \checkmark \end{array}$$

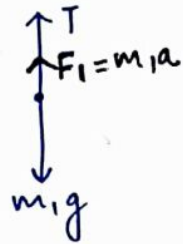
② $\boxed{F_I = ma}$ acting leftwards is an imaginary (Pseudo force) which is not actually acting on the body.

Case I



Considering Dynamic
equim.

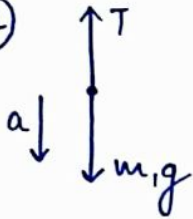
(m1)



$$T_1 + m_1 a = m_1 g - \underline{\underline{\text{D'Alemb}}}$$

Inertia force

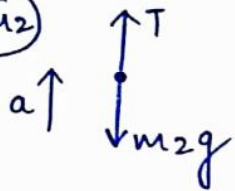
(m1)



$$m_1 g - T = m_1 a - \underline{\underline{\text{2nd}}}$$

effect

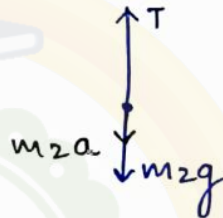
(m2)



$$T - m_2 g = m_2 a - \underline{\underline{\text{2nd}}}$$

effect

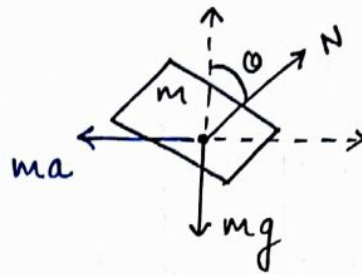
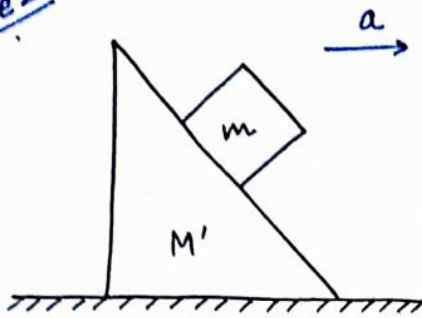
(m2)



$$T = m_2 a + m_2 g - \underline{\underline{\text{D'Alemb}}}$$

Inertia force

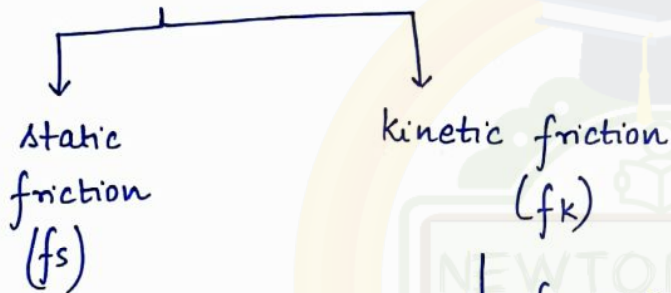
Case 2 →



$$N \cos \theta = mg - \underline{\underline{1^{st}}}$$

$$N \sin \theta = \underbrace{ma}_{\text{Inertia force}} - \underline{\underline{\text{D'Alemberts}}}$$

* FRICTION :→



$$0 \leq f_s \leq (f_s)_{\max}$$

$$(f_s)_{\max} \propto N$$

$$(f_s)_{\max} = \mu_s N$$

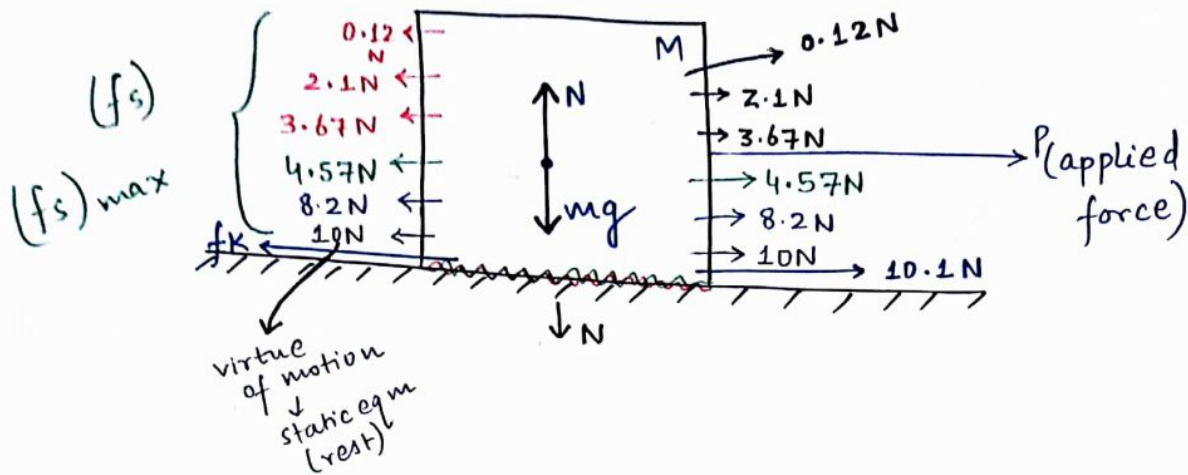
$\mu_s = \text{coeff. of static friction}$

$$f_k = \text{constant} = \mu_k N$$

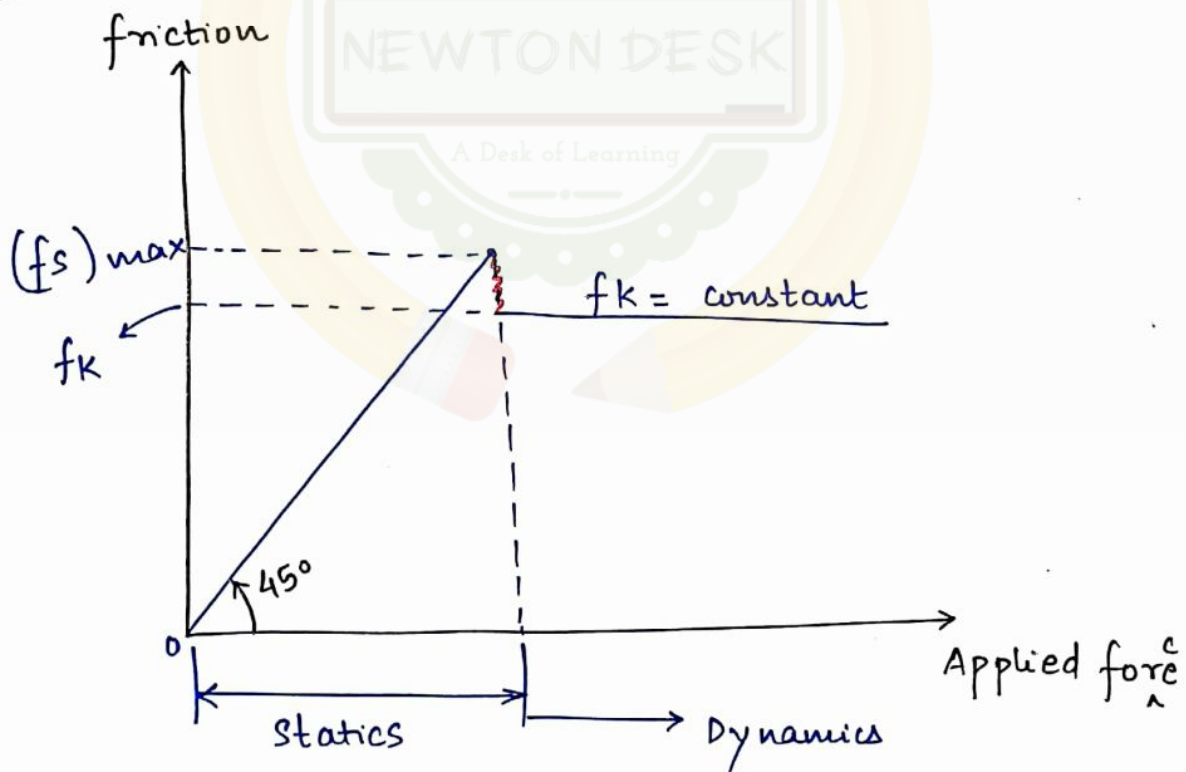
$\mu_k = \text{coeff. of kinetic friction}$

$$(f_s)_{\max} > f_k$$

$$\mu_s > \mu_k$$



* Static friction :- When 2 bodies in contact trying to slide with respect to each other but they are not sliding then there exist static friction whose values varies from 0 to a maximum value $(f_s)_{\max}$ depending on the value of applied force.

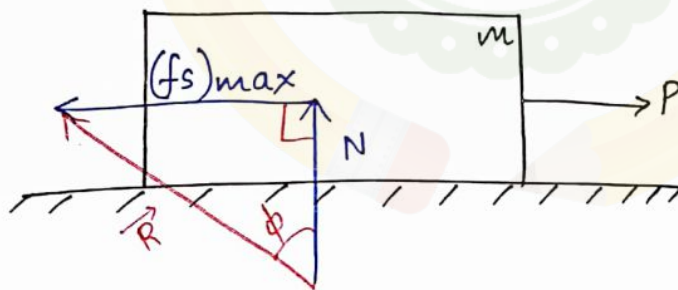


NOTE :- If the value of applied force is less than (or) equal to $(f_s)_{\max}$, then there exist static friction whose value will be exactly equal to the applied load & will act opposite to the applied force.

* Kinetic friction :- when there is relative motion, there exist kinetic friction whose value remains constant and equal to $\mu_k N$ and will act opposite to the relative motion.

* Angle of Friction, ϕ :- when the body is at verge of motion ($(f_s)_{\max}$ is acting), the angle made by contact force with normal reaction is called angle of friction.

$$\tan \phi = \mu_s$$



$$\tan \phi = \frac{(f_s)_{\max}}{N} = \frac{\mu_s N}{N} = \mu_s$$

if μ_s and μ_k are not given separately

i) $\mu_s = \mu_k = \mu$

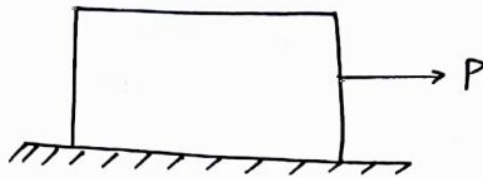
ii) $0 \leq f_s \leq (f_s)_{\max} = \mu N = f_k$

iii) $\tan \phi = \mu$

Practical Q's in Real life
 input force applied
 if μ_s in no picture
 only in gradually applied load for verge of motion.

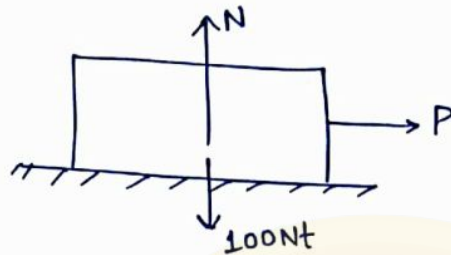
Q Find the Frictional Force developed in the system shown in figure below:- for $P = 14\text{ Nt}$, 50 Nt and 60 Nt

$$\begin{aligned} \text{wt} &= 100\text{ Nt} \\ \mu_s &= 0.5 \\ \mu_k &= 0.3 \end{aligned}$$



Nt $\rightarrow$ Newton

Sol



$$(f_s)_{\max} = \mu_s N = 0.5(100\text{ N}) = 50\text{ Nt}$$

$$f_k = \mu_k N = 0.3(100\text{ N}) = 30\text{ Nt}$$

(i.) if $P = 40\text{ Nt}$

$$P < (f_s)_{\max}$$

$$f = f_s = P = 40\text{ Nt}.$$

(ii.) if $P = 50\text{ Nt}$

$P = (f_s)_{\max}$ (Body is at verge of motion)

$$f = (f_s)_{\max} = 50\text{ Nt}.$$

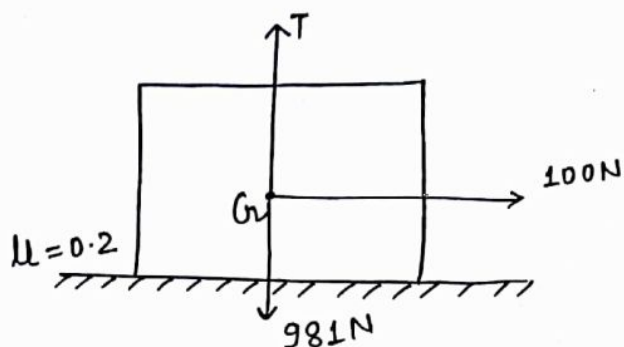
(iii.) if $P = 60\text{ Nt}$

$$P > (f_s)_{\max}$$

$$f = f_k = \mu_k N = 30\text{ Nt}.$$

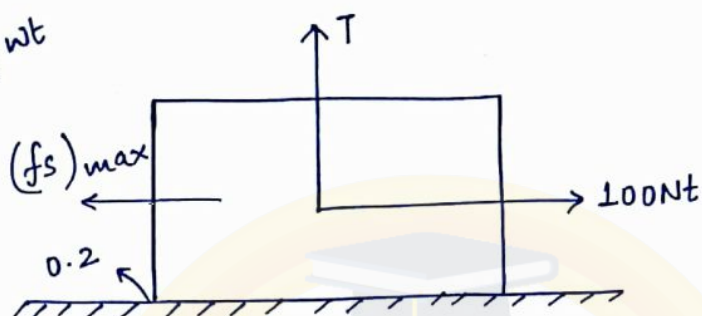
WK 3.10

Pg. 81



Sol

weight = $981 \text{ N} = wt$



$$100 = (fs)_{\max} = \mu N$$

$$100 = 0.2 N$$

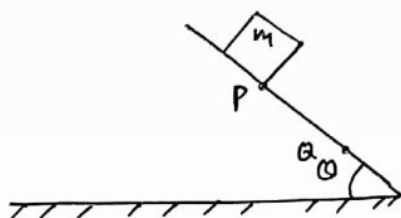
$$\Rightarrow N = 500 \text{ Nt} \quad (1)$$

$$T + N = 981 \quad (2)$$

$$T = 981 - N = 981 - 500 = 481 \text{ Nt}$$



WK Q3.9



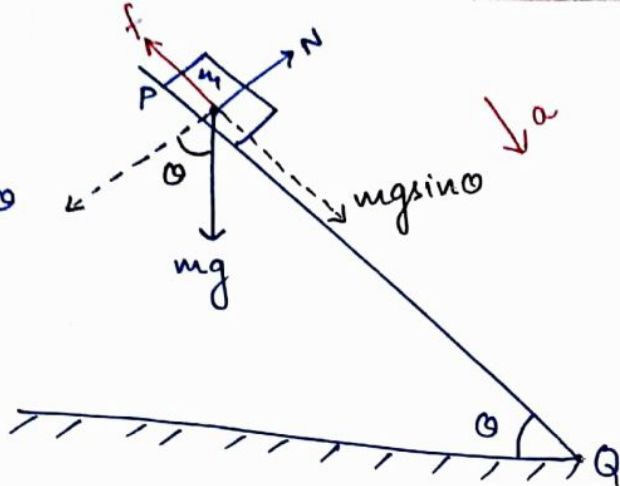
$$\mu, \mu < \tan \theta$$

$$P \rightarrow Q \quad \boxed{PQ = s}$$

$$t = ?$$

Sol

$u=0$
 $\mu < \tan \theta$
 $PQ = S$



released / rest
 $\downarrow$
 $u=0$

$mg \sin \theta$ ← effective force

$$N = mg \cos \theta \quad \text{--- 1st}$$

$$mg \sin \theta > (f_s)_{\max} = \mu N$$

$$mg \sin \theta > \mu mg \cos \theta$$

$$\boxed{\tan \theta > \mu}$$

motion kar raha hai.

$$mg \sin \theta - f_k = Ma \quad \text{--- 2nd}$$

$$mg \sin \theta - \mu mg \cos \theta = ma$$

$$\boxed{a = g \cos \theta [\tan \theta - \mu]}$$

constant

hence, $s = \cancel{ut} + \frac{1}{2} at^2$

$$t = \sqrt{\frac{2s}{a}}$$

$$\boxed{t = \sqrt{\frac{2s}{g \cos \theta (\tan \theta - \mu)}}$$

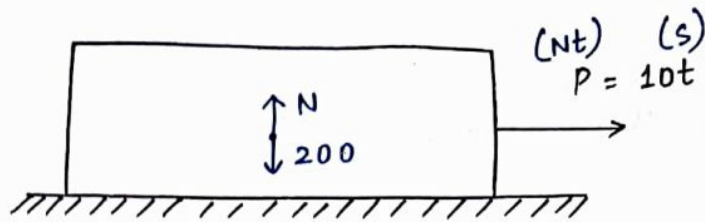
Q3.12

Pg 82

$$Wt = 200 \text{ Nt}$$

$$\mu_s = 0.4$$

$$\mu_k = 0.2$$



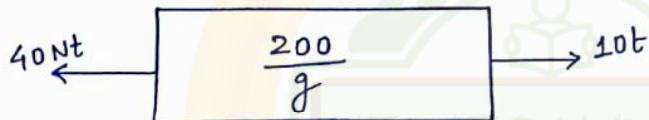
$$(f_s)_{\max} = \mu_s N = 0.4 \times 200 = 80 \text{ Nt.}$$

$$f_k = \mu_k N = 0.2 \times 200 = 40 \text{ Nt.}$$

at $t = 8 \text{ sec}$, block will be at verge of motion

$$V_{t=8s} = 0$$

from $t = 8s$ to $10s$



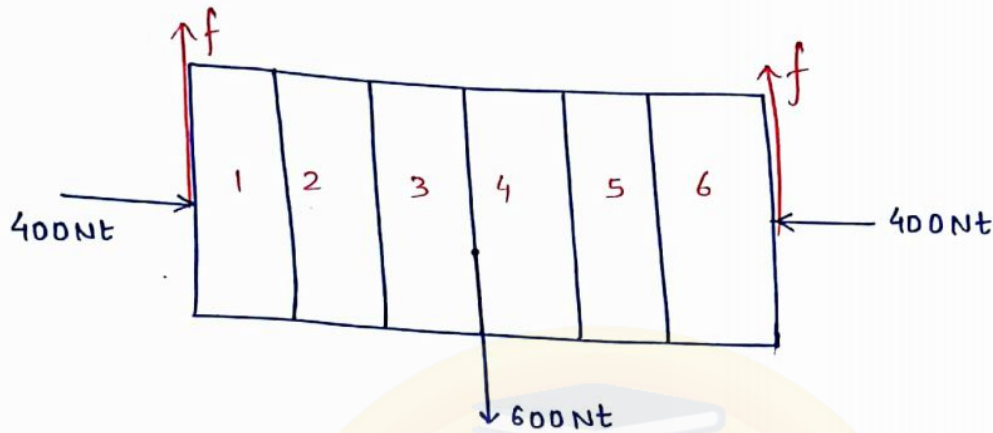
$$a = \frac{10t - 40}{m} = \frac{10t - 40}{\left(\frac{200}{g}\right)} = \frac{dv}{dt}$$

$$\Rightarrow \int_8^{10} (10t - 40) dt = \int_0^{V_{10}} \frac{200}{g} dv$$

$$\Rightarrow \boxed{V_{10} = 4.905 \text{ m/s}}$$

Q 6 identical Books each of weight 100 Nt are lifted with hands by applying compressive force of 400 N as shown in fig. if the Books are in the verge of slipping then μ b/w books and hands is?

Sol



$$2f = 600$$

$$f = 300 = \mu N$$

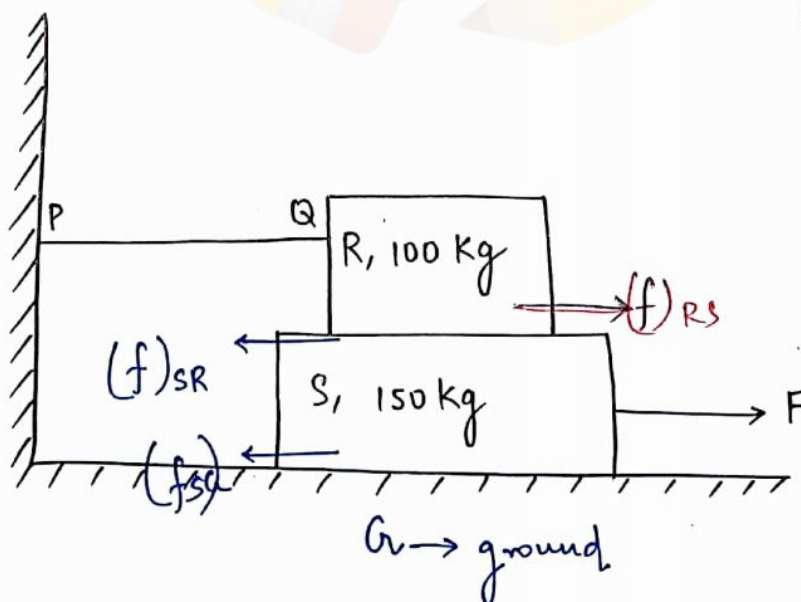
$$\frac{300}{400} = \mu$$

$$\mu = 0.75$$

(massless $\hookrightarrow$ झरोगी नहीं.)

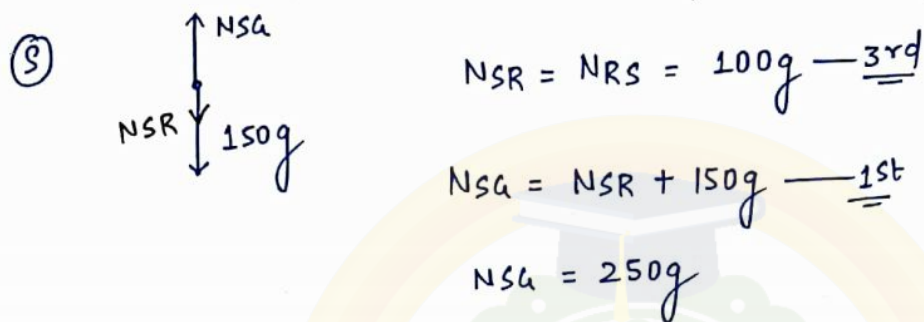
$$\mu_s = 0.4$$

WK
Q3.13

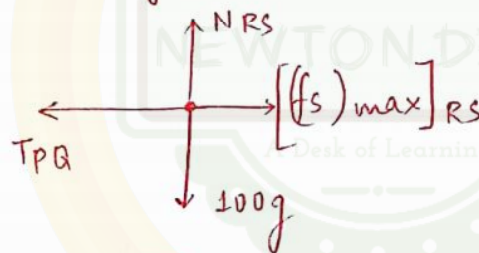


$$F_{\min} = [(f_s)_{\max}]_{SR} + [(f_s)_{\max}]_{SLR}$$

$$F_{\min} = \mu_s N_{SR} + \mu_s N_{SLR} = 0.4 \times 350 \times 9.81 = 1.37 \text{ kN}$$

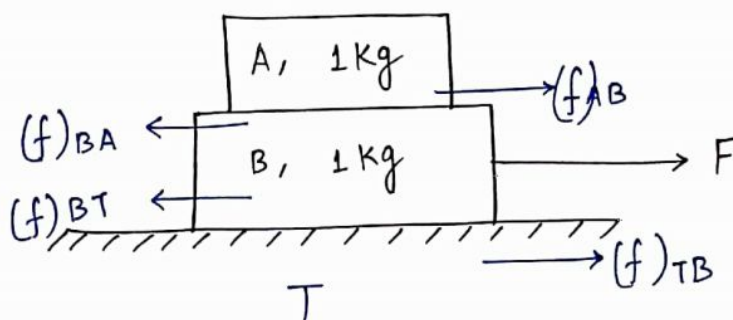


if Tension in string $\rightarrow ?$



$$TPQ = \mu \times 100g$$

(3.)
(Q7)



$$\mu = 0.3$$

$$[(f_s)_{\max}]_{BA} = \mu N_{BA} = 0.3 \times 10 = \underline{3 \text{ N}}$$

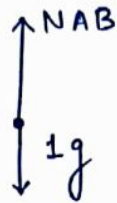
$$[(f_s)_{\max}]_{BT} = \mu N_{BT} = 0.3 \times 20 = \underline{6 \text{ N}}$$

will be at verge of motion.

At $F = 6 \text{ N}$, Both A & B together

Taking $g = 10 \text{ m/s}^2$

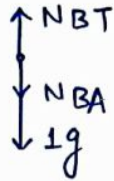
(A)



$$N_{AB} = 10 \text{ Nt} - \underline{1^{\text{st}}}$$

$$N_{BA} = N_{AB} = 10 \text{ Nt} - \underline{3^{\text{rd}}}$$

(B)



$$N_{BT} = N_{BA} + 1g$$

$$N_{BT} = N_{BA} + 10 - \underline{1^{\text{st}}}$$

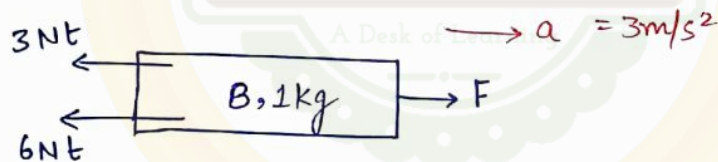
$$N_{BT} = 20 \text{ Nt.}$$

(A)

A, 1 kg $\rightarrow 3 \text{ Nt} = [(f_s)_{\text{max}}]_{AB}$

$$(a_A)_{\text{max}} = \frac{3}{1} = 3 \text{ m/s}^2$$

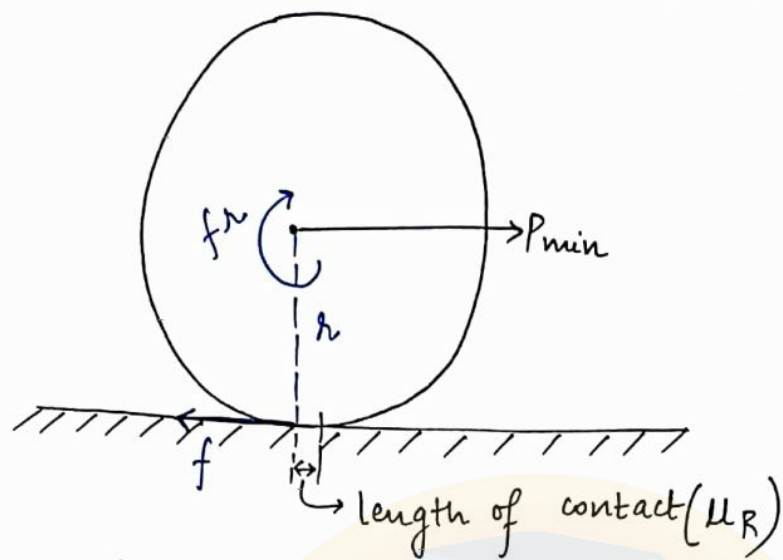
(B)



$$F - 9 = 1 \times 3$$

$$F = 12 \text{ Nt}$$

* Rolling Resistance (f_R) →



$$P_{min} = f_R = \frac{\mu_R W}{R}$$

W → weight 'or' load on Roller

R → Radius of Roller

μ_R → coefficient of Rolling resistance (mm)

Q. A wheel of radius 600 mm diameter on a horizontal steel rail carries a load of 500 N, if coefficient of rolling resistance is 0.3 mm, the force in Nt necessary to roll the wheel along the rail is ?

Sol.

$$D = 600 \text{ mm} ; R = 300 \text{ mm}$$

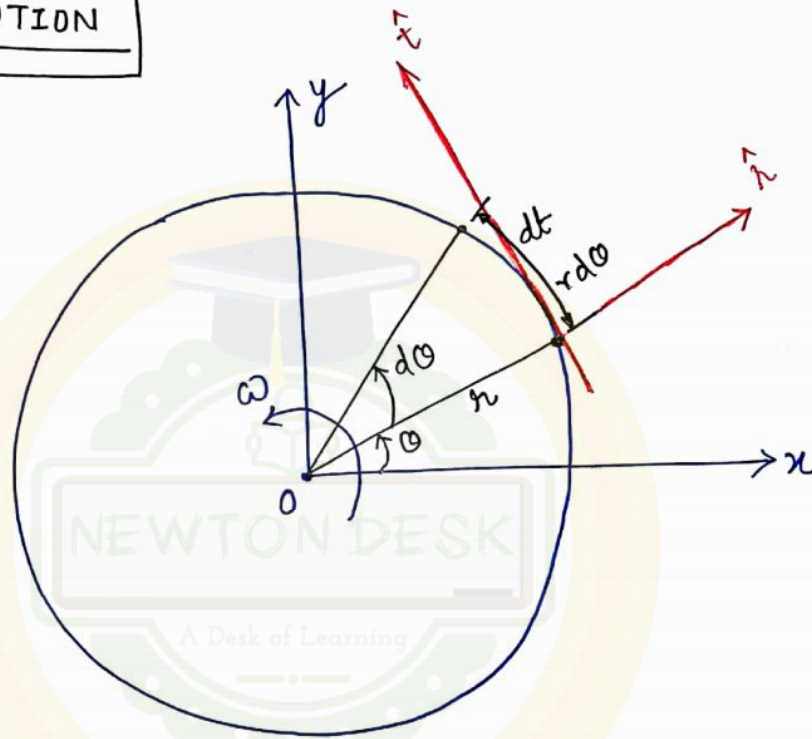
$$\mu_R = 0.3 \text{ mm}$$

$$W = 500 \text{ N}$$

$$P_{\min} = fR = \frac{\mu_R W}{R} = \frac{0.3 \times 500}{300} = 0.5 \text{ Nt}$$

* CIRCULAR MOTION

Kinematics



ω = angular velocity

$$= \frac{d\theta}{dt} \text{ (rad/s)}$$

$$\alpha = \frac{d\omega}{dt} \text{ (rad/s}^2\text{)}$$

$\vec{\omega}$ and $\vec{\alpha}$ $\rightarrow$ vector quantities

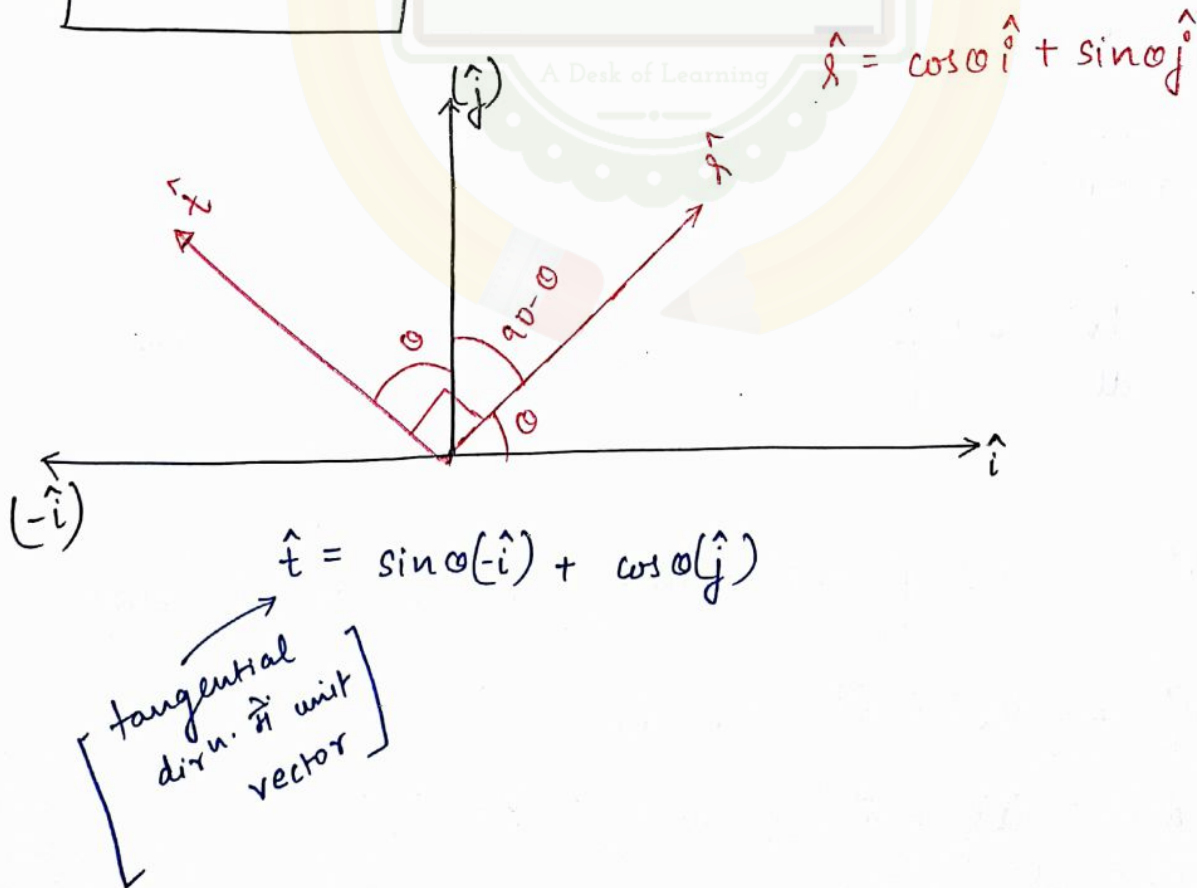
Dirn $\rightarrow \vec{\omega}$ and $\vec{\alpha}$ will be $\perp$ to plane of ~~circle~~ circle, will pass through centre to and will be given by Right hand thumb rule.

$$\underline{\text{Linear speed}} = \frac{\text{Distance}}{\text{time}} = \frac{r d\theta}{dt} = r\omega \text{ (m/s)}$$

If $\vec{\alpha}$ is constant $\Rightarrow$

1. $\omega = \omega_0 + \alpha t$
2. $\theta = \omega_0 t + \frac{1}{2} \alpha t^2$
3. $\omega^2 = \omega_0^2 + 2\alpha\theta$

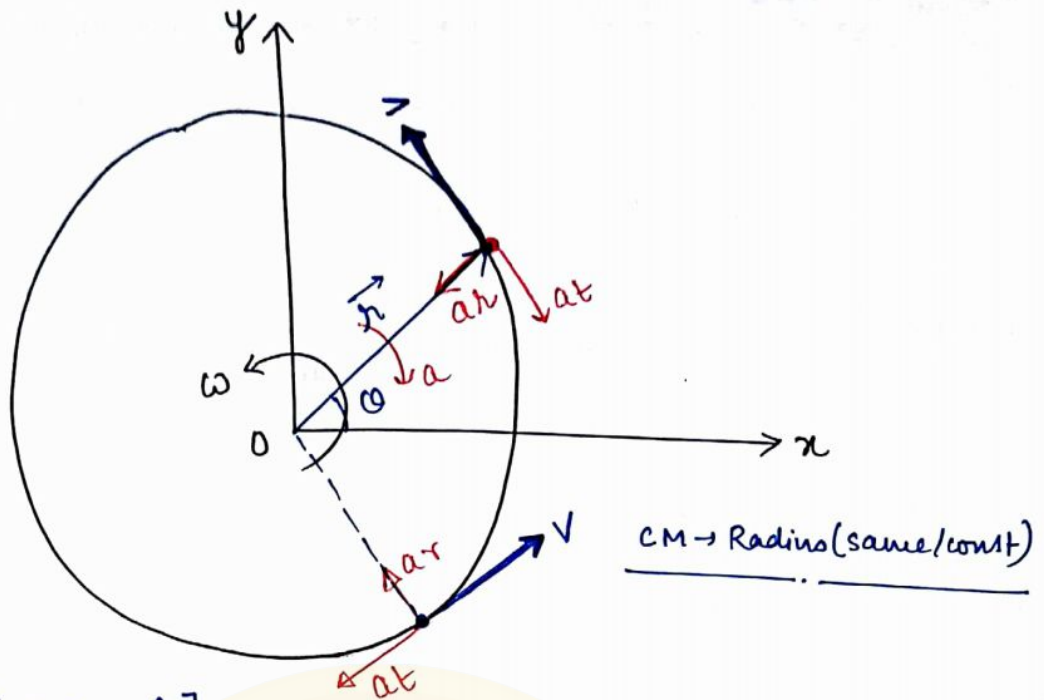
$$\boxed{(2\pi) \left(\frac{N}{60} \right) = \omega}$$



$\vec{v}, \vec{a}$

$$\vec{v} = \frac{d\vec{r}}{dt}$$

$$\vec{r} = |\vec{r}| \hat{n}$$



$$\vec{r} = r [\cos\theta \hat{i} + \sin\theta \hat{j}]$$

$$\vec{v} = \frac{d\vec{r}}{dt} = r \left[-\sin\theta \frac{d\theta}{dt} \hat{i} + \cos\theta \frac{d\theta}{dt} \hat{j} \right]$$

$$\vec{v} = r\omega [-\sin\theta \hat{i} + \cos\theta \hat{j}]$$

$$\boxed{\vec{v} = r\omega (\hat{t})}$$

$$\vec{a} = \frac{d\vec{v}}{dt} = r \left[\omega \left(-\cos\theta \frac{d\theta}{dt} \hat{i} - \sin\theta \frac{d\theta}{dt} \hat{j} \right) + (-\sin\theta \hat{i} + \cos\theta \hat{j}) \frac{d\omega}{dt} \right]$$

$$\vec{a} = r\omega^2 [-\cos\theta \hat{i} - \sin\theta \hat{j}] + r\alpha (-\sin\theta \hat{i} + \cos\theta \hat{j})$$

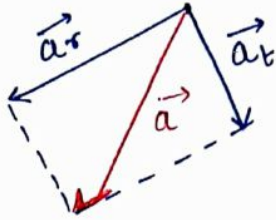
$$\vec{a} = r\omega^2 (-\hat{r}) + r\alpha (\hat{t})$$

$$\vec{a} = \vec{a}_r + \vec{a}_t \rightarrow \begin{array}{l} \text{causes change in} \\ \text{direction of velocity} \end{array} \quad \begin{array}{l} \text{causes change in} \\ \text{mag. of vel.} \end{array}$$

$$a_r = \lambda \omega^2 = \frac{v^2}{r}$$

$$v = \lambda \omega$$

$$\frac{v}{\lambda} = \omega$$



$$|\vec{a}| = \sqrt{a_r^2 + a_t^2}$$

* Uniform Circular Motion: →

$$v = \lambda \omega = \text{Constant} \Rightarrow \omega = \text{Constant} \Rightarrow \alpha = 0 \Rightarrow \vec{a}_t = 0$$

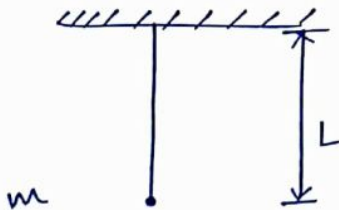
$$\vec{a} = \vec{a}_r$$

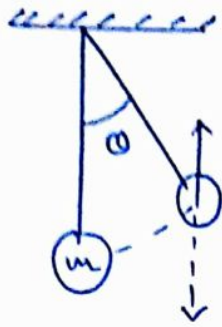
Note:

Q A particle of mass 'm' is suspended from a ceiling through a string of length 'L' and moves in a horizontal circle of Radius 'r'. Find :- (a) speed of the particle.

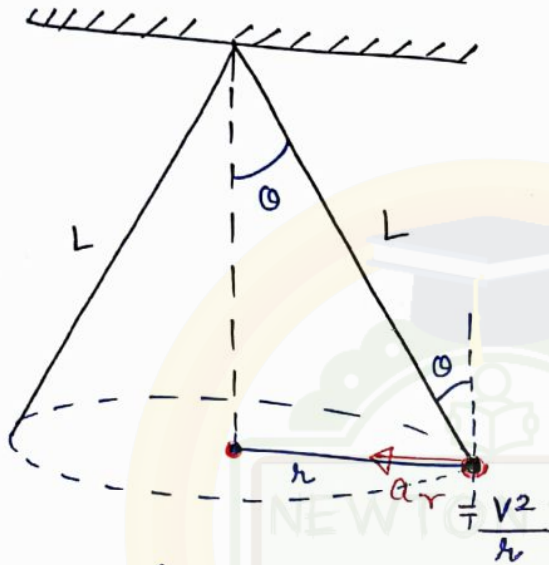
(b) Tension in the string.

Sol

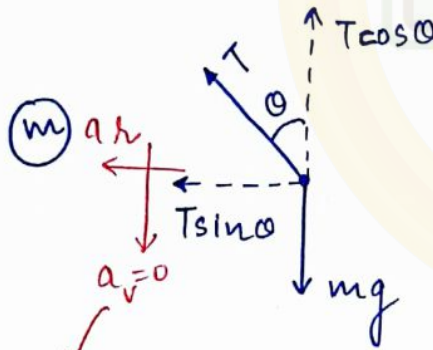




Sol :-



$$\sin \theta = \frac{r}{L}$$



$$T \cos \theta = mg \quad \text{--- 1st}$$

$$T \sin \theta = m|\vec{a}_h| = \frac{mv^2}{r} \quad \text{--- 2nd}$$

acc_r in vertical

centripetal force

actual forces ka Resultant

FR (net Resultant force.)

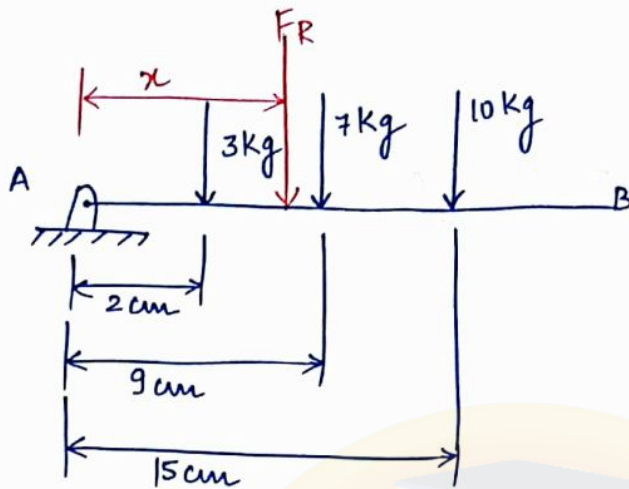
effect

$$T = \frac{mg}{\cos \theta}$$

$$\tan \theta = \frac{v^2}{rg}$$

$$\Rightarrow V = \sqrt{rg \tan \theta}$$

Q1.3



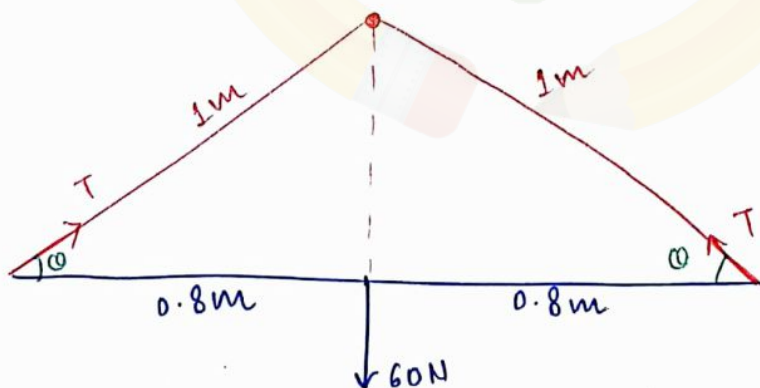
$$F_R = 3 + 7 + 10 = 20 \text{ kg}$$

$$\sum \vec{M}_A = 3 \times 2 + 7 \times 9 + 10 \times 15 = F_R \times x$$

$$6 + 63 + 150 = F_R x = 20 \times x$$

$$x = 10.95 \text{ cm}$$

Q1.8



$$2T \sin \theta = 60 \text{ N}$$

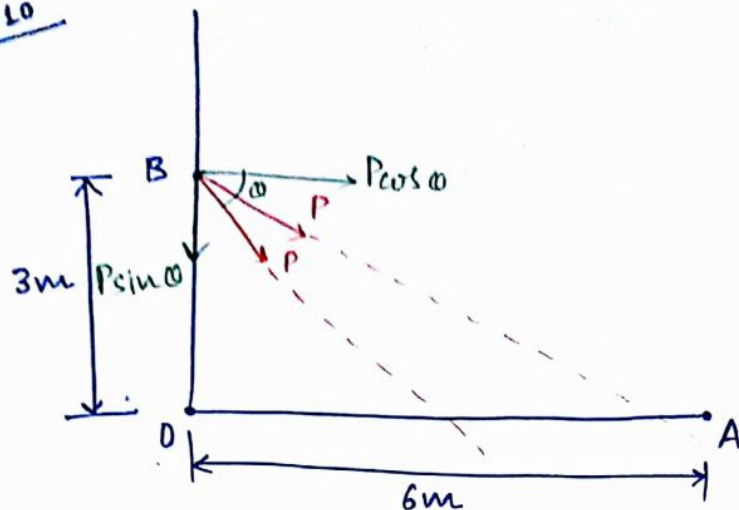
$$T = 30 / \sin \theta$$

$$\cos \theta = \frac{0.8}{1} = 0.8$$

$$\sin \theta = \sqrt{1 - 0.8^2} = 0.6$$

$$T = 50$$

L. 10



$$M_O = 180 = 3P \cos \theta \quad \text{--- (1)}$$

$$M_A = 90 = 6P \sin \theta - 3P \cos \theta \quad \text{--- (2)}$$

$$M_O = 180 \text{ Nm}; \text{ CW}$$

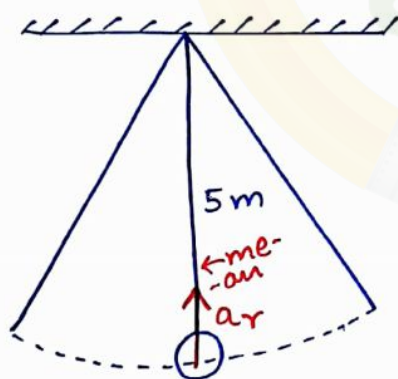
$$M_A = 90 \text{ Nm}; \text{ ACW}$$

$$M_B = 0$$

6/10/2016 Q> A simple pendulum of length 5m with a bob of mass 1kg is in simple harmonic motion as it passes through its mean position, the bob has a speed of 5m/sec. The net force on the bob at the mean position is?

← mean position

Sol>



Newton's
2nd Law
of motion

FBD of Bob

(Bob)



$$T - mg = ma_1 \quad \text{--- 2nd}$$

↑
centripetal force

$$T - mg = \frac{mv^2}{r}$$

$$F_{\text{net}} = \frac{mv^2}{r} = \frac{1 \times 5^2}{5} = 5 \text{ Newton (Nt)}$$

Q. A stone of mass 'm' is tied to the end of a string of length 'l' and whirled in a vertical circle at constant speed. The Tension in the string is max^m. when the stone is —

(a) at top of circle.

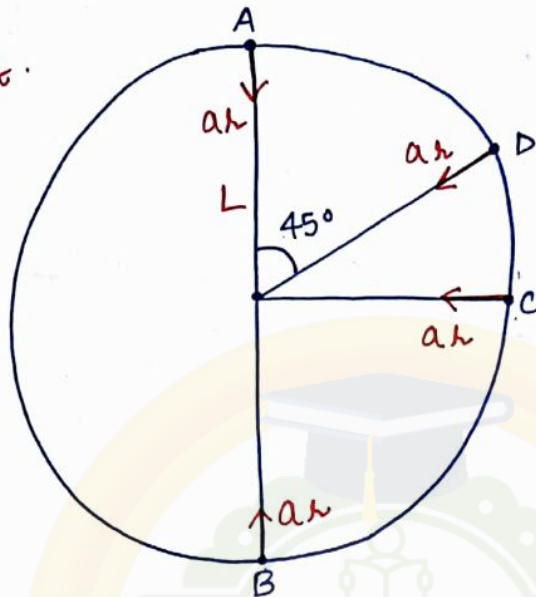
(b) at bottom of circle.

(c) half way down from top.

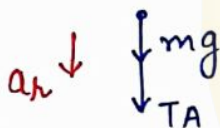
(d) Quarter way down from top.

Ans.

$v = \text{const.}$
 $a_h = \frac{v^2}{L} = \text{const.}$



(A)

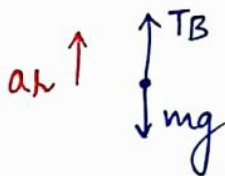


$$T_A + mg = ma_h \quad \text{--- 2nd}$$

$$T_A + mg = \frac{mv^2}{L}$$

$$T_A = \frac{mv^2}{L} - mg = T_{\min}$$

(B)

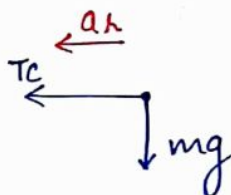


$$T_B - mg = ma_h \quad \text{--- 2nd}$$

$$T_B - mg = \frac{mv^2}{L}$$

$$T_B = \frac{mv^2}{L} + mg = T_{\max}$$

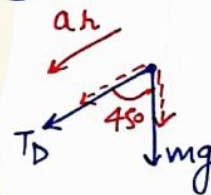
(C)



$$T_C = ma_h \quad \text{--- 2nd}$$

$$T_C = \frac{mv^2}{L}$$

(D)

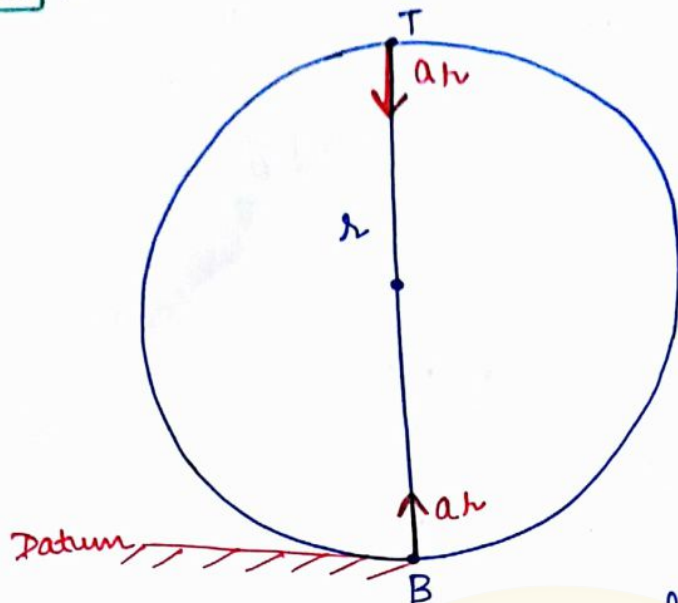


$$T_D + mg \cos 45^\circ = ma_r \quad \text{--- 2nd}$$

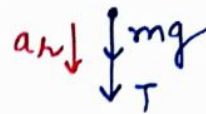
$$T_D + mg \cos 45^\circ = \frac{mv^2}{L}$$

$$T_D = \frac{mv^2}{L} - mg \cos 45^\circ$$

* Minm. velocities at Top and Bottom to complete a vertical circle :-



Top



$$T + mg = ma_h - \text{2nd radial accn.}$$

$$T + mg = \frac{mv^2}{r}$$

$$\overset{\text{min}}{A} + \overset{\text{min}}{B} = \overset{\text{min}}{C} \text{ const.}$$

for velocity to be minimum,

$$T = 0$$

$$\cancel{mg} = \frac{\cancel{m}v_{\min}^2}{r}$$

$$v_{\min} = \sqrt{gr}$$

Now, Bottom

$$\underset{\text{gravitational}}{(GPE)_B} + (K.E)_B = (GPE)_T + (KE)_T$$

$$0 + \frac{1}{2} \cancel{m} v_B^2 = \cancel{m} g (2r) + \frac{1}{2} \cancel{m} v_T^2$$

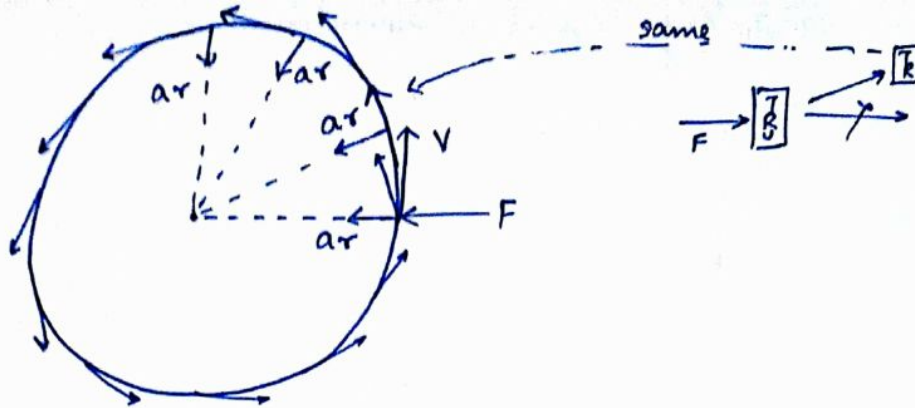
$$v_B^2 = 4gr + v_T^2$$

$$\overset{\text{min}}{C} = \overset{\text{const.}}{A} + \overset{\text{min}}{B}$$

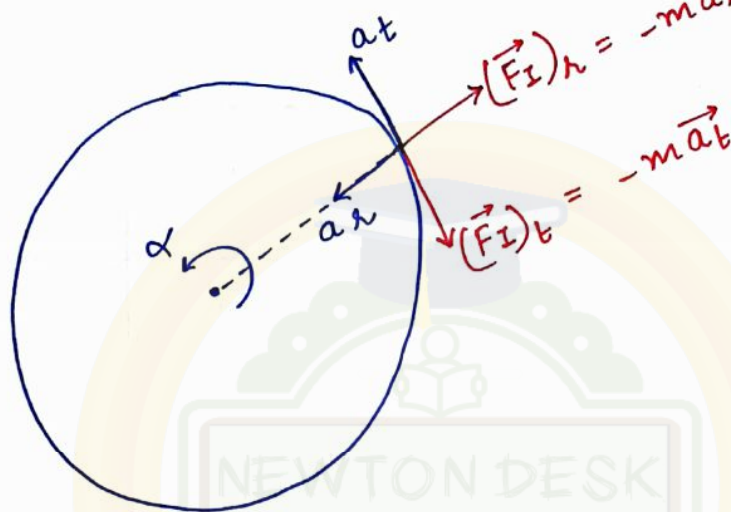
$$v_B^2 = 4gr + gr$$

$$(v_B)_{\min} = \sqrt{5gr}$$

NOTE :-



*** D'Alemberts Analysis in Circular Motion :-**



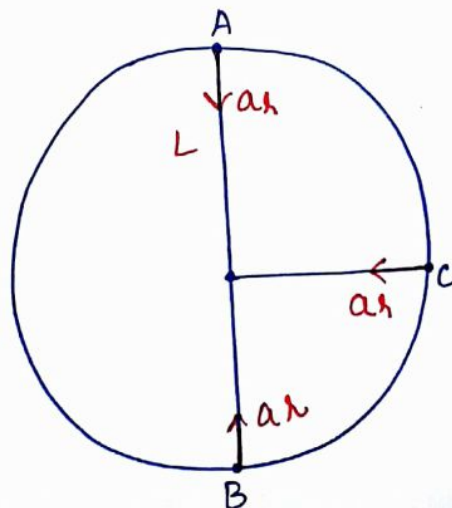
$$(\vec{F}_R)_r + (\vec{F}_I)_r = 0$$

centripetal force centrifugal force

$$(\vec{F}_R)_t + (\vec{F}_I)_t = 0$$

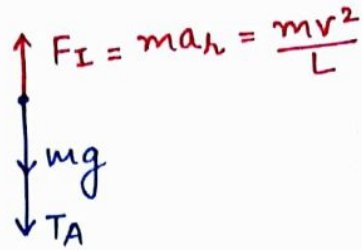
Case →

$v = \text{const.}$
 $a_t = \frac{v^2}{L} = \text{const.}$



Considering dynamic equilibrium

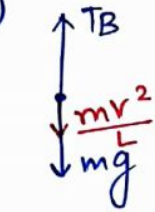
(A)



$$T_A + mg = \frac{mv^2}{L} \quad \text{--- D'Alemb}$$

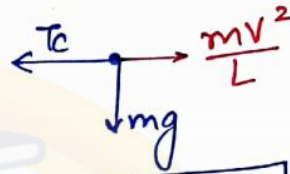
Inertia
force

(B)



$$T_B = mg + \frac{mv^2}{L}$$

(C)



$$T_C = \frac{mv^2}{L}$$

→ D'Alemb

NEWTON DESK

A Desk of Learning

* Linear Momentum, $\vec{P} = m\vec{V}_{cm}$

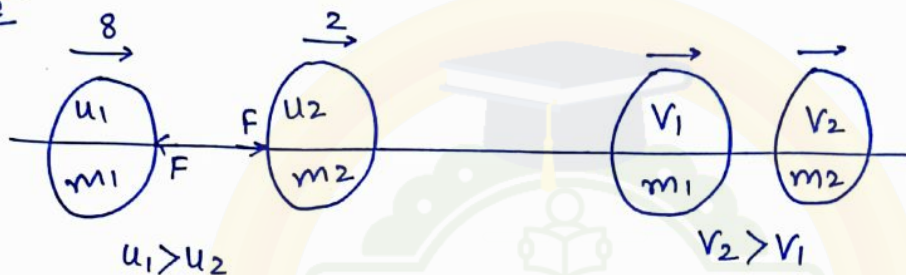
$$\frac{d\vec{P}}{dt} = \frac{d(m\vec{V}_{cm})}{dt} = m \frac{d\vec{V}_{cm}}{dt} = m \vec{a}_{cm} = (\vec{F}_R)_{ext} \quad \text{2nd}$$

$$\boxed{\frac{d\vec{P}}{dt} = (\vec{F}_R)_{ext}} \rightarrow \text{2nd}$$

* Conservation of $\vec{P} \Rightarrow \vec{P} = \text{constant} \Rightarrow d\vec{P} = 0$

only when $(\vec{F}_R)_{ext} = 0$

Exam:-
Ple



$$m_1 + m_2$$

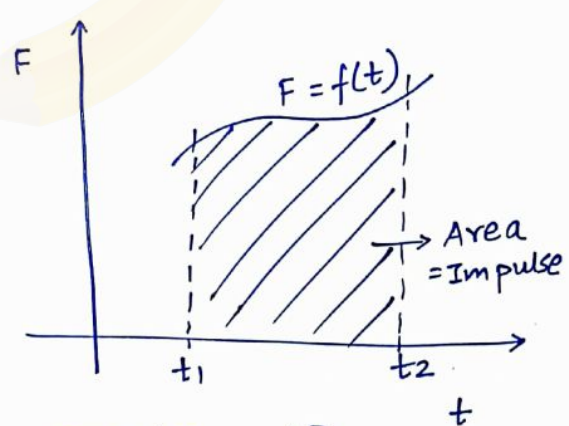
$$\boxed{m_1 u_1 + m_2 u_2 = m_1 v_1 + m_2 v_2} \leftarrow \text{vector form}$$

* Impulse $\rightarrow$

$$\vec{F} = \frac{d\vec{P}}{dt}$$

$$\boxed{\vec{F} dt = \frac{d\vec{P}}{\text{change in } \vec{P}}}$$

Impulse
(vector)



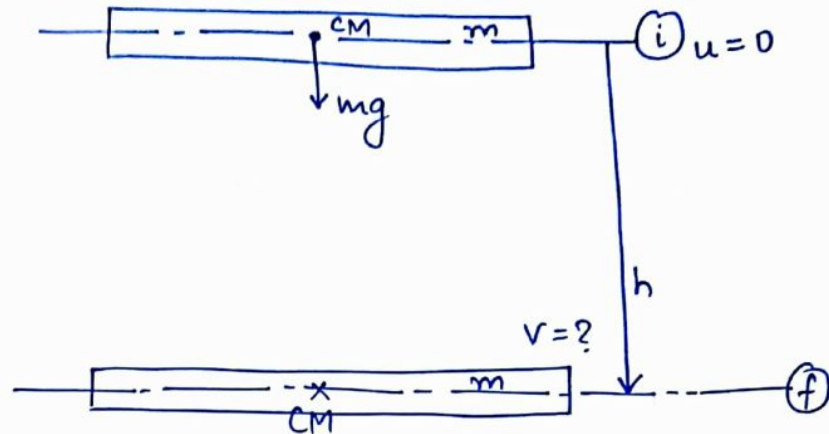
$$\text{Impulse} = \Delta P$$

$$\int_{t_1}^{t_2} F dt = \Delta P$$

* Work - Energy Theorem $\rightarrow$

$$\text{Total workdone} = \Delta(K.E.) = (K.E.)_f - (K.E.)_i$$

Case 1 $\rightarrow$



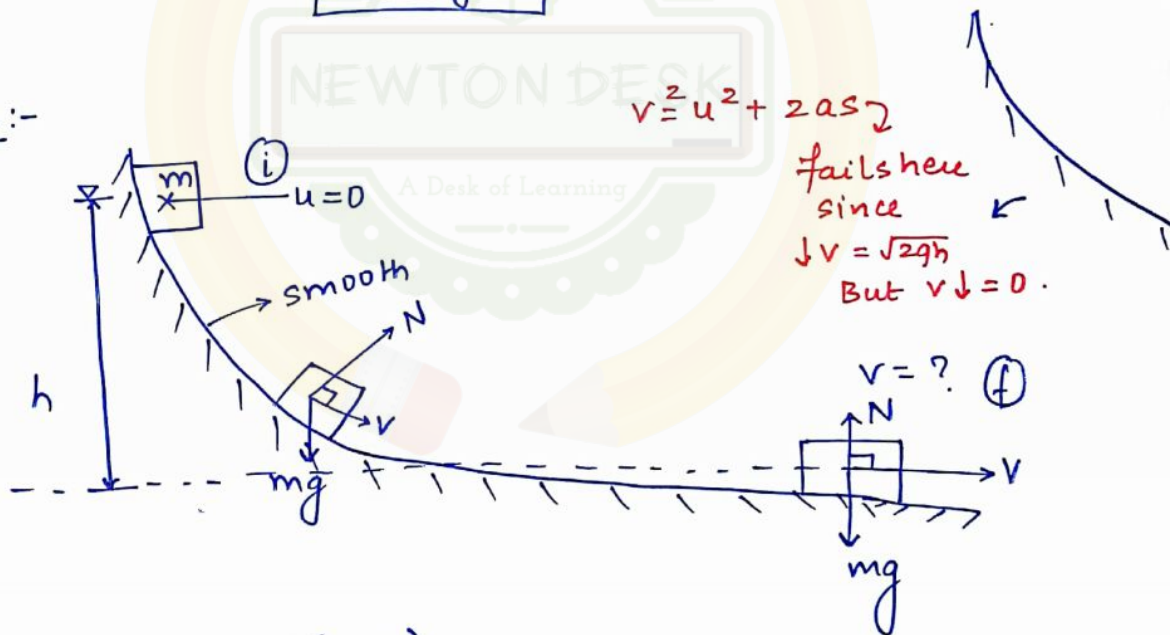
(m)

$$\text{TWD} = \Delta(K.E.)$$

$$Wmg = mgh = \frac{1}{2}mv^2 - 0$$

$$v = \sqrt{2gh}$$

Case 2:-



$$v^2 = u^2 + 2as$$

fails here
since
 $v = \sqrt{2gh}$
But $v \downarrow = 0$.

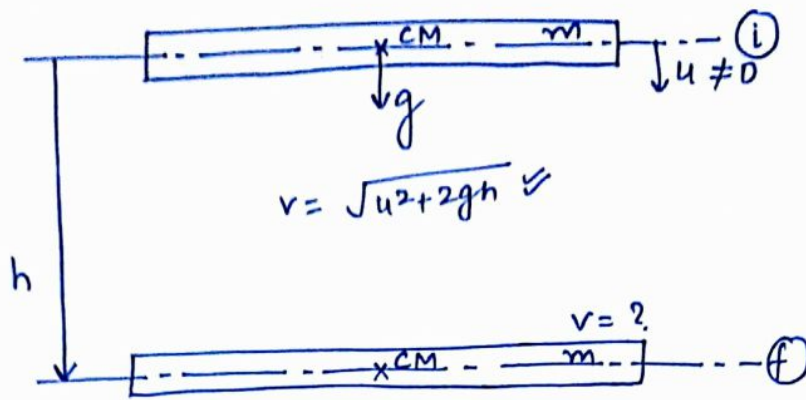
(m)

$$\text{TWD} = \Delta(K.E.)$$

$$Wmg + W_N = mgh = \frac{1}{2}mv^2 - 0$$

$$\sqrt{2gh} = v$$

Case 3 :-



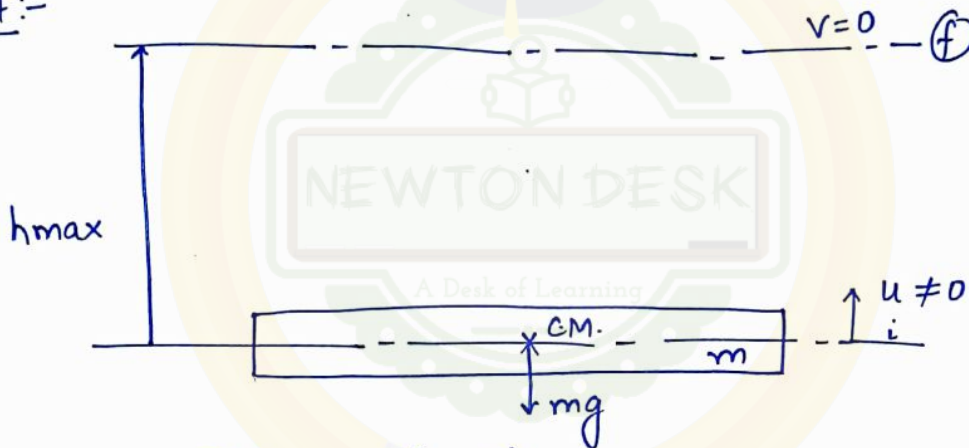
$$v = \sqrt{u^2 + 2gh}$$

$$TWD = Wmg = \Delta(K.E.)$$

$$mgh = \frac{1}{2}mv^2 - \frac{1}{2}mu^2$$

$$v = \sqrt{u^2 + 2gh}$$

Case 4 :-



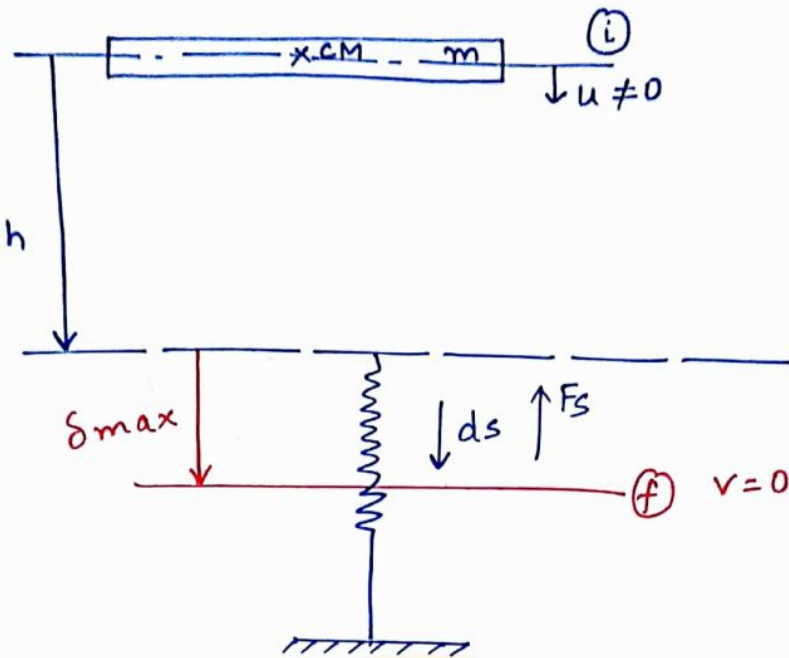
$$TWD = \Delta(K.E.)$$

$$Wmg = mgh_{max} = 0 - \frac{1}{2}mu^2$$

$$h_{max} = \frac{u^2}{2g}$$

Case 5 :-

$\delta_{\max} = ?$



(m)

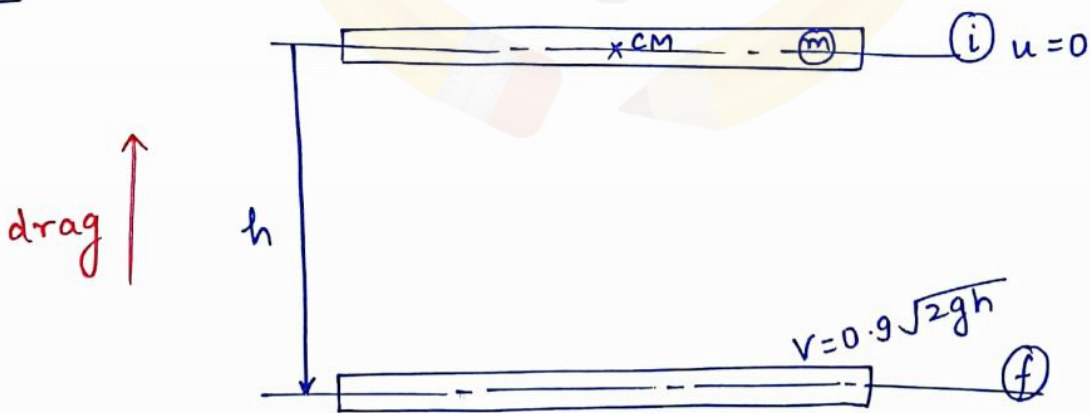
$$TWD = \Delta(K.E.)$$

$$W_{mg} + W_{F_s} = mg(h + \delta_{\max}) - \frac{1}{2} K \delta_{\max}^2 = 0 - \frac{1}{2} m u^2$$

spring force

$$\Rightarrow \boxed{mg(h + \delta_{\max}) + \frac{1}{2} m u^2 = \frac{1}{2} K \delta_{\max}^2}$$

Pr (1)



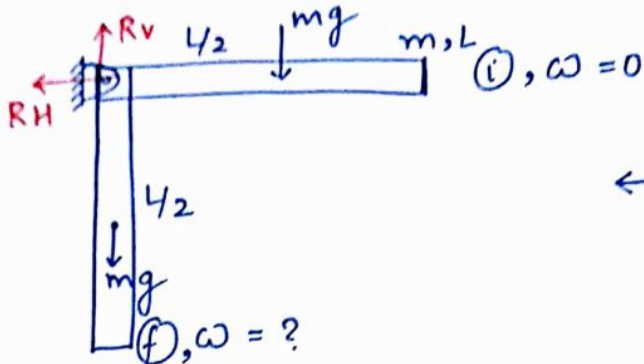
$$W_{\text{drag}} = ?$$

Sol.

$$\begin{aligned} \text{TWD} &= Wmg + W_{\text{drag}} \\ &= mgh + W_{\text{drag}} = \frac{1}{2} m(0.81) \times 2gh - 0 \\ &\quad \frac{1}{2} mv^2 \end{aligned}$$

$W_{\text{drag}} = -0.19mgh$

Pb ②



← Pure Rotational Motion

Sol

$$\text{TWD} = \Delta(K.E.)$$

$$Wmg = mg \times L/2 = \frac{1}{2} I \omega^2 - 0$$

$$mg \times L/2 = \frac{1}{2} \times \frac{mL^2}{3} \times \omega^2$$

$\omega = \sqrt{\frac{3g}{L}}$

* COLLISIONS/Impacts :-

It is a short time phenomenon in which momentum of the individual Body undergoing ^{collision} changes significantly.

- ① Perfectly Elastic ($e = 1$).
- ② Perfectly Inelastic/Plastic ($e = 0$).
- ③ Partially Elastic ($0 < e < 1$).

① Perfectly Elastic ($e=1$) $\rightarrow$ recovery of shape

when the initial K.E. = final K.E., such a collision is called perfectly elastic collision.

{ collide $\rightarrow$ deform $\rightarrow$ Regain }
so K.E. conserved



$m_1 + m_2$

1st order eqn. $m_1 u_1 + m_2 u_2 = m_1 v_1 + m_2 v_2$ — (1) ✓

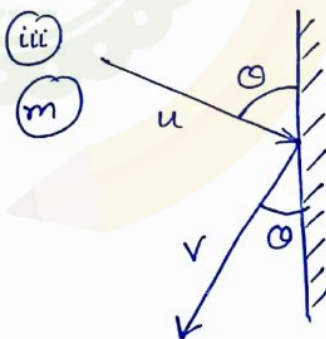
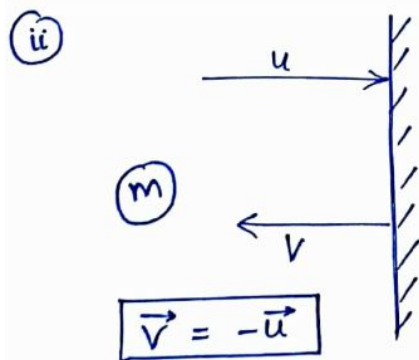
2nd order eqn. $\frac{1}{2} m_1 u_1^2 + \frac{1}{2} m_2 u_2^2 = \frac{1}{2} m_1 v_1^2 + \frac{1}{2} m_2 v_2^2$ — (2)

equating or from (1) & (2)

$u_1 - u_2 = v_2 - v_1$ — (3) ✓
velocity of approach before collision = velocity of separation after collision

Special cases for $e=1$

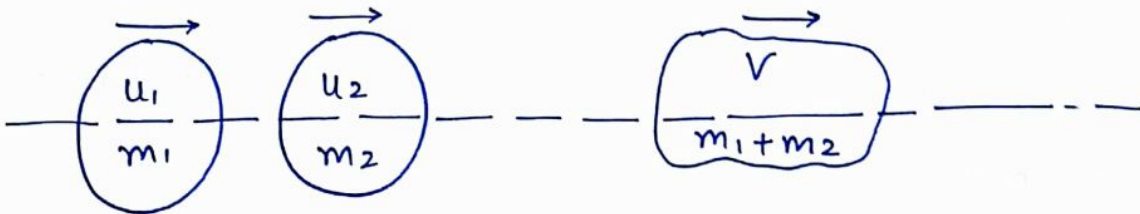
i) if $m_1 = m_2$
 $u_1 = v_2$
 $u_2 = v_1$



② Perfectly Inelastic / Plastic ($e = 0$) $\Rightarrow$

when two perfectly inelastic bodies moving along the same line collide, they stick to each other.

collide $\rightarrow$ deform
(plastic)

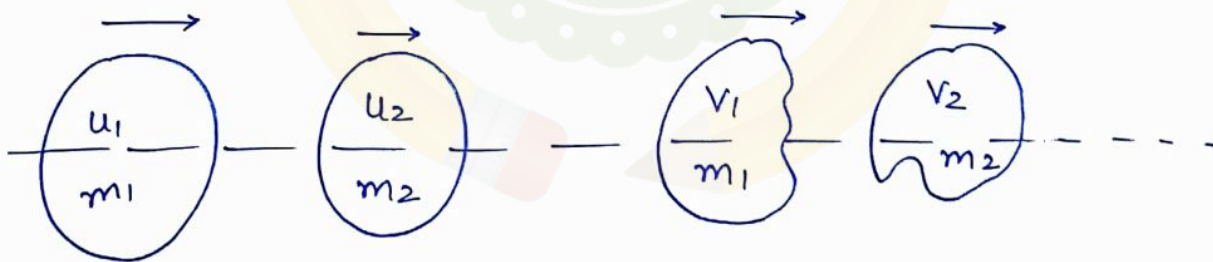


$m_1 + m_2$

$$m_1 u_1 + m_2 u_2 = (m_1 + m_2) V$$

$$(K.E.)_{\text{loss}} = (K.E.)_i - (K.E.)_f$$

③ Partially Elastic ($0 < e < 1$) $\Rightarrow$



$$u_1 > u_2$$

$m_1 + m_2$

$$m_1 u_1 + m_2 u_2 = m_1 v_1 + m_2 v_2 \quad \text{--- (1)}$$

$$\text{Vel. of sep.} = e (\text{Vel. of approach})$$

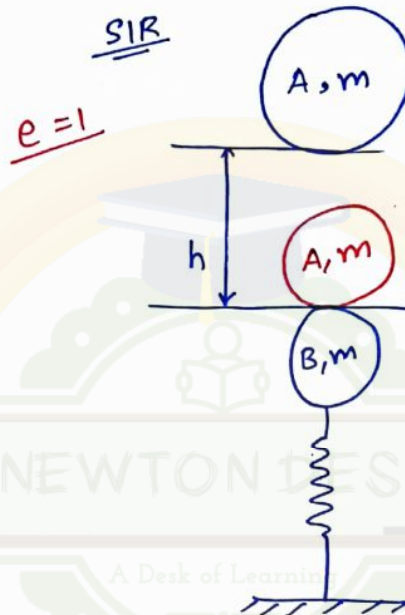
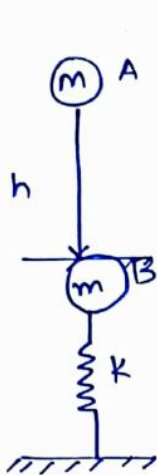
$$v_2 - v_1 = e (u_1 - u_2) \quad \text{--- (2)}$$

$\Rightarrow e = \text{Coefficient of restitution.}$

$\Rightarrow e = \frac{\text{Velocity of separation}}{\text{Velocity of approach}}$

Q A Ball 'A' of mass 'm' falls under gravity through a height h and strikes another Ball 'B' of mass 'm' which is supported at rest by a spring of stiffness 'k'. The velocity of Ball 'A' immediately after the collision is (assume perfect elastic collision) —.

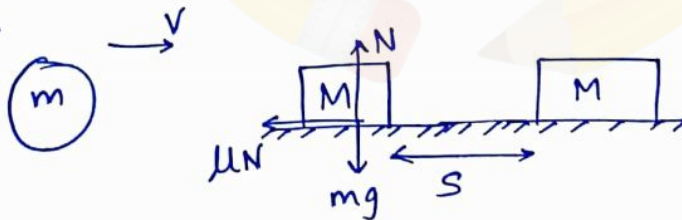
Sol



$$u_A = \sqrt{2gh} = v_B$$

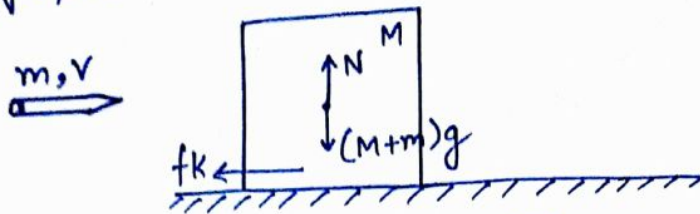
$$u_B = 0 = v_A \quad \underline{\text{Ans}}$$

WB 5.09



(u)

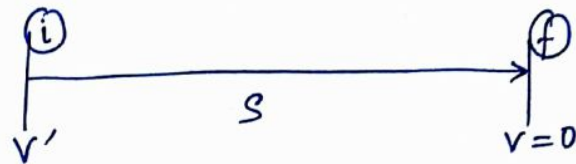
Sol $\frac{1}{2} m v^2 \neq \frac{1}{2} (M+m) (v')^2$



$(m+M)$

$$m v + 0 = (M+m) v'$$

$$v = \left(\frac{M+m}{m} \right) v' \quad \text{--- (1)}$$



$$TWD = \Delta(K.E.)$$

$$TWD = \cancel{W_{(M+m)g}} + \cancel{W_N} + W_{f_k} = -f_k s$$

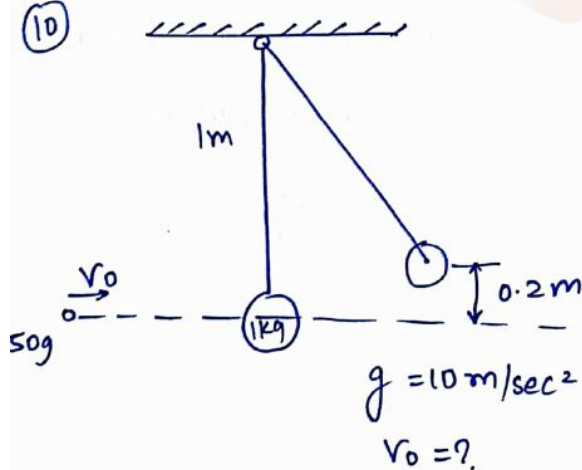
$$\times \mu (M+m) g s = 0 \times \frac{1}{2} (M+m) (v')^2$$

$$v' = \sqrt{2 \mu g s} \quad \text{--- (2)}$$

from (1) & (2)

$$v = \left(\frac{M+m}{m} \right) \sqrt{2 \mu g s}$$

(10)



$(m+M)$

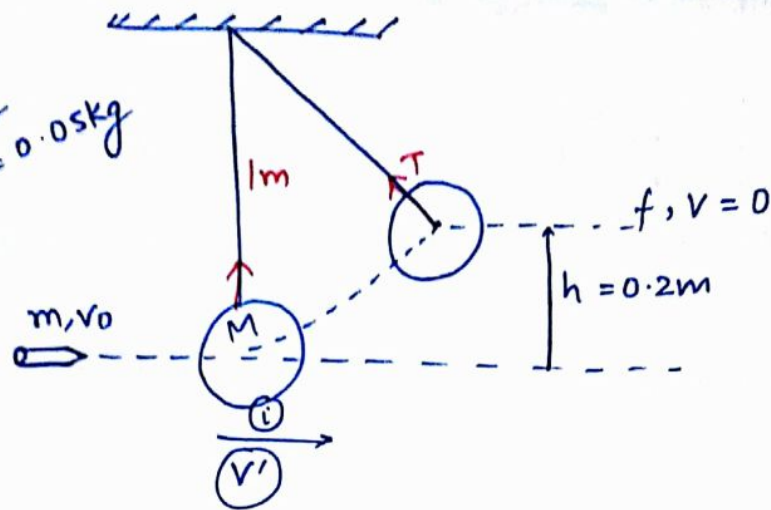
$$m v + 0 = (M+m) v'$$

$$v = \left(\frac{M+m}{m} \right) v'$$

Q 5.10 →

$$M = 1\text{kg}$$

$$m = 50\text{g} = 0.05\text{kg}$$



$(M+m)$

$$m v_0 + 0 = (M+m) v'$$

$$v_0 = \left(\frac{M+m}{m} \right) v' \quad \text{--- (1)}$$

$$TWD = W_{(M+m)g} + W_T = \Delta(K.E.)$$

$$(M+m)gh = 0 + \frac{1}{2} (M+m) (v')^2$$

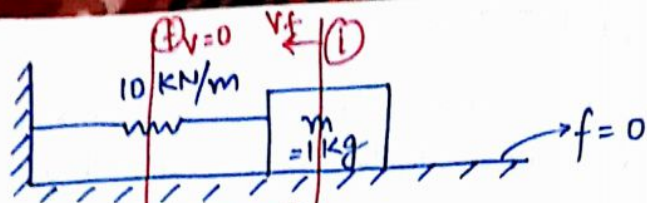
$$v' = \sqrt{2gh} = \sqrt{2 \times 10 \times 0.2} = 2\text{m/s} \quad \text{--- (2)}$$

from (1) & (2)

$$v_0 = \frac{1.05}{0.05} \times 2 = 42\text{ m/s}.$$

Q7 A single degree of freedom having mass 1kg and stiffness 10kN/m, initially at rest is subjected to an impulsive force of 5kN for 10^{-4} sec. The amplitude in mm of the resulting free vibrⁿ is —.

Sol $\vec{F} = \frac{d\vec{P}}{dt}$



$$F_{\text{im}} = 5 \times 10^3$$

$$t = 10^{-4} \text{ sec}$$

$$\text{Impulse} = Ft = 5 \times 10^3 \times 10^{-4}$$

$$= \Delta P = P_f - P_i \rightarrow 0$$

$$\Rightarrow 0.5 = m v_f$$

Before collision

$$\frac{0.5}{1} = v_f = 0.5 \text{ (m/sec)}$$

$$\frac{1}{2} m v_f^2 = \frac{1}{2} k x_{\text{max}}^2$$

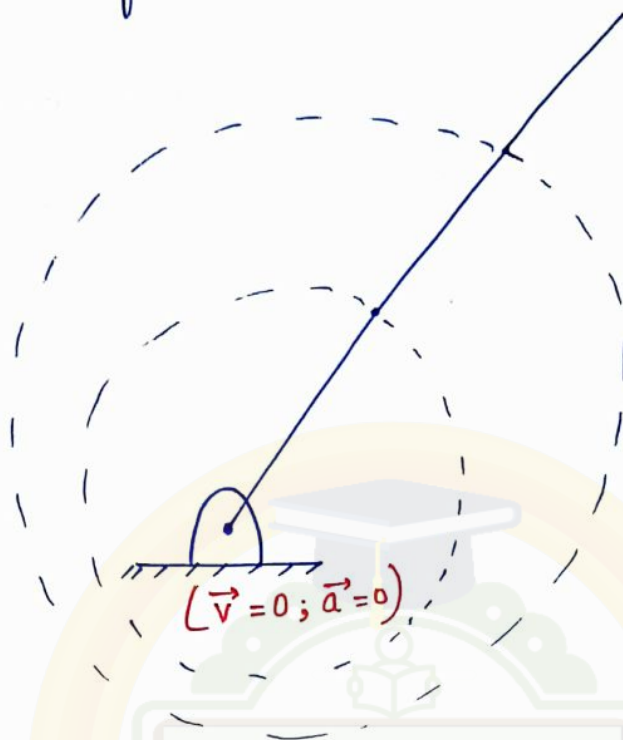
$$1 \times 0.5^2 = 10 \times 10^3 \times x_{\text{max}}^2$$

$$x_{\text{max}} = 0.005 \text{ m} = \underline{\underline{5 \text{ mm}}}$$

07/10/2016

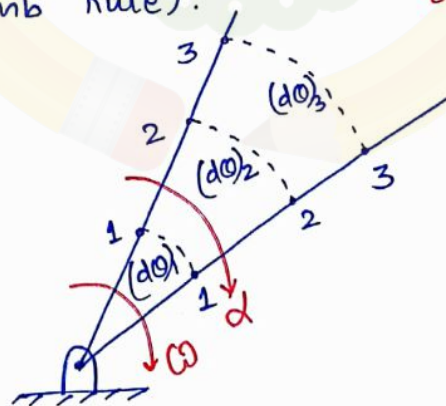
Rotation

In Rotation, all the particles will be in circular motion and their centres will lie on a line which will / should remain fixed called axis of Rotation.



Kinematics $\Rightarrow$

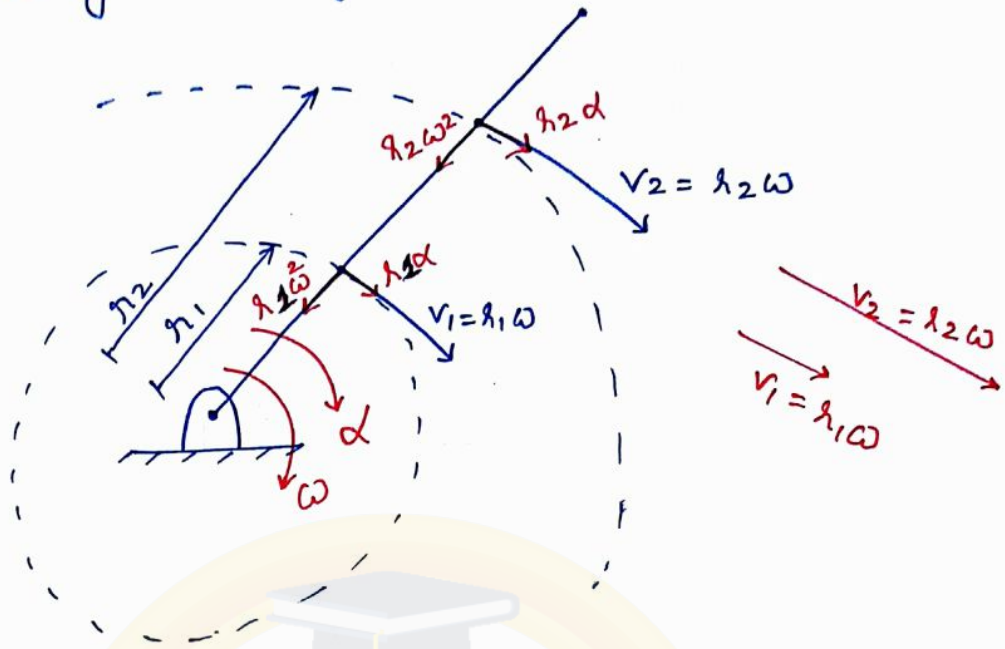
$\vec{\omega}$ and $\vec{\alpha}$ of each and every particle on a Rotating Body remains same and will be along axis of rotation (decided by Right hand thumb Rule).



$$(d\theta)_1 = (d\theta)_2 = (d\theta)_3 = d\theta \rightarrow \text{during time } dt$$

$$\left. \begin{aligned} \vec{\omega}_1 &= \vec{\omega}_2 = \vec{\omega}_3 = \vec{\omega} \\ \vec{\alpha}_1 &= \vec{\alpha}_2 = \vec{\alpha}_3 = \vec{\alpha} \end{aligned} \right] \text{ at an instant}$$

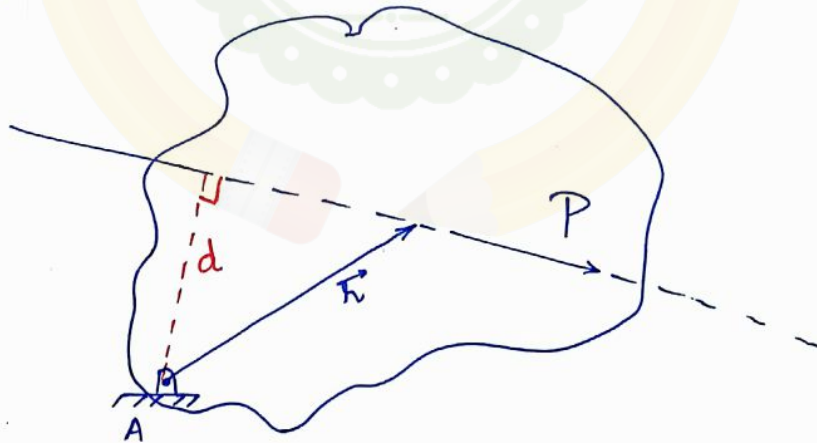
velocity vector and accel. vector ($\vec{v}$ & $\vec{a}$) on a rotating body is dependent on its location from axis of rotation and can be calculated by considering its circular motion.



* Moment of Linear Momentum (Angular Momentum): $\vec{L} \Rightarrow$

$$\vec{L} = \vec{r} \times \vec{P}$$

$$\vec{P} = m \vec{V}_{cm}$$



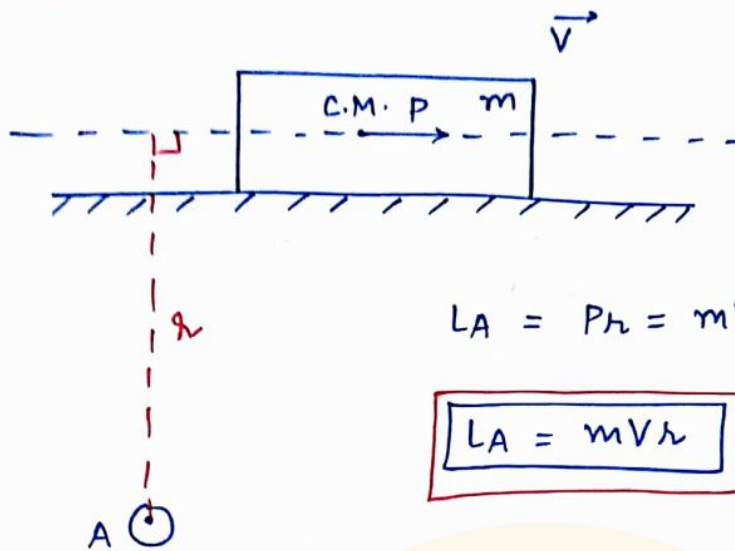
$$\vec{L}_A = \vec{r} \times \vec{P}$$

$$L_A = (P)d = m v_{cm} d$$

Dirⁿ $\Rightarrow$ $\perp^r$ inward through A

(i) Recti. Trans.

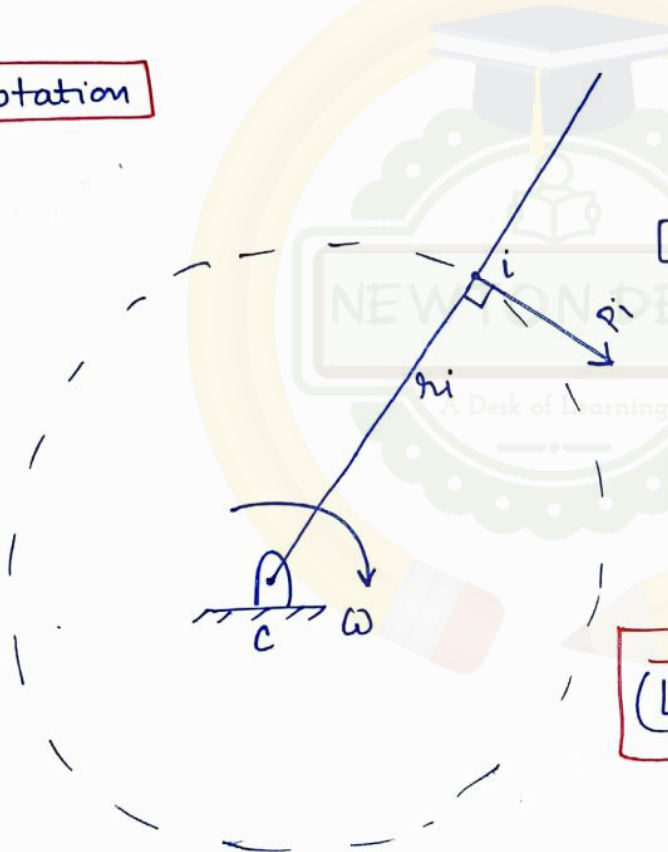
$$\vec{P} = m \vec{V}_{CM} = m \vec{V}$$



$$L_A = P h = m V_{CM} h$$

$$L_A = m V h$$

(ii) Rotation



$$[\vec{L}_i]_C = \vec{r}_i \times \vec{p}_i \quad \left| \quad \vec{p}_i = m \vec{v}_i \right.$$

$$[L_i]_C = p_i r_i = m_i v_i r_i = m_i r_i^2 \omega$$

$$\sum_i [L_i]_C = [L_{Body}]_{axis} = \sum m_i r_i^2 \omega$$

$$(\vec{L})_{axis} = I_{axis} \vec{\omega}$$

$$\left[\because I_{axis} = \sum m_i r_i^2 \right]$$

* Kinetic Energy of a Rotating Body:-

$$(KE)_i = \frac{1}{2} m_i v_i^2 = \frac{1}{2} m_i h_i^2 \omega^2$$

$$\sum_i (KE)_i = (KE)_{\text{Body}} = \frac{1}{2} \omega^2 \sum_i m_i h_i^2$$

$$\boxed{KE = \frac{1}{2} I \omega^2}$$

Similarly, $(\vec{T}_{\text{axis}})_{\text{ext.}} = I_{\text{axis}} \vec{\alpha}$

$$\frac{d\vec{L}}{dt} = \frac{d(\vec{h} \times \vec{P})}{dt} = \frac{d\vec{h}}{dt} \times \vec{P} + \vec{h} \times \frac{d\vec{P}}{dt}$$

$$\frac{d\vec{L}}{dt} = \cancel{\vec{v} \times \vec{P}}^0 + \vec{h} \times \vec{F} = \vec{T}$$

$$\boxed{\frac{d\vec{L}}{dt} = \vec{T} = \vec{T}_{\text{ext}}}$$

Conservation of $\vec{L}$ $\Rightarrow \vec{L} = \text{constant} \Rightarrow d\vec{L} = 0$

only when $\vec{T} = 0$

$$\Rightarrow \vec{T} = \frac{d\vec{L}}{dt}$$

$$\Rightarrow \boxed{\vec{T} dt} = \boxed{d\vec{L}}$$

↓
Angular
Impulse

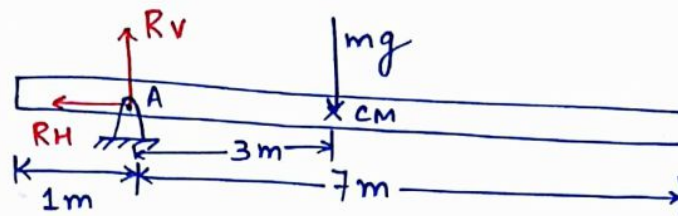
Power = rate of doing work

$$= \frac{\vec{F} \cdot d\vec{s}}{dt} = \boxed{\vec{F} \cdot \vec{v}}$$

$$\text{and } = \frac{T d\theta}{dt} = T\omega = \boxed{\frac{T 2\pi N}{60}}$$

Q. A uniform slender rod of mass 3 kg and length 8 m rotates in a vertical plane hinged at 1 m from one of its ends as shown in figure. Find angular acceleration of rod at the position shown in figure.

$$m = 3 \text{ kg}$$



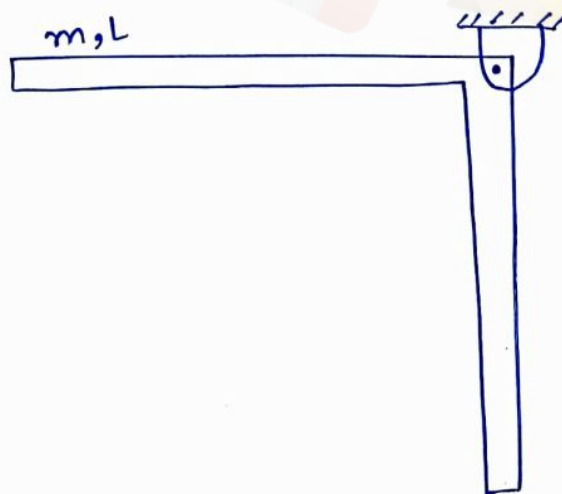
$$\tau_A = I_A \alpha$$

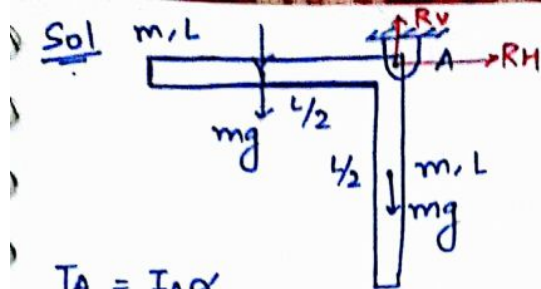
$$mg \times 3 = [I_{CM} + m \times 3^2] \times \alpha$$

$$3 \times 9.81 \times 3 = \left[\frac{3 \times 8^2}{12} + 3 \times 3^2 \right] \times \alpha$$

$$\Rightarrow \alpha = 2.05 \text{ rad/sec}^2$$

Q. A uniform L shaped member with each of its limb has mass 'm' and length 'l' as shown in figure. Find angular accⁿ of the member at the position shown.





$$T_A = I_A \alpha$$

$$\Rightarrow mg \times \frac{L}{2} = \left[\frac{mL^2}{3} + \frac{mL^2}{3} \right] \times \alpha$$

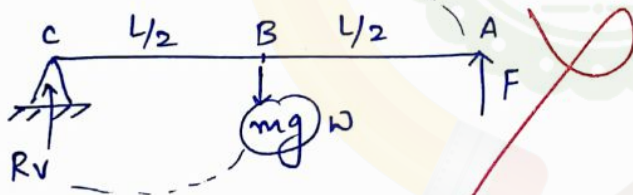
$$\Rightarrow \cancel{mg} \times \cancel{L}/2 = \frac{2\cancel{m}\cancel{L}^2}{3} \times \alpha$$

$$\Rightarrow \boxed{\alpha = \frac{3g}{4L}}$$

Q> A uniform beam of weight 'W' and length 'L' is supported as shown in figure. The moment force 'F' is removed, the R_{xn} at hinge is :-



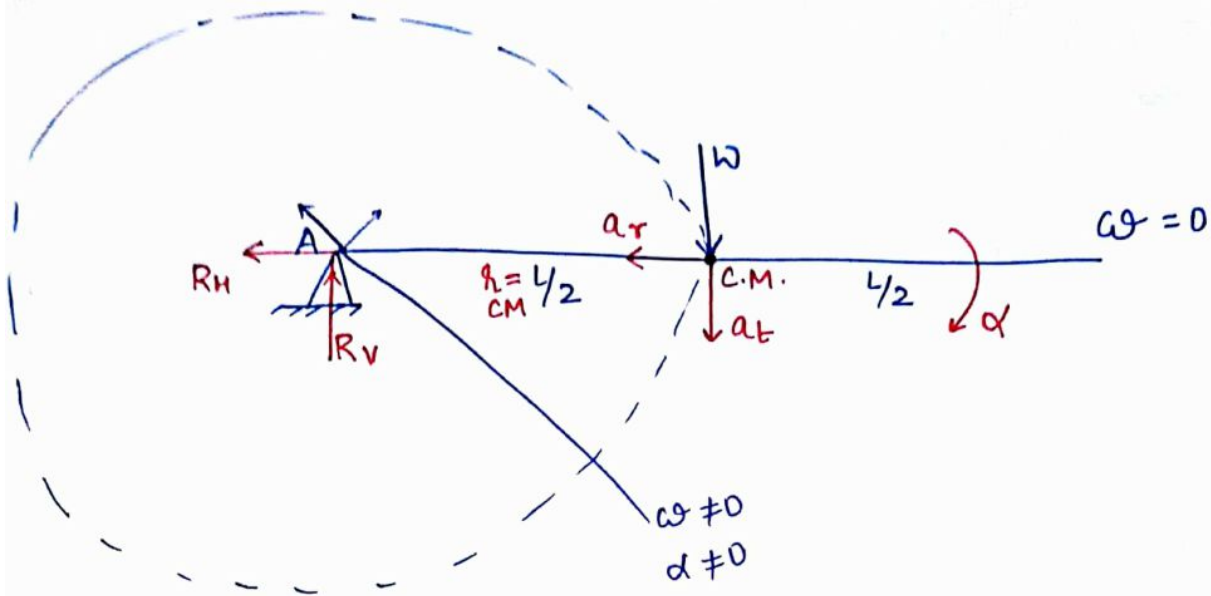
Sol



$$\cancel{W}L/2 + RL = 0$$

$$\cancel{W}L/2 = R\cancel{L}$$

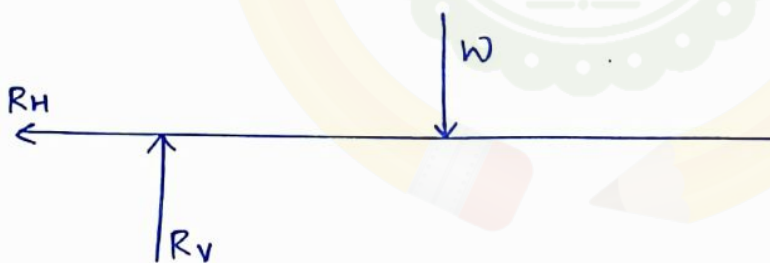
$$R = W/2$$



$$\tau_A = I_A \alpha$$

$$W \times L/2 = \frac{W}{g} \frac{L^2}{3} \times \alpha$$

$$\alpha = \frac{3g}{2L}$$



$$W - R_V = m a_{cm(y)} \quad \text{--- 2nd}$$

$$W - R_V = \frac{W}{g} (a_{cm})_t = \frac{W}{g} r_{cm} \alpha = \frac{W}{g} \times L/2 \times \frac{3g}{2L} = \frac{3W}{4}$$

$$R_V = W - \frac{3W}{4} = W/4$$

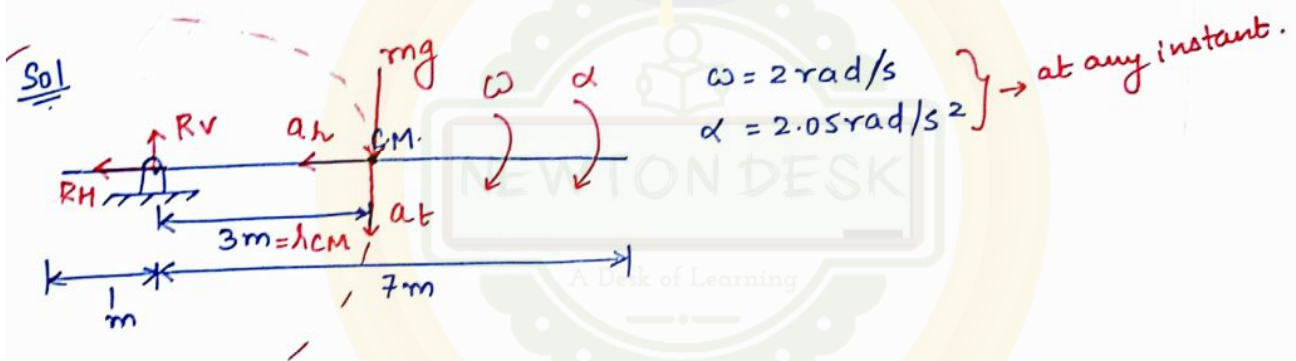
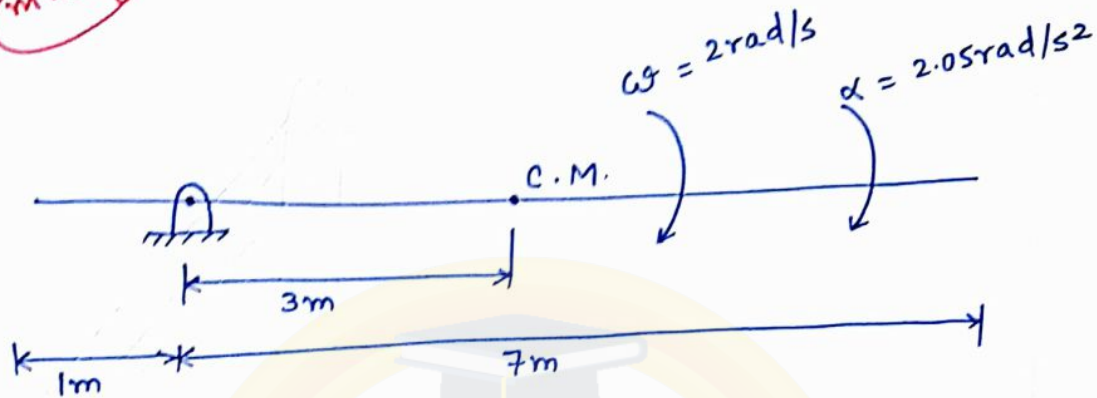
$$R_H = m a_{CM}(-x) \quad \text{--- 2nd}$$

$$R_H = m(a_{CM})_x = m l_{CM} \omega^2$$

$$R_H = 0 \quad [\because \omega = 0]$$

Q. Find Reaction at hinge for the system shown in figure.

$$m = 3 \text{ kg}$$



$$mg - R_v = m a_{CM}(y) \quad \text{--- 2nd}$$

$$mg - R_v = m(a_{CM})_y = m l_{CM} \alpha$$

$$3 \times 9.81 - R_v = 3 \times 3 \times 2.05 \Rightarrow R_v = 10.98 \text{ N}$$

$$R_H = m a_{CM}(-x) \quad \text{--- 2nd}$$

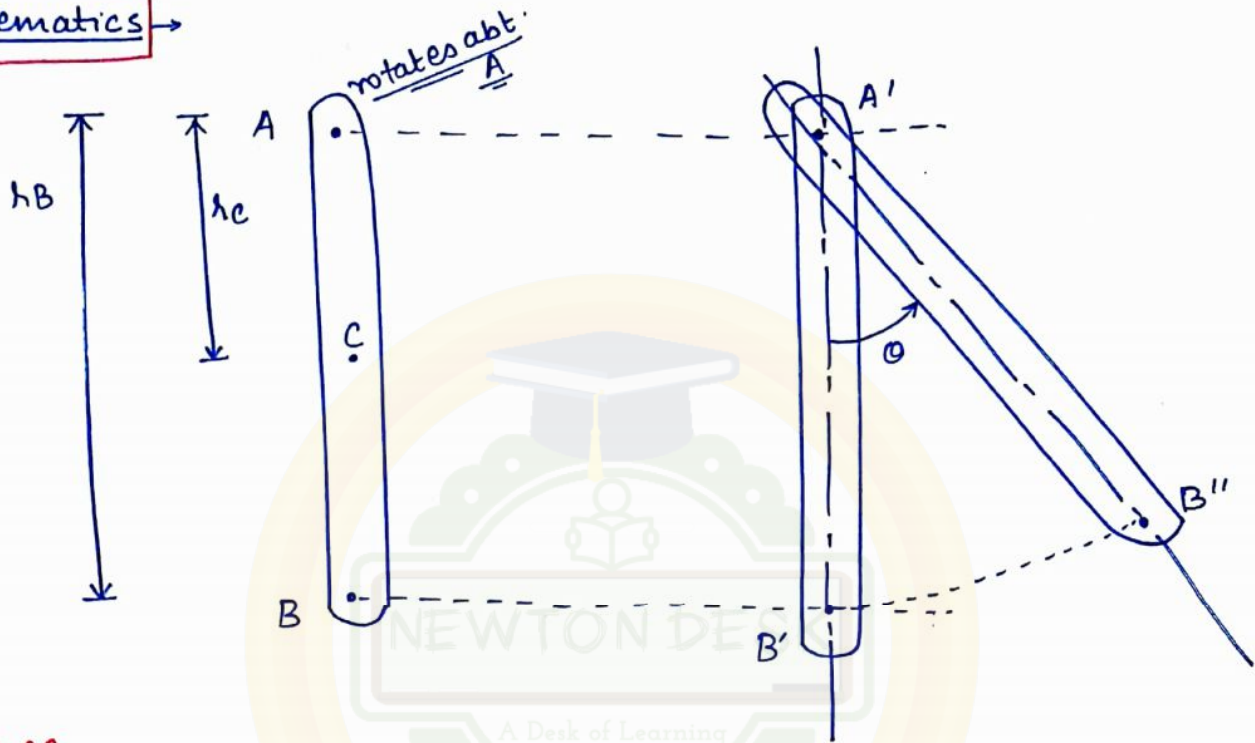
$$R_H = m(a_{CM})_x = m l_{CM} \omega^2 = 3 \times 3 \times 2^2 = 36 \text{ N}$$

* **GENERAL MOTION** :- If a Rigid body rotates as well as translates simultaneously, such a motion is called General Motion.

Example - (1) Rolling Motion.

- (2) Motion of connecting rod.
- (3) Motion of ladder, etc.

Kinematics →



Hence,

$$[\vec{S}, \vec{V}, \vec{a}]_{\text{General motion}} = [\vec{S}, \vec{V}, \vec{a}]_{\text{Trans.}} + [\vec{S}, \vec{V}, \vec{a}]_{\text{Rotation}}$$

$$\vec{S}_B = \vec{S}_A + \vec{S}_{B/A} \quad [\vec{S}_{B/A} = \lambda_B \vec{\omega}]$$

$$\vec{V}_B = \vec{V}_A + \vec{V}_{B/A} \quad [\vec{V}_{B/A} = \lambda_B \vec{\omega}]$$

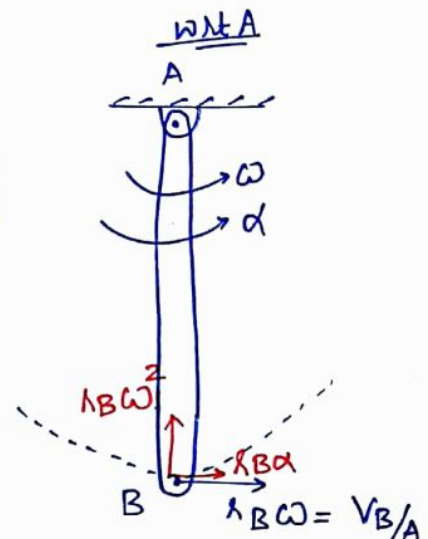
$$\vec{a}_B = \vec{a}_A + \vec{a}_{B/A} \quad [\vec{a}_{B/A} = \lambda_B \vec{\omega}^2 + \lambda_B \vec{\alpha}]$$

simil.

$$\vec{S}_C = \vec{S}_A + \vec{S}_{C/A} \quad [\vec{S}_{C/A} = \lambda_C \vec{\omega}]$$

$$\vec{V}_C = \vec{V}_A + \vec{V}_{C/A} \quad [\vec{V}_{C/A} = \lambda_C \vec{\omega}]$$

$$\vec{a}_C = \vec{a}_A + \vec{a}_{C/A} \quad [\vec{a}_{C/A} = \lambda_C \vec{\omega}^2 + \lambda_C \vec{\alpha}]$$



Rolling $\Rightarrow$

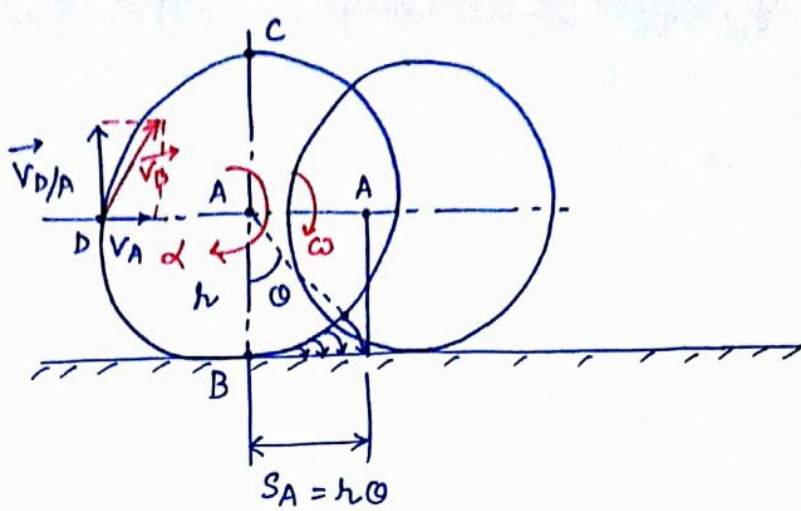
$$\vec{S}_A = \hbar \hat{i}$$

$$\vec{v}_A = r \frac{d\theta}{dt} \hat{i}$$

$$\vec{V}_A = R \omega \hat{i}$$

$$\vec{a}_A = \frac{r d\omega}{dt} \hat{i}$$

$$\vec{a}_A = h\alpha \hat{i}$$



$$\vec{V}_B = \vec{V}_A + \vec{V}_{B/A} = 1\omega \hat{i} + 1\omega(-\hat{i}) = 0$$

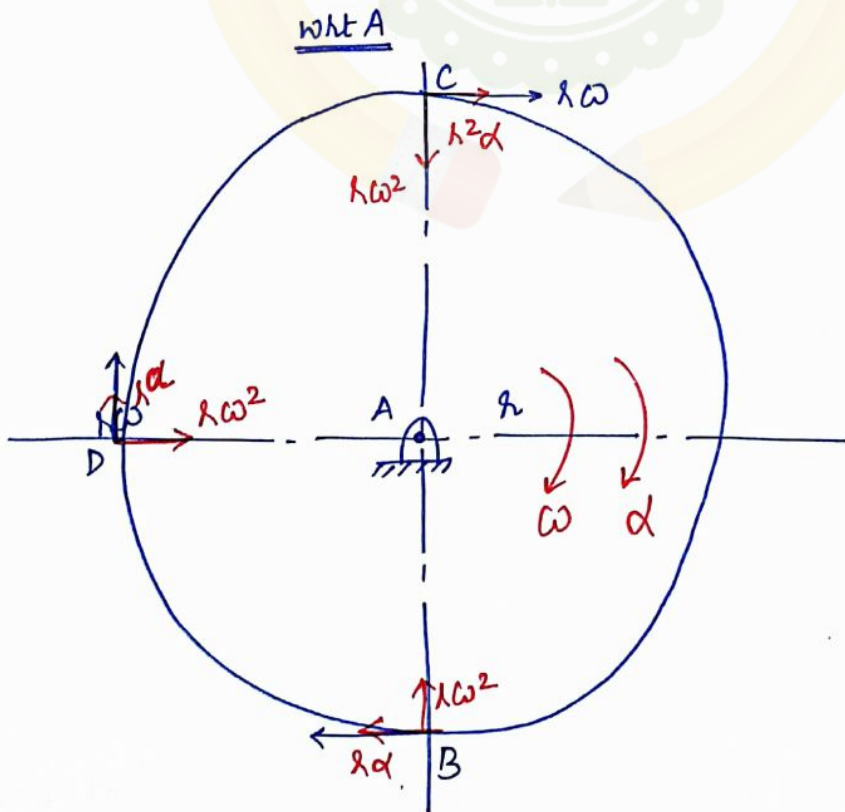
$$\vec{V}_C = \vec{V}_A + \vec{V}_{C/A} = \lambda \omega \hat{i} + \lambda \omega (\hat{i}) = 2\lambda \omega \hat{i}$$

$$\vec{V}_D = \vec{V}_A + \vec{V}_{D/A} = r\omega\hat{i} + r\omega(\hat{j})$$

$$\vec{a}_B = \vec{a}_A + \vec{a}_{B/A} = 1\alpha \hat{i} + 1\alpha(-\hat{i}) + 1\omega^2(\hat{j}) = 1\omega^2(\hat{j}) \neq 0$$

$$\vec{a}_C = \vec{a}_A + \vec{a}_{C/A} = \lambda \alpha \hat{i} + \lambda \alpha (\hat{i}) + \lambda \omega^2 (-\hat{j}) = 2\lambda \alpha \hat{i} + \lambda \omega^2 (-\hat{j})$$

$$\vec{a}_D = \vec{a}_A + \vec{a}_{D/A} = r\alpha \hat{i} + r\omega^2(\hat{i}) + r\alpha(\hat{j})$$



(i) if $s_A = r\omega \rightarrow$ pure rolling (DOF = 1) $\rightarrow$ free rolling
In 1 rev, $\omega = 2\pi$, $s_A = 2\pi r$

(ii) if $s_A > r\omega \rightarrow$ skidding (DOF = 2)

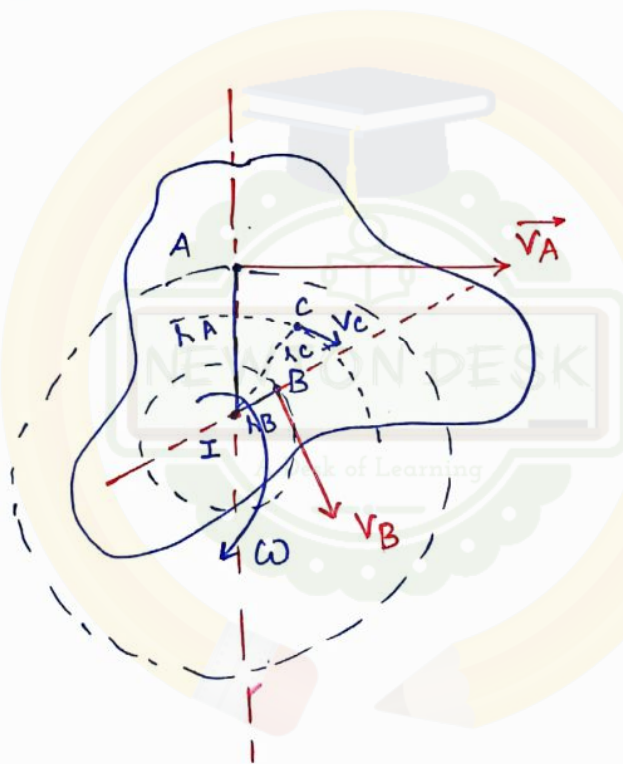
(iii) if $s_A < r\omega \rightarrow$ slipping (DOF = 2)

* **Instantaneous Centre / Axis** $\rightarrow$ It is a point / line in space about which a body in general motion can be assumed as in pure rotation to find the velocities.

Note $\rightarrow$ (i) I-centre has 0 velocity But not 0 acceleration.

* **Location of I-Cs** $\rightarrow$

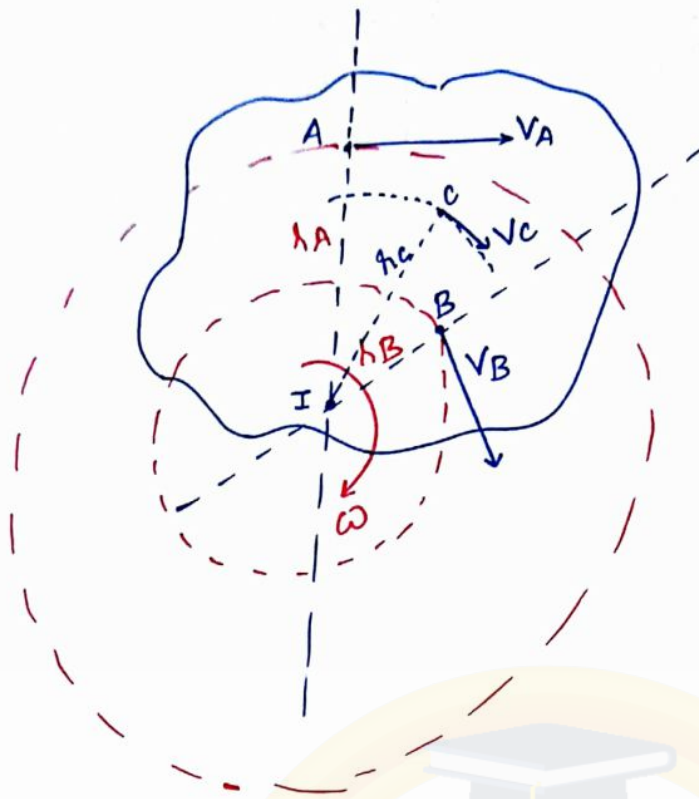
Case 1 :-



$$V_A = r_A \omega$$

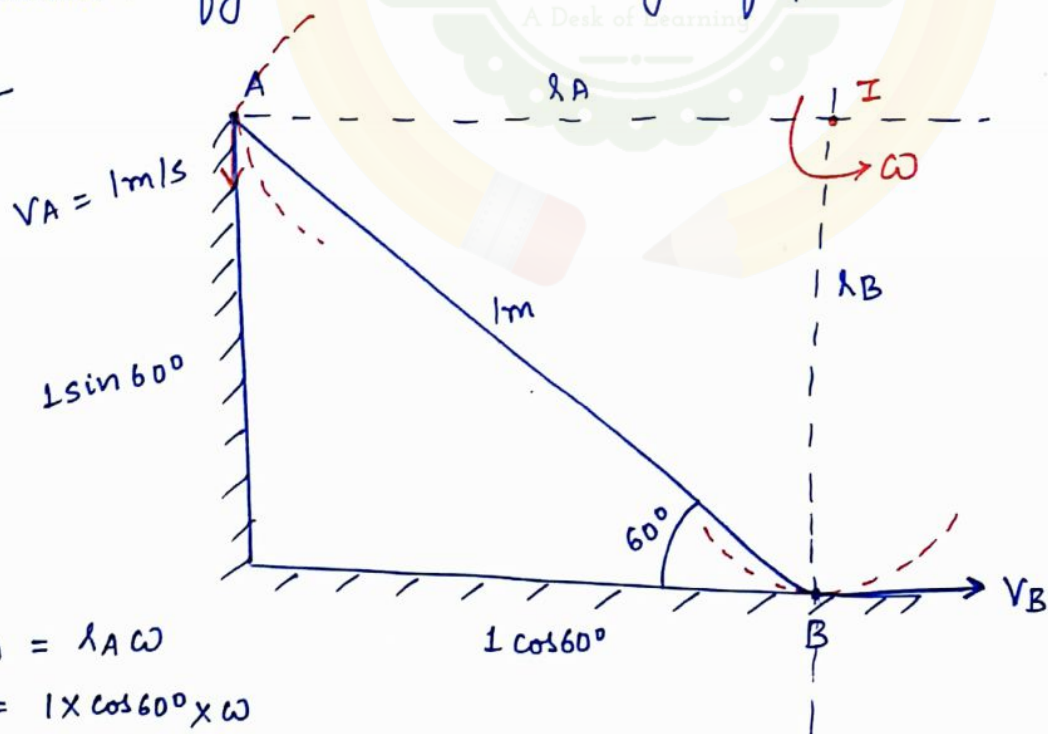
$$V_B = r_B \omega$$

$$V_C = r_C \omega$$



Q. A ladder AB of length 1m is sliding in a corner as shown in figure. at an instant; when the ladder makes an angle 60° with the horizontal, velocity of pt. A is 1m/sec as shown in figure. Then the velocity of pt. B is

Sol



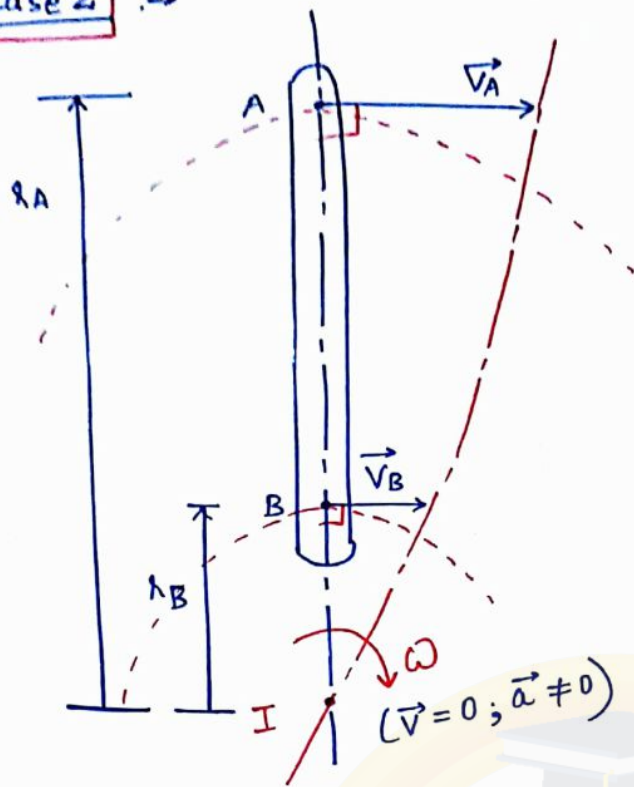
$$V_A = \lambda_A \omega$$

$$1 = 1 \times \cos 60^\circ \times \omega$$

$$\omega = 2 \text{ rad/s}; \text{ AC } \omega$$

$$V_B = \lambda_B \omega = 1 \times \sin 60^\circ \times 2 = \sqrt{3} \text{ m/s}$$

Case 2 :→



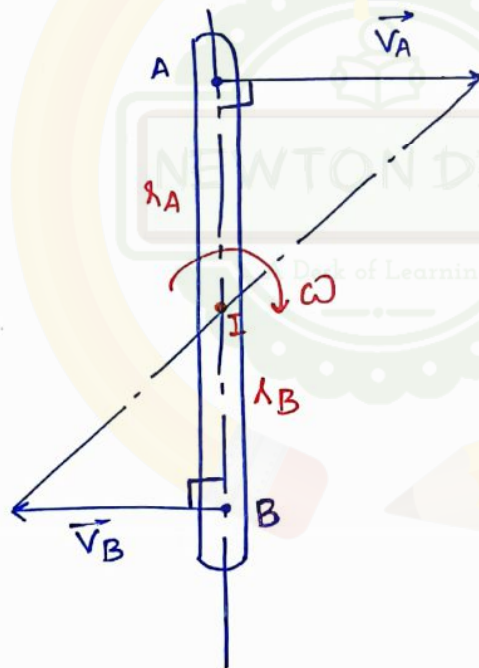
$$v_A = \lambda_A \omega$$

$$v_B = \lambda_B \omega$$

$$\frac{v_A}{v_B} = \frac{\lambda_A}{\lambda_B} \quad \text{--- (1)}$$

$$AB = \lambda_A - \lambda_B \quad \text{--- (2)}$$

Case 3 :→



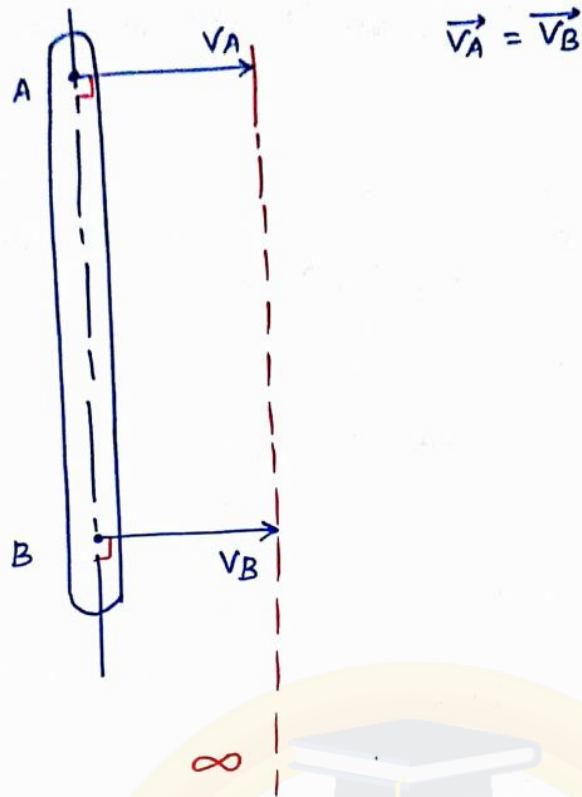
$$v_A = \lambda_A \omega$$

$$v_B = \lambda_B \omega$$

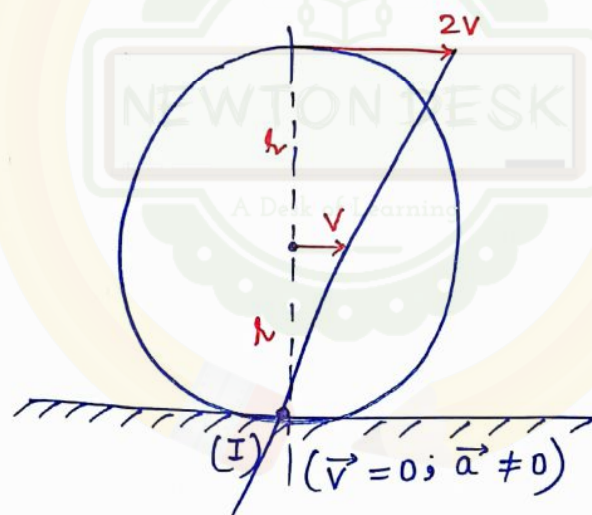
$$\frac{v_A}{v_B} = \frac{\lambda_A}{\lambda_B} \quad \text{--- (1)}$$

$$AB = \lambda_A + \lambda_B \quad \text{--- (2)}$$

Case 4 $\rightarrow$



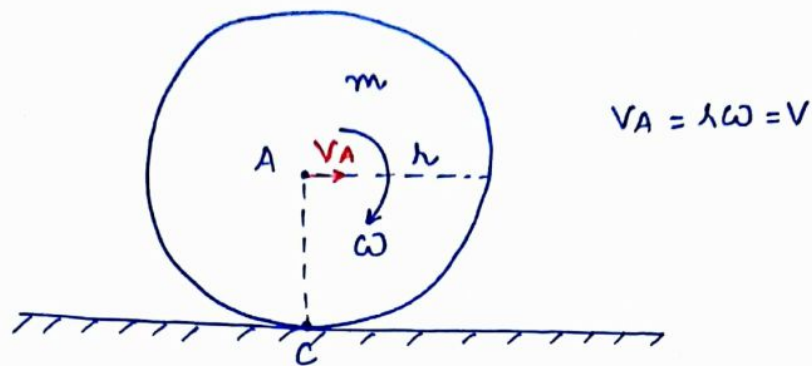
Application $\rightarrow$



$$(KE)_{\text{Rolling}} = (KE)_{\text{trans.}} + (KE)_{\text{Rotation}}$$

"(N.P.)"

$$(K.E)_{\text{Rolling}} = (K.E)_{\text{Trans.}} + (K.E)_{\text{Rotation}}$$



$$(K.E) = \frac{1}{2} m v^2 + \frac{1}{2} I_A \omega^2$$

$$= \frac{1}{2} m h^2 \omega^2 + \frac{1}{2} I_A \omega^2$$

$$= \frac{1}{2} \omega^2 [m h^2 + I_A]$$

$$= \frac{1}{2} I_C \omega^2$$

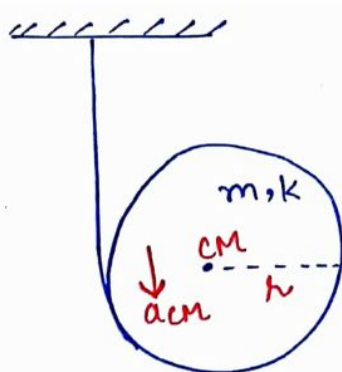
$$(K.E)_{\text{rolling}} = \frac{1}{2} I_C \omega^2$$

$$I_C = I_C \omega$$

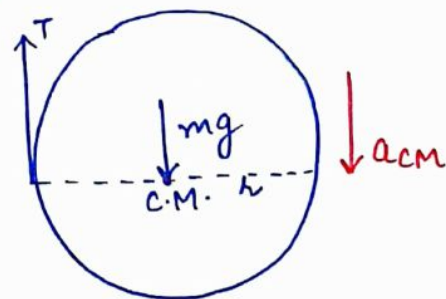
Q A reel of mass 'm', radius 'r' is rolling down smoothly from rest with one end of the thread wound on it held in the ceiling as shown in fig. Neglect wrapped thickness of thread. Find :- (a) Tension in the thread.

(b) acceleration of COM of reel.

Sol



wrapped



$$mg - T = m a_{cm} \text{ (y)} \quad \text{--- 2nd}$$

$$mg - T = m h \alpha \quad \text{--- (1)}$$

$$[\text{Torque}]_{cm} = I_{cm} \alpha$$

$$Tl = mk^2 \alpha$$

$$\alpha = \frac{Tl}{mk^2} \quad \text{--- (2)}$$

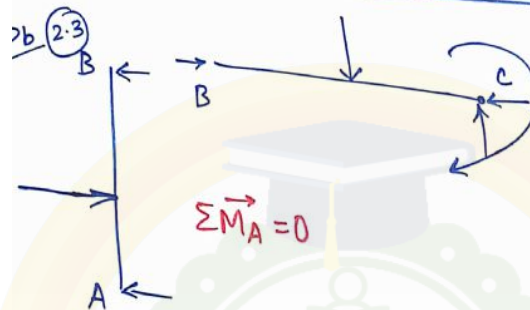
from (1) & (2)

$$mg - T = \frac{ml^2 T}{mk^2}$$

$$T = \frac{mgk^2}{l^2 + k^2}$$

$$a_{cm} = l\alpha = \frac{Tl^2}{mk^2} = \frac{mgk^2}{l^2 + k^2} \times \frac{l^2}{mk^2}$$

$$a_{cm} = \frac{gl^2}{l^2 + k^2}$$



NEWTON DESK

A Desk of Learning